

INVESTOR BRIEF · DRAFT V1 · 3 OCT 2026

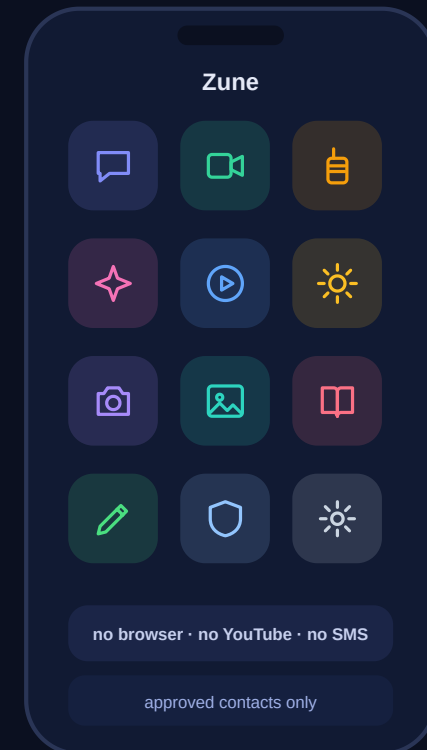
Zune

working name (codename; trademark clearance pending)

A kids-only Android.
No browser. No YouTube exit. No cellular calling.
Safe by construction, not by policy.

A custom Android 17 OS image and a browser-based parent portal, flashed onto qualified Pixel phones. Text, voice and video only with parent-approved contacts, a kid-safe AI tutor, curated learning videos, weather lessons and more.

India first English only ~200-user first phase Software-led, Pixel first Own hardware later



A phone where the unsafe things simply do not exist

What we are building, why it can win, what could kill it

What it is

A custom **Android 17 (AOSP)** OS image for children plus a **parent web portal**. The child's phone has **no web browser, no route to YouTube, no cellular calls or SMS**. It has WhatsApp-style 1:1 messaging, voice and video calls with parent-approved contacts, a kid-safe AI assistant (text + images), curated learning videos, weather with lessons, walkie-talkie, camera, photos, journal, notebook and a book reader.

Sold as **software plus a flash-and-hand-over service** on the customer's own qualified phone. **Version 1:** we flash and re-lock the phone and hand it back (about 200 users, India, Pixel 10a/9a). Later: customer self-install, then **our own hardware**.

0

browsers and routes to YouTube
enforced in the OS image

Android 17

AOSP android-17.0.0_r1
released 2026-06-16

~200

users in the first phase
customer-owned Pixels, India

Pixel 10a/9a

first supported phones
re-locked with our own key

2 stages

S1: everything exists, S2:
better
no dates promised yet

21

research reports written
12 through a skeptic pass

The bet

Parents should not have to police a general-purpose phone. Remove the dangerous capabilities from the OS, and give parents a portal for everything that remains.

Why it can win

- **Enforcement in the image, not an app.** No browser and no YouTube route exist on the device; the bootloader is re-locked so only our signed images boot.
- **Contact graph, not phone numbers.** 1:1 text, voice, video and walkie-talkie only with contacts both families approve; no groups, no strangers.
- **The moat is the whole product, not the lock.** Kid-only contact graph, AI tutor, curated learning content and portal. A no-browser Pixel is already sold by Pinwheel (US) and a \$199 phone by Freckle.
- **Software-led model.** No hardware inventory; recurring revenue from cloud services (portal, AI, comms, content).
- **Timing.** Android 17 ships a native supervision framework; child-safety law is tightening worldwide.

What could kill it

- **Google firmware licence** for commercially re-distributing Pixel firmware is unread and unresolved. **High**
- **Security patching.** AOSP source drops twice a year; public fixes reach downstream months late (~127-day median seen). **High**
- **India child-privacy law (DPDP s.9(3)).** Under-18 is a child; parent-visible monitoring has no clear exemption. Counsel first. **High**
- **Re-lock support.** Only Pixel has vendor-documented custom-key re-lock; Snapdragon phones are unproven. **High**
- **Willingness to pay in India.** ~Rs 1,250/month proposed vs JioShield at Rs 1,000/year; ~40,000 parents must be reached to sign ~200. **High**
- **Killable dependencies.** YouTube must be optional (Videos work without it); the LLM vendor's approval for a child-facing product is gate one. **Med**

Details, mitigations and evidence quality: Risk register and Evidence status pages.

Every kids' phone today is policy on top of an open phone

Where Zune sits, and why now

1. Screen-time and filter apps

Family Link, Qustodio, Net Nanny, Bark app

- ✓ Policy layer on an open phone
- ✗ Child can still reach a browser and YouTube
- ✗ Defeated by reset, safe mode, sideloading, other apps
- ✗ Parents police an adult-grade device

2. Managed kids' phones

Pinwheel, Bark Phone, Gabb, Troomi

- ✓ Rebadged commodity phones plus managed software
- ✗ Browser-free by policy, not by image (our reading; to validate)
- ✓ Subscription \$15-20/mo; devices \$100-240 (secondary sources)
- ✗ Pinwheel does not install on your own phone

3. Zune

Capabilities are absent from the OS image

- ✓ No browser or YouTube exit in the image; re-locked bootloader
- ✓ Contact graph both families approve; no strangers, no groups
- ✓ Parent portal in any browser; signed policy; offline-safe
- ✓ Software-led: customer brings the phone

WHERE ENFORCEMENT LIVES

Layer	Filter apps	Managed kids' phones*	Zune
App (filters, screen time)	enforces here	enforces here	enforces here
Device policy (MDM / supervision)	policy only	enforces here	enforces here
OS image (what exists on the phone)	stock / not ours	stock / not ours	enforces here
Bootloader (what can boot)	stock / not ours	stock / not ours	enforces here
Updates (who ships fixes)	stock / not ours	stock / not ours	enforces here

*Our reading of how managed kids' phones work (stock firmware plus managed-device software; R02 F9, inferred). To validate.

WHY NOW

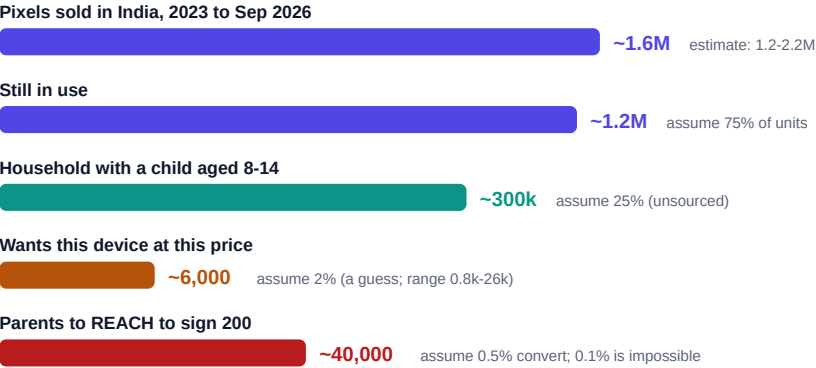
- **Android 17** adds a native supervision framework (SupervisionManager, ROLE_SUPERVISION) we can anchor on. R01 F6
- **AI tutors** are now credible for children when wrapped in age policy, moderation and parent visibility. R07
- **Regulation** is moving toward duty of care for minors (India DPDP; US and EU rules). R11, R19
- **AOSP is open** and Pixel offers a documented route to custom images re-locked with our own key. R02

Competitor facts come from secondary sources (R02 F9); the Market page validates and extends them.

Market: heavy parental concern, a thin paying niche

Honest sizing; analyst market-size numbers are not investor-grade, so we show the funnel instead

FINDING ~200 FAMILIES (R20 + MARKET RESEARCH; EVERY STEP LABELLED)



- **Path A (owner already has a Pixel):** ~200 is possible only through targeted outreach; the pool is thousands nationally, not hundreds of thousands.
- **Path B (parent buys a Pixel for the child):** year-1 cost about Rs 59-71k (~\$610-730) against Rs 1,000/yr JioShield or free Family Link.
- **Bottom line:** the binding constraint is willingness to pay ~Rs 1,250/month, not Pixel supply. Treat the 200-user phase as a test of price.

~168M

children aged 8-14 in India (Zune core band)
estimate
estimate from age bands

22%

of India's 2025 phone units cost over INR 30k
reported
Counterpoint; Pixel was ~2% of >INR 45k

~0.43M

Pixels sold in India per year (2025) **estimate**
no published figure; analyst-share arithmetic

\$1.6-2.8B

global parental-control software, 2025
reported
analyst range; not investor-grade

SELECTED SIZING AND ANCHORS

Metric	Value	Source	Basis
Children aged 8-14 (Zune core band)	~168.5M	Own arithmetic on UN WPP 2024 ...	estimate
Children under 18	~460M (about 31.5% of population)	UNICEF India Annual Report 2025	secondary
Households with at least one smartphone	85.5%	MoSPI/NSO Comprehensive Modular ...	primary
Premium share (above INR 30,000)	22% of units in 2025, +11% YoY, the fastest-growing band ...	Counterpoint Research (via ...)	secondary
Estimated Pixel units sold in India	2025: ~0.43M (range 0.3-0.5M); cumulative Jan 2023-Sep 2026 ~1.6M (range ...)	Own arithmetic (see funnel)	estimate
Pixel models officially sold in India, INR launch prices	Pixel 9a: Rs 49,999 (8/128GB), Rs 56,999 (8/256GB), launched 16 Apr 2025.	BusinessToday; Smartprix ...	secondary
Parental control software, global	2025: \$2.8B (Grand View Research; \$6.1B by 2033, 10.4% CAGR) ...	Grand View Research ...	secondary
Real revenue: Qustodio and Qoria	Qustodio consumer ARR ~US\$21.1M (+26% in the half; ~34% annualised).	Qoria ASX H1 FY26 results ...	primary
WTP anchor: US subscriptions	Gabb \$19.99-24.99/mo + phone \$149.99-199.99 ...	Company pricing as reported by ...	secondary
WTP anchor: India	JioShield Rs 1,000/year (launched 26-29 Sep 2026) ...	Deccan Herald (JioShield) ...	secondary
Telecom ARPU (anchor for monthly spend)	Airtel Rs 264, Jio Rs 215.6, Vi Rs 195 per month (Q1 FY27)	Bharti Airtel Q1 FY27 release ...	primary

Competition: the concept is validated, the lock is copyable

Closest analogues, substitutes and tailwinds (prices from company pages where available; many from reviewers)

COMPETITORS AND SUBSTITUTES

Competitor	Where	Price	Funding	Gap vs Zune
Pinwheel	US (UK launch too)	Devices \$119 (Genesis 4), \$199 (Slim 6), \$299 (Plus 5) ...	\$3.5M seed (Jan 2022: Drumbeat ...)	This is the closest analogue to Zune's Pixel + custom-OS plan, so 'Pixel + no browser' is not a moat.
Bark (Bark Phone and Bark parental control)	US	Bark Phone service \$39/mo (talk, text, Wi-Fi) to \$79/mo (unlimited ...)	~\$66M: Series B \$10M (Jul 2020) ...	Allows social apps and monitors them, which is the opposite philosophy to Zune's no-browser design.
Gabb	US	Gabb Phone 4 \$149.99, Phone 4 Pro \$199.99 (TechCrunch Jul 2026: from ...)	~\$30.4M over 4 rounds (Dealroom): \$3.0M AIM ...	Locked to a US carrier plan. Philosophy is close to Zune's (no browser) but with no AI assistant ...
Troomi	US	Samsung A17 5G \$249.95 (or \$6.94/mo); plans \$24.95 (Do) ...	~\$11.3M; last round ~\$11M (Aug 2023) ...	Small, US-only. Higher tiers add a browser and apps, which dilutes the safety pitch.
Freckle	US (preorders opened ...)	\$199 device + \$25/mo plan; \$25 preorder deposit.	Backed by Alexis Ohanian's Seven Seven Six ...	Very new and unproven; hardware plus carrier dependency; US only. Its philosophy (no browser ...)
HMD Fusion X1 with Xplora	Europe/UK	EUR 269.99 / GBP 229 (~Rs 26,461 per a conversion in the source) ...	n/a	Parent-managed but still a general phone (apps, browser possible). Xplora scale: 476,000 subscriptions in 2025, +33%.
Light Phone III and Wisephone II (Techless)	US/global	Light Phone III \$699 pre-order, \$799 retail.	Light Phone funding is unreliable.	Not child-specific. Indicates niche willingness to pay \$400-800 for a restrictive phone ...
Google Family Link / Apple Screen Time	Global (incl. India)	Free.	n/a	Layered on a full phone with browser and YouTube. Easy for kids to work around.
Qustodio	Global (India via app)	Basic \$59.95/yr (5 devices); Complete \$104.95/yr ...	Acquired by Family Zone (now Qoria) in 2022 ...	Software-only, so it cannot remove the browser or YouTube. Consumer ARR only ~US\$21M after ~15 years.
Net Nanny / Kidslox / Aura Family	US/global	Net Nanny \$39.99 (1 device) to \$89.99/yr (20 devices) ...	Aura: \$200M Series F at ~\$2.5B (Oct 2021) ...	Software add-ons, with no controlled OS. Aura's value is identity protection, so kids controls are a bundle feature.
Fully Kiosk Browser and generic Android kiosk/launcher apps (QuadNode, Kids Place, AirDroid, FlashGet, Findmykids)	Global/India	Fully Kiosk PLUS: one-time EUR 8.90 / \$10.99 per device.	n/a	Technically easy to copy, which proves the moat must be content, communication, AI and portal, not the lock.
JioShield (new) / Jio Safety Shield on JioBharat and JioPhone Prima	India	Rs 1,000/year; the 10th-anniversary offer gives 10 x Rs 100 recharge ...	n/a	Weakness is for Zune to exploit: it is a control layer on a normal phone, which keeps browser and YouTube.
India kids smartwatches and phones (imoo, boAt, Noise, Easyfone Star)	India	imoo Z1 from Rs 9,490 (Rs 6,499 on one Amazon deal) ...	n/a	Calls and location only, so no AI, learning video or messaging, and a small screen.

WHAT IT MEANS FOR ZUNE

- **Not a moat:** Pinwheel already sells Pixel 9a phones with a no-browser OS at \$599 + \$14.99/mo; Freckle (1 Oct 2026) sells a \$199 no-browser phone. The moat must be communication graph, AI tutor, learning content and portal.
- **Price anchor:** JioShield launched 26-29 Sep 2026 at Rs 1,000/year; Zune's unvalidated ~Rs 1,250/month is ~15x that.
- **Cold start:** Zune kids only talk to approved Zune contacts; Indian children use WhatsApp, so recruit in school or parent clusters.
- **Capital:** Pinwheel raised ~\$6.4M in total; Qustodio's consumer ARR is ~\$21M after ~15 years.

TAILWINDS

- **India: Supreme Court and Centre move to bar** 28 Sep 2026: a bench headed by CJI Surya Kant asked the Centre to consider statutory rules (18-year minimum for ...)
- **India: DPDP Act 2023 and DPDP Rules 2025.** A 'child' is under 18. Rules notified 13 Nov 2025 (G.S.R. 846(E)). Verifiable parental consent before processing ...
- **India: Economic Survey 2025-26.** 29 Jan 2026: flags digital addiction as a public-health issue; recommends age-based limits on social media ...
- **India: Karnataka under-16 and Andhra Pradesh** 6 Mar 2026: Karnataka's CM announced an under-16 ban in the state budget ...
- **India: Madras High Court (Dec 2025).** Madurai Bench (S. Vijayakumar v. Union of India) suggested the Union explore an Australia-style under-16 framework and ...
- **India: school phone restrictions.** State-led, no national rule. Himachal Pradesh bans phones in all schools from 1 Mar 2026 (Rs 500 fine and ...)
- **Australia: under-16 social-media minimum age.** Online Safety Amendment (Social Media Minimum Age) Act passed 28 Nov 2024; in force 10 Dec 2025.

Source: market research agent (sources named per figure in the data file); unverified items listed in Evidence status. 14 research gaps recorded.

Confidential. Figures are labelled sourced / reported / estimate; see Evidence status.

One safe phone for a child, one portal for the parent

Stage 1 scope: everything exists in its simplest form (D1-D21)

 Messenger 1:1 text + emoji; approved contacts; no groups, no links	 Voice & video calls 1:1 over the internet; approved only. Voice reuses the video stack (to confirm)	 Walkie-talkie Push-to-talk with approved contacts
 AI assistant Chat UI; text + image input; age policy; parent-visible	 Learning videos Curated learning video; YouTube tier gated by compliance review	 Weather Live forecast with 'why is it...?' lessons
 Camera Kid camera; no geotags	 Photos Private by default; parent-shared	 Book reader EPUB; parent-approved library
 Journal & notebook Private text, drawing, handwriting	 Settings (minimal) Wi-Fi and mobile data as standard; nothing else needed	 Never on device Browser, YouTube, app store, cellular calls, SMS

Parent portal (any browser)

- Approve contacts (both families for cross-family)
- Read messages and AI chats; review calls metadata
- Set screen-time and quiet hours
- Control AI and learning-video settings
- Pair and manage devices (QR claim code)
- Remote lock; activity feed; approvals queue
- *Later:* parent-side voice and video calling

PRINCIPLES

- English only; India first
- Parent-visible by design (age-appropriate disclosure to the child)
- Safe when the network is down (signed, cached policy)
- Nothing the child does can add an app or open the web

SAFETY DEFAULTS

- Every contact needs parent approval; cross-family contacts need both families
- No groups, links, attachments or discovery in messaging (Stage 1)
- No recording or covert listening on calls or walkie-talkie
- Stage 1 locations: parent-set places only

STAGE 2: MAKE IT BETTER

- Richer content, games, sensors, climate stories
- Parent-side voice and video calling
- Customer self-install and more qualified devices
- More languages (Hindi and regional)

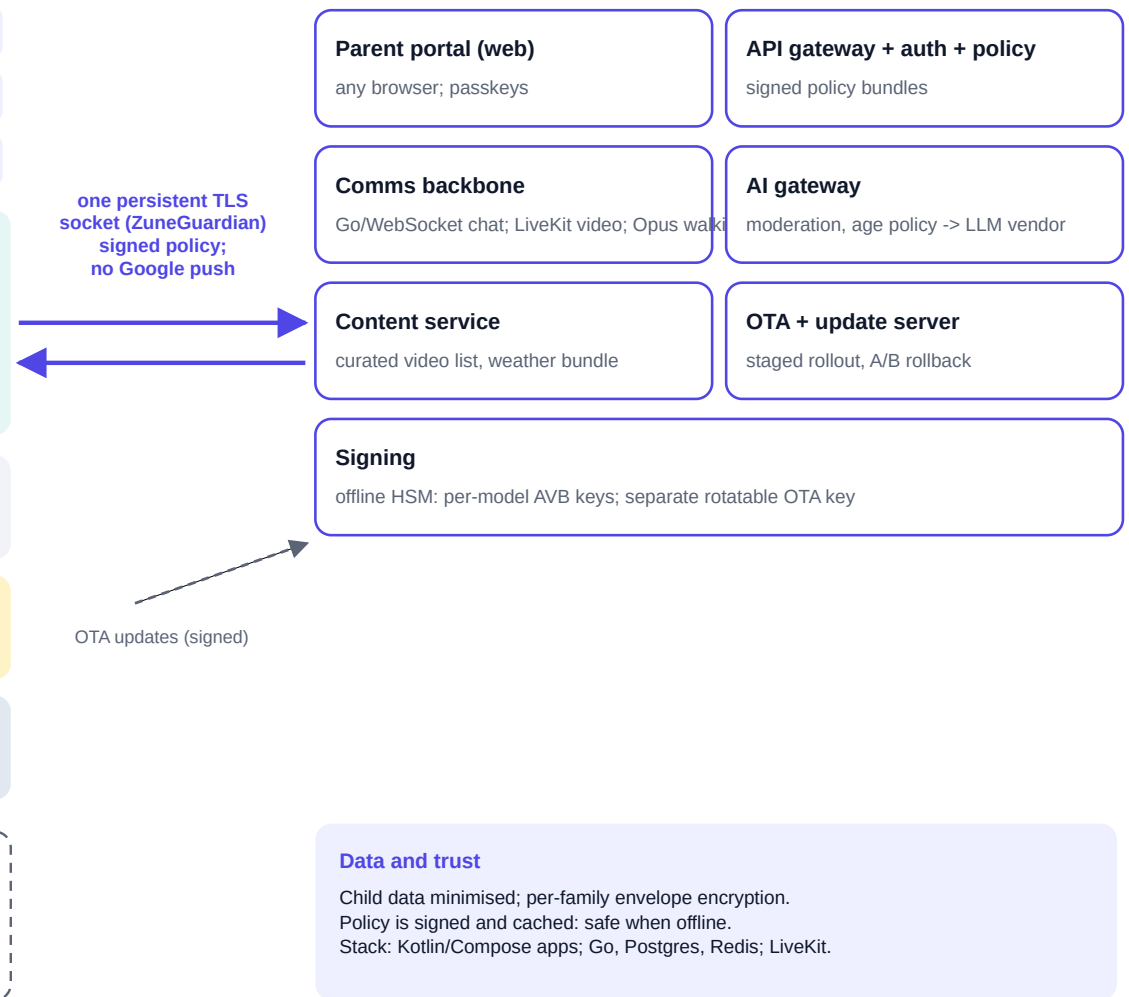
How it fits together

Phone, cloud and factory: three planes with one signed trust chain

ON THE CHILD'S PHONE



CLOUD



Source: reports R01, R03, R05, R09, R10, R13, R16, R18. Technology choices are recommendations, not yet built.

Confidential. Figures are labelled sourced / reported / estimate; see Evidence status.

No-browser is enforced in the locked image, not by deleting an app

R04 unverified

No-browser lockdown: threat model and enforcement

DECISION

Compile 'no browser' into a locked, verified image: user build, Zune AVB key, WebView caller gate, no http(s) handler, INTERNET allowlist. Policy only tunes it.

WHY IT MATTERS

The product's core promise depends on it. Real kid-control bypasses are hidden-browser bugs, so the defensible claim is narrow: no browser app, no way to type a URL.

KEY POINTS

- AOSP 17 is not browser-free: CarrierDefaultApp (disabled http/https wildcard handler, JS WebView), OsuLogin and HTMLViewer are escape hatches. R04 F1
- AOSP 17 ships no WebView implementation. Vanadium (hardened Chromium, GPL-2.0 patches) or an own Chromium build means owning a Chromium update pipeline. R04 F2
- WebView gate: system_server already sees caller UID, so a ~20-line allowlist restricts WebView to two Zune packages (feasibility inferred). R04 F3
- App-level origin allowlists leak: shouldInterceptRequest misses redirects, blob:, javascript: and assets; Network Security Config has no allowlist. R04 F4
- eBPF denies INTERNET at socket creation per app. Stage 1 grants it to about 8 packages; the EPUB reader gets none. R04 F6

23

Bypass vectors modelled
estimate
Count of the report's table; likelihood and residual ratings

~20 lines

WebView caller-gate patch
estimate
Inferred from reading system_server code; not built or teste

under 5

Patches to AOSP (target)
estimate
Design target, not measured

TOP BYPASS VECTORS AND RESIDUAL RISK

Vector	Likelihood	Mitigation	Residual
http(s) link tapped in any app	High	No handler (CI gate), baked Intent Firewall rule	Low
Hidden WebView hosts: CaptivePortalLogin, OsuLogin, HTMLViewer, CarrierDefaultApp	Medium	Remove from product; WebView caller allowlist; captive_portal_mode=0	Low
Videos WebView: YouTube UI, end cards	High	One iframe, no navigation, proxy/allowlist	Medium
AI assistant as web gateway	High	No browse tools, strip URLs, server filters	Web-derived content still reaches child
ADB, Developer options, trade-in mode	High	user build, ro.adb.secure=1, trade-in flag off	Low
OEM unlock, fastboot, stock reflash	High	OEM-unlock bit off; locked bootloader, Zune AVB key	Low on Pixel; higher on BROM-exploitable SoCs
Recovery factory reset	High	Cannot block; device inert until re-paired; backend alert	Parent policy lost; no browser gained

Superseded: Hardware-sales framing ('sell pre-flashed devices') is reframed by D15/D16: we sell the OS image and flash customer-supplied phones. Vector #6 'SMS unreadable (D6)' is moot under D19 (no SMS).

High

Chromium/WebView is the largest attack surface and imposes a permanent update cadence; GPL-2.0 patch obligations need legal review.
Commit to an update SLA at Chromium cadence; choose Vanadium prebuilt or own build; legal review.

Medium

Embedded YouTube UI exposes links we cannot remove, capping Videos containment at medium.
Single iframe with no navigation, proxy allowlist; Chromium allowlist patch or licensed/native video in Stage 2.

Medium

ECH, QUIC and HTTPS-RR erode DNS/SNI enforcement; AOSP always-on VPN lockdown has known leaks.
Treat DNS as a layer; use per-app INTERNET as base; gate content at the single-iframe Videos page.

Stage 1: Image: user build, ro.adb.secure=1, locked bootloader with Zune AVB key, OEM-unlock forced off, no Developer options, fw.max_users=1. Browser2, QuickSearchBox, CaptivePortalLogin, CarrierDefaultApp, HTMLViewer, CertInstaller and similar excluded ... Full scope: detail sheet.

Google's parental controls aren't in AOSP, so Zune builds engine, backend and portal.

R05unverified

Parental-controls platform: portal, on-device enforcement, sync

DECISION

Build ZuneGuardian, a system app that is Device Owner and holds the SYSTEM_SUPERVISION role, enforcing signed cloud policy locally. Device Owner is a one-way door awaiting founder approval.

KEY POINTS

- AOSP ships supervision plumbing (SupervisionService, role, bound app service) with all config hooks empty; consumer features and Family Link backend are Google's. R05 F1
- SYSTEM_SUPERVISION role is system-only and exclusive; grants SUSPEND_APPS, lock-task, time, safe-boot and factory-reset controls, but not WIPE_DATA, private DNS or VPN. R05 F2
- Platform policy is thin: only per-app ALLOWED/BLOCKED/TIME_LIMIT, and TIME_LIMIT suspends immediately (open TODO). Zune must own the schedule, contact and content engine. R05 F3
- Cloud is authority; device enforces signed JWS ES256 bundles locally. Past expiry plus 14-day grace, it falls to guardians and emergency only. R05 4.4
- Sync is one persistent TLS WebSocket as a doorbell (policy.update, then HTTPS GET); polling and alarms as fallback. No FCM, no UnifiedPush. R05 4.5

\$0.25-0.50/month

Infra cost per device at 1k devices

estimate

\$250-500/mo total; excludes LiveKit, AI, SMS, carrier, staff

\$0.09-0.22/month

Infra cost per device at 10k devices

estimate

\$0.9-2.2k/mo; logs and Postgres writes dominate

\$0.06-0.15/month

Infra cost per device at 100k devices

estimate

\$6-15k/mo; about 100k sockets

Superseded: US assumptions (A1 replaced by D18, India): AWS us-east-2 region, COPPA and California AB 1043 legal hooks, 911 emergency number, America/Chicago example, USD framing.

Medium

GrapheneOS trees are newer than android-17.0.0_r1, and Google's source was unreachable; cited files may differ.

Re-verify every cited file on android-17.0.0_r1 once AOSP is reachable.

Medium

Role grants may miss needed restrictions (private DNS, VPN, OEM unlock, wipe); supervision API is flagged and still moving.

Device Owner fills gaps; EnforcementBackend interface isolates API; validate on Cuttlefish; CI runs every rebase.

High

Server holds children's location and transcripts; a breach is existential. 24/7 SOS staffing is a dependency.

Per-family KMS envelope encryption with crypto-shredding, retention limits, dual-approved break-glass staff access, hash-chained audit log.

Stage 1: Device Owner plus supervision role at setup; QR pairing; signed policy bundles for screen time, downtime, app rules and contact allowlists; parent-PIN approvals; remote lock; browser portal with passkeys; one active device per child. Full scope: detail sheet.

Kid messaging and video are designed around parent-approved contacts and parent-readable threads.

R13

partly verified

Kid messenger, 1:1 video calling and shared comms backbone

DECISION

Build one first-party comms service (Go, WebSocket, Postgres) with a parent-approved contact graph; server-readable text-and-emoji messenger and LiveKit 1:1 video; no recording, no E2EE in Stage 1.

WHY IT MATTERS

Cross-family contact is where child-safety risk concentrates; the report calls one breach of children's chats existential, so trust model and moderation limits decide viability.

KEY POINTS

- One first-party comms service (Go, one WebSocket per device, Postgres, Redis); Matrix and Signal rejected, XMPP kept as the 'buy' fallback. R13 S1, 3.1
- Parent-approved contact graph governs everything; cross-family links need both guardians via invite code. No phone matching, address-book upload or kid-searchable directory. R13 S1, 4.1
- Messages are server-readable with per-family envelope encryption; both families' guardians read threads, kid is told. No E2EE in Stage 1. R13 S1, 3.2
- Text-only, 500 characters, no links, images or GIFs; age-banded emoji allowlist; server rules block URLs, numbers, handles and flag bullying or secrecy. R13 4.3
- Video on LiveKit SFU (Apache-2.0) at 360p/24 fps; server can end calls; locked-down tokens; no recording or covert listening in Stage 1. R13 F5, 4.4

MESSENGER STACK OPTIONS

Option	For	Against	Verdict
Matrix (Synapse/Conduit + rust-sdk)	Receipts, offline, multi-device built in	Room-based authz; AGPL or paid licence; heavy to strip	Reject for Stage 1
XMPP (Prosody + Smack)	Receipts, archive, stream management; permissive licences	Parent graph and moderation still custom; second runtime	Best 'buy' fallback
Signal protocol	Strong E2EE	libsignal is AGPL-3.0; blocks parent review and moderation	Reject
MQTT	Tiny keepalive	No store, ACLs or history	Not needed
Custom WebSocket + Postgres	Exact semantics; in-line moderation; one runtime	We own reliability: idempotency, ordering, resync	Recommended

\$50/mo incl. 150k participant-min; \$0.0005 overage

LiveKit Cloud Ship plan reported
Aggregator pages only; livekit.com blocked; \$0.12/GB egress

\$500/mo incl. 1.5M min; \$0.0004 overage

LiveKit Cloud Scale plan reported
\$0.10/GB; region pinning, SOC 2 report and BAA reportedly Sc

360p / 24 fps, ~0.64 Mbps incl. audio

Video bitrate cap estimate
Design cap; hardware codec behaviour to be benched per devic

Superseded: US-only launch (A1) and US legal framing (COPPA, 2258A/NCMEC, TAKE IT DOWN, CA AB 1043): superseded by D18 (India, DPDP, under-18).

Medium

Persistent-connection battery and carrier NAT timeouts are unmeasured; ntfy evidence says short heartbeats keep the modem awake.

Start probing at minutes, not 30 s; measure on Pixel 10a/9a first (D17); fall back to longer-interval design.

High

Android 17 may silently drop incoming audio and block camera/mic from a boot-started service; inferred, not tested.

Keep room client in the visible call app; test with 'adb shell cmd audio set-enable-hardening enable' first.

High

Server-readable store of children's chats is a high-value target; the report calls one breach existential.
Per-family envelope encryption with KMS keys, break-glass staff access, audit logs.

Stage 1: In-app 1:1 text+emoji messenger and 1:1 video between approved contacts; contact graph with dual-guardian approval; server-side moderation and flags; one device WebSocket with signed policy; portal thread viewer, approve/revoke, End call. Full scope: detail sheet.

Walkie-talkie designed as our own Opus relay, avoiding a third-party voice processor; untested.

R08unverified

Walkie-talkie (push-to-talk) and safe kid-to-kid voice

DECISION

Build push-to-talk as Opus frames over Zune's own authenticated WebSocket relay, cloud-only, with parent-visible metadata only in v1; LiveKit SFU is Plan B.

WHY IT MATTERS

Push-to-talk carries children's voice, so transport choice sets privacy exposure (no third-party voice processor), per-kid cost and what parents can see. The report is an unbuilt design.

KEY POINTS

- Transport: Opus frames over Zune's own authenticated WebSocket instead of a WebRTC SFU; half-duplex 1:1 speech needs no SFU, ICE or UDP. R08 S1
- Audio: Opus VOIP wideband at 16 kbps, 20 ms frames bundled in threes, 120-200 ms jitter buffer; about 300-480 ms mouth-to-ear (unmeasured). R08 S4 Audio
- Android 17 forces a split: transmit starts from the visible Walkie activity; incoming playback runs in a privileged system service with MODIFY_AUDIO_SETTINGS_PRIVILEGED. R08 F1, F2
- No Google services: Nearby Connections is unusable; Wi-Fi Aware and BLE 'near me' mode deferred to Stage 2; v1 is cloud-only. R08 F4, F5
- Waking a no-GMS device needs a persistent socket plus battery exemption; Doze allows one exact alarm per 9 minutes. R08 F3

16 kbps Opus wideband

Speech bitrate sourced
libwebrtc source recommends 16-20 kbps; Android 10+ has a Me

~300-480 ms (SFU ~185-335 ms)

Mouth-to-ear latency, own relay estimate
Unmeasured; judged acceptable for half-duplex

29.6 kbit/s (13.3 MB/h)

Bandwidth per direction, continuous talk estimate
WebSocket/TLS/TCP, 60 ms bundles; WebRTC at 20 ms packets is

Superseded: Telephony paragraph (carrier VoLTE voice/SMS, native 911 dialer) and SOS one-tap cellular call to a guardian: superseded by D19 (optional portal SMS to a guardian number is server-side; confirm).

Medium

TCP head-of-line stutter on weak cellular could make walkie audio glitchy.
Test on target carriers; if more than a few percent glitch, switch to LiveKit Plan B (Opus-only, SPEECH preset).

High

Android 17 native audio-playback enforcement is unread; incoming audio could be silently dropped on device.
Test on device or Cuttlefish with 'adb shell cmd audio set-enable-hardening enable' before other media work.

Medium

Persistent-socket battery and carrier NAT behaviour unmeasured; report assumed Snapdragon devices, D17 starts on Pixel.
Measure on Pixel 10a/9a first; ntfy reports 0-1% battery in 17 h on one phone (n=1).

Stage 1: MVP: 1:1 kid-to-kid between approved contacts; hold-to-talk, 30 s cap, floor control with busy/unavailable tones; Opus relay on the existing socket; no ring-through, no stored audio; metadata log, rate limits, 'Tell a grown-up', SOS alert
... Full scope: detail sheet.

Cloud AI tutor, not companion; Anthropic's written approval for children is gate one.

R07unverified

Kid-safe AI assistant: architecture, models, safety, policy

DECISION

Stage 1: cloud assistant via Zune AI Gateway (Claude Sonnet 5.5, Haiku 4.5 guards, zero data retention), on-device voice, tutor-not-companion design. On-device LLM only as Stage 2 offline fallback.

WHY IT MATTERS

The AI assistant (D11) carries per-child running cost, child-safety exposure and single-vendor dependence. Anthropic's approval and a release-blocking safety eval gate launch; image input is the highest-risk part.

KEY POINTS

- Anthropic allows minors' API products with age gating, moderation, monitoring and AI disclosure; under-13s not spelled out, so get written confirmation. R07 F1
- Zero data retention (ZDR) is available on request for the Messages API; flagged content may still be kept up to 2 years. R07 F2
- Gemini API is out: terms bar clients directed at or likely accessed by under-18s. OpenAI is the backup, needing ZDR first. R07 F3
- Gateway pipeline: PII scrub and Haiku input screen, Sonnet 5.5 call grounded in curated packs, output screen before display or speech. R07 §4
- Voice is on-device from day one: push-to-talk, sherpa-onnx, Moonshine or Whisper STT, Kokoro or Matcha TTS; AOSP ships no usable speech stack. R07 F6

\$1.42

COGS per typical child (240 turns/month) estimate

Sonnet 5.5 plus guards; \$1.10 with 25% FAQ cache hits; \$0.78

\$2 in / \$10 out

Claude Sonnet 5.5 price per MTok sourced

Cache read \$0.20; read on Anthropic pricing page

\$1 in / \$5 out

Claude Haiku 4.5 price per MTok sourced

Cache read \$0.10

Superseded: US launch assumption (A1) replaced by D18 (India first): COPPA, FTC 6(b), CA SB 243, WA/OR laws, GUARD Act, CAAB 1043, US-only inference, US crisis lines (988, Childhelp) ...

High

Anthropic may decline, add conditions, audit or suspend; single-vendor dependency.

Provider abstraction; OpenAI or self-hosted fallback; written confirmation, ZDR and DPA before launch.

High

Child jailbreaks, multi-turn drift; self-harm false negatives are the worst case, false positives frustrate families.

Layered guards, scripted T0 responses, release-blocking red-team evals, kill switch, human Trust & Safety queue.

High

Image input: PII, self-generated sexual imagery, CSAM reporting duty; abuse disclosures where parent is the risk.

Objects/homework only in v1, on-device gate, EXIF stripped, ephemeral; abuse policy written by counsel and child-safety expert.

Stage 1: Cloud gateway (Go) with Claude Sonnet 5.5 main model and Haiku 4.5 input/output guards under ZDR; on-device push-to-talk voice; four modes (Ask Why, Homework Helper, Story Maker, Look at This) across three age bands; limited image input ... Full scope: detail sheet.

YouTube is a gated, killable add-on; Videos must work without it.

R06

unverified

Curated learning video: legality, technology, alternatives

DECISION

Conditional go: offline licensed content plus paid partners are the base; YouTube is an optional made-for-kids-only tier behind a kill switch, dropped if refused or silent at 8 weeks.

WHY IT MATTERS

The founder's core video promise (D2, D3) rests on a platform that can revoke access at will and whose rules collide with our containment design. The fallback protects launch.

KEY POINTS

- No third-party YouTube Kids API; the sanctioned route is the IFrame player in a WebView, which needs no Play Services. R06 §1, F5
- Four YouTube rules collide with our containment: no overlays on the player, no disabling links, no ad blocking, no offline caching. R06 §2.2
- Every embedded video must be checked as Made-For-Kids (MFK); we publish MFK-only for ages up to 12 to avoid mislabel exposure. R06 F1, §2.4
- D3 topic requests are answered from our own catalog index, not live YouTube search, which allows about 100 searches a day fleet-wide. R06 §2.3
- Tier 2: one isolated WebView page on our own HTTPS origin, one iframe, navigation blocked in our client, remote kill switch. R06 §4.1

YOUTUBE RULES THAT COLLIDE WITH THE ZUNE DESIGN

YouTube rule	Zune element hit	Verdict
No overlays in front of the player	Touch shield (report 04)	Drop the shield; block navigation in WebView client
No removing, obscuring or disabling player links	D2: no path to youtube.com	Grey area; needs written YouTube answer
No modifying or blocking ads	Network allowlist could drop ad hosts	Allowlist must include Google ad hosts
No download, cache or offline copies	Offline videos	No offline YouTube; Tier 1 covers offline
No background or audio-only player	Listen mode	Not offered
No charging users to watch embedded content	Product is sold; D3 gating	Keep YouTube free; PIN and screen-time locks grey
Derived-metrics policy for custom scores	ML age-suitability pre-filter	Unclear; make all labels human editorial

~100

Live YouTube searches per day, fleet-wide reported
search.list costs 100 units in its own 10,000/day bucket sin

\$10M

Disney settlement for mislabelled child-directed videos reported
Order entered 2025-12-23; law-firm summaries

45 / 80 editor-days

Launch catalog curation effort estimate
~1,500 videos at ~35 videos per editor-day, single / double

Superseded: US launch assumption (A1) is replaced by D18 (India first): decision 3 'US (Common Core/NGSS) or UK (Oak)?' is overtaken; COPPA/FTC exposure and US-standards schema fields (CCSS, NGSS ...

High

YouTube can cut referrer access for our origin at any time; only recourse is an appeals form.
Remote kill switch; Videos works on day one from Tier 1 and Tier 3 content.

High

YouTube may require clickable player links, incompatible with D2; ad clicks open advertiser pages we must refuse.
Ask YouTube in writing (Q2, Q3); founder decides whether D2 or YouTube Videos wins; Tier 1 unaffected.

Medium

Mislabelled child-directed videos carry enforcement risk (Disney \$10M); legal status of YouTube data collection via our embed unresolved.
MFK-only up to age 12, per-video nightly MFK check, counsel sign-off; never claim "100% safe".

Stage 1: Tier 1 offline player on Media3 with a ~40 h signed starter pack from open-licensed sources; human-labelled catalog index (age, topic) answering D3 requests; gated Tier 2 YouTube WebView page, MFK-only, behind a remote kill switch ... Full scope: detail sheet.

Weather app teaches with authored lessons and parent-set places; alerts are US-only so far

R14 partly verified

Weather app that teaches children about weather

DECISION

Backend-proxied weather via Open-Meteo paid plan (MET Norway fallback); parent-set places; authored explainers and safety cards; no LLM in app. NWS/WEA alerts are US-only: re-plan for India.

WHY IT MATTERS

Weather carries a vendor-terms dependency, the child-location question and safety alerts, where a wrong design is legal or cost risk. Beta cost is small; scale needs Enterprise or self-hosting.

KEY POINTS

- Open-Meteo's free tier is non-commercial only: Zune needs a paid plan (Standard 1M, Professional 5M, Enterprise 50M+ calls/month) or self-hosting. R14 S2.1, claim 1
- Google Weather API excluded: terms bar apps whose primary purpose is weather information; corrected to a 'primary purpose' judgement. R14 S2.1, claim 2
- Devices call only Zune's backend; vendors see one server identity and ~5 km grid coordinates, never device or account IDs. R14 S1, S4.1
- Stage-1 location is parent-set in the portal. GrapheneOS's network location calls Apple's undocumented endpoint; Mozilla Location Service is dead since 2024. R14 S2.2, claim 5
- Education is authored, not LLM-generated: 20 explainers x 3 age bands plus 6 safety cards, triggered on device by live conditions. R14 S4.4

Standard 1M, Professional 5M, Enterprise 50M+ per month

Open-Meteo plan call budgets
sourced

Read from pricing page source; call weighting (variables x m)

about \$29 and \$99 a month

Open-Meteo first two plan prices
reported

Search summary only; pricing table unreadable.

about 2.6M (10k devices), 12.6M (100k) a month

Corrected request volume
estimate

Place-cell and refresh assumptions are inference; AQI separa

WEATHER DATA SOURCES FOR A COMMERCIAL CHILD PRODUCT

Source	Commercial terms	Cost (beta / scale)	Verdict
Open-Meteo	Global; paid plan grants commercial licence; data CC BY 4.0	Standard ~\$29, Professional ~\$99 (reported); 100k: Enterprise or self-host	Primary
MET Norway Locationforecast	Global, CC BY 4.0, free incl. commercial; unique User-Agent mandatory	\$0 at fallback volumes	Fallback (terms from search summary)
NWS api.weather.gov	US only; public domain with conditions; limits undocumented	\$0 at any scale	US alerts only; stale under D18
OpenWeather One Call 3.0	Global; ODbL-based; on-screen attribution	About EUR 1.8k (10k) / 8.8k (100k) a month	Reject (cost)
Tomorrow.io	Free plan bars commercial use for evaluation accounts	From about \$50 (reported)	Reject
WeatherAPI.com	Free tier reportedly allows commercial use (unverified)	\$7/3M, \$25/5M, \$65/10M calls (reported)	Plan B; read ToS first
Google Weather API	Terms bar use where weather is the primary purpose	About \$0.15/1,000 after 10k free	Excluded (judgement)

Superseded: US launch assumptions (A1 superseded by D18 India): NWS alerts, WEA design, US-English/imperial default (decision 8), COPPA framing, NGSS standards, ZIP. Spanish as first Stage-2 locale conflicts with D20 (English only).

Medium

Open-Meteo can change terms effective immediately and end a subscription; exact prices unverified.

Vendor adapter, MET Norway fallback, stale-while-error, self-host path (AGPL, 8-16 GB RAM).

High

Alert design (NWS, WEA) assumes a US launch; this report has nothing for India's alerts.

Reconcile through reports 19 and 20; NWS is US-only.

High

Child can disable cell-broadcast alerts; carrier acceptance on a non-stock OS and non-tappable links untested.

DISALLOW_CONFIG_CELL_BROADCASTS plus RRO flags; test on Android 17; ask carrier at certification.

Stage 1: Report's scope: current + 48 h + 7 days forecast, NWS alert banner and parent notice, 20 explainers x 3 bands, 6 safety cards, 4 Compose visuals, launcher tile, Journal note, parent place/units/alert settings, MET Norway fallback, CellBroadcast overlay. Full scope: detail sheet.

Kid apps built in-house on Compose: no store, no GMS, roughly 80 person-weeks (estimate)

R09 unverified

First-party app suite: stack, in-tree vs APK, per-app design

DECISION

Recommended: at most six OS-coupled components in-tree; every other app in a Gradle monorepo, separately signed, silently updated. Compose UI, Radium EPUB reader without INTERNET access.

WHY IT MATTERS

Sets build cost, update path and attack surface for every on-device app. No store UI and few INTERNET holders leave fewer ways to break out or leak.

KEY POINTS

- Two tiers: at most six OS-coupled components built in-tree with the platform key; every other app is Gradle-built, separately signed, silently updated. R09 S1 pt1, F3
- One stack: Kotlin, Compose, in-house ZuneKit design system with three age-band profiles, Hilt, Room; minSdk and targetSdk both 37. R09 S1 pt2, S4.2
- Launcher written from scratch in Compose (5 pw). Launcher3 forks need a rebase every Q2/Q4 drop; Lawnchair is based on Android 16. R09 S3, F7
- EPUB: Radium Kotlin Toolkit 3.4.0 (BSD-3, WebView-based, reflowable and fixed-layout). KORreader, crengine, FBReader, Libreria rejected as copyleft or proprietary. R09 F6-F7, S3
- Radium's EPUB navigator is a WebView that intercepts every navigation and serves resources in-process, making a no-INTERNET Reader plausible (inference). R09 F6, S4.3

STAGE-1 APP SUITE AND EFFORT

App	Tier	Stage-1 MVP	pw
Core, ZuneKit, CI pipeline	n/a	Policy client, attested HTTP client, egress lint, design system	8
ZuneHome + ZuneSetup	A	Approved-app grid, bedtime screen; QR pairing, age band, kid name	9
Camera + Photos	B	CameraX photo/video, no geotags; MediaStore grid, explicit send-to-parent	8
Journal	B	Text, drawing, voice note; Room + SQLCipher; private, child-shares or parent-visible	6
Notebook	B	Typed + Jetpack Ink handwriting, checklists, PDF export to parent only	6
Reader	B	Radium EPUB, dyslexia fonts, read-aloud, bookmarks; no INTERNET	6
Assistant UI + Walkie UI	B	Chat bubbles, image attach (EXIF stripped); one hold-to-talk button	9

61.5 person-weeks

Stage-1 app effort estimate
Per-app estimates summed; excludes Messenger, calls, Weather

about 80 pw

With test, accessibility, usability
estimate
5 Android engineers for 16-20 weeks beside the platform trac

3.4.0 (2026-09-11), BSD-3, minSdk 24

Radium Kotlin Toolkit sourced
Read in LICENSE, CHANGELOG, README.

Superseded: Spanish in v1.1 (decision 6) no longer fits India-first (D18); D20 sets English only for v1. 'US only' public-domain book screening and COPPA/US-hosted Sentry framing assume a US launch (A1 superseded by D18 ...

Medium

Radium inherits the WebView attack surface; 3.x navigators have churned between versions.

Pin versions, budget 1 pw per upgrade, intercept all navigation, no INTERNET permission in Reader.

Medium

Re-flash or repair under D16 wipes a child's private Journal.

Stage 1: parent-triggered encrypted export; automatic backup deferred to v1.1.

Medium

Public-domain classics carry dated content; editorial review is a recurring cost.

Editorial screening; a content editor is required headcount, not optional.

Stage 1: Core libraries and CI pipeline, then ZuneHome, ZuneSetup, Camera, Photos, Journal, Notebook, Reader, Assistant UI, Walkie UI, Videos UI, Clock, Calculator and Recorder, all on-device in simplest form. Full scope: detail sheet.

Child sees about 11 Settings rows; stock AOSP Settings stays hidden as engine room

R16 partly verified

Minimal Settings app: what stays, goes, is parent-gated

DECISION

Option D+: new Compose ZuneSettings (about 11 rows) deep-links to stock Wi-Fi, mobile data and Bluetooth screens; stock Settings stays as default-deny engine room. Parent PIN gates risky rows.

WHY IT MATTERS

D14 requires standard Wi-Fi and data, the hardest code in Settings. This design reuses it and defaults everything else to off, but leaves known residual gaps.

KEY POINTS

- Stock Settings 17 is huge: 437 manifest entries, 334 exported, 213 ENTRY_FRAGMENTS, 237 Catalyst annotations, 134 config bools (GrapheneOS tree). R16 F1 [CORRECTED]
- Full rewrite rejected: framework and SystemUI hard-reference Settings classes and send 57 distinct Settings.ACTION_* intents. R16 F4 and verification
- Stock Settings is trimmed by a generated sysconfig component-override: about 40 of 437 stay enabled; new upstream ones default to off. R16 F2, section 5
- Two user restrictions are traps: DISALLOW_CONFIG_WIFI and DISALLOW_CONFIG_MOBILE_NETWORKS blank the whole Wi-Fi and mobile pages, contradicting D14. Never set them. R16 F3 [CORRECTED]
- config_show_wifi_settings=false does NOT hide the 'Wi-Fi preferences' row (WEP, Wi-Fi Direct, install certificates); hiding it needs a small Java patch. R16 F2 [CORRECTED]

SETTINGS APPROACH OPTIONS

Option	Effort	Rebase per drop	Tamper resistance	Verdict
A. Stock Settings, overlays only	Low	Low	Weak (Catalyst/SPA pages, adult homepage)	Part of D+
B. Slim fork, delete screens	3-4 weeks	1-3 weeks	Good	Reject
C. New Compose app	4-6 months	Low, but system-API drift	Best	Reject: worst crash safety
D+. ZuneSettings front, stock engine room	8-9 weeks	2-4 days (likely understated)	Good; strong with Stage 2 gate	Recommend

437 (334 exported)

Stock Settings manifest activity entries sourced
XML-parsed recount of GrapheneOS Settings; LineageOS 428

~23

Stock top-level Settings rows sourced
From top_level_settings.xml

~11

Child-visible ZuneSettings rows estimate
Design target

Superseded: Launch locales en-US and es-US (decision 6, row 21) superseded by D20 (English only). US legal basis (CVAA, FCC e-label, WEA, 911 and 'National Alerts') assumed a US launch: superseded by D18 (India first) ...

Medium

Nothing checked against Google's android-17.0.0_r1; GrapheneOS and LineageOS trees may differ in Settings and flags.

Re-verify on Google's tag when reachable; week-1 Cuttlefish spike on component-override, restrictions and SystemUI.

Medium

Catalyst screens bypass ENTRY_FRAGMENTS; whether a disabled host activity makes a row a no-op or a crash is unverified.

Mandatory tap-every-row crawl in CI plus a catch-all intent router to prevent ActivityNotFoundException crashes.

Medium

Stage 1 leaves Wi-Fi advanced fields, Wi-Fi preferences row, captive-portal buttons and a Wi-Fi share QR (may expose password) visible.

Allow a Stage-1 Java patch of about 30 lines to hide them; add a gate patch in Stage 2.

Stage 1: ZuneSettings (Compose, platform-signed) owns android.settings.SETTINGS: home plus 8 native pages (Sound, Display, Accessibility-lite, Language [English only, D20], Battery, Storage, Emergency, About/Legal) and deep links to stock Wi-Fi ... Full scope: detail sheet.

Build on pinned AOSP Android 17; security-patch lag is the central risk.

R01

partly verified

AOSP base release, branches, cadence and build host

DECISION

Base on AOSP tag android-17.0.0_r1 (CP2A.260605.016), zero-fork-first with our own device/zune and vendor/zune trees; plan one rebase onto the Q4-2026 AOSP drop before beta freeze.

WHY IT MATTERS

No GMS means no Play system updates, and Zune should plan for no partner early access. Mainline and WebView fixes ship only through Zune's own OTA.

KEY POINTS

- Base is AOSP tag android-17.0.0_r1, build CP2A.260605.016, API 37, security patch level 2026-06-05; GrapheneOS and LineageOS 24.0 pin it. R01 F1
- Observed: AOSP source lands about Q2 and Q4. GrapheneOS had no AOSP rebase Jan-May 2026; QPR1 (Sep 2026) is not in AOSP. R01 F2
- A 49-commit batch including security fixes, authored Mar-Jul 2026, landed in GrapheneOS on 2026-09-09; median lag 127 days. R01 F2
- Android 15 security tags had 2-3 month gaps in 2025; mirror shows no Android 16 security branch five months after release. R01 F2
- AOSP carries a native supervision framework (SupervisionManager, Settings Supervision screen); three config hooks are empty for Zune's agent to fill. R01 F6

BASE OPTIONS CONSIDERED

Option	Verdict	Why
A. Pure AOSP android-17.0.0_r1 plus Zune trees	Chosen	Clean provenance, least diff; we own patching, WebView, OTA, kernel
B. Fork GrapheneOS	Reuse ideas, not the tree	Hardened, Pixel-focused; 129 own repos; early access partial, not transferable
C. LineageOS base	Only if hardware needs its device trees	108 own repos and app set; not locked down
D. Wait for Q4-2026 drop	Rejected	Idle about 2-3 months; rebase onto it instead
E. Track android-latest-release	Never in CI	Unreproducible; base shifts at each drop

android-17.0.0_r1 (CP2A.260605.016, SDK 37)

AOSP base tag sourced
Read in GrapheneOS and LineageOS manifests; googlesource its

2026-06-05

Security patch level of tag sourced
Strong but indirect: inferred from first downstream diff; ta

127 days (range 65-178)

Median fix lag, author to GrapheneOS sourced
49-commit batch merged 2026-09-09; 'first public' date is in

Superseded: R01 Decision 5 (telephony/SMS in or out) is answered by D19: out. R01's hardware caveats pre-date D17 (Pixel first) and D21 (about 200 Pixel users); nothing else is overturned.

High

Public AOSP patches can trail authoring by 2.5-6 months and arrive batched; Mainline and WebView updates are Zune's to ship.

Monthly patch-ingest pipeline; OTA SLA: Critical within 14 days of bulletin, others 30; about 1 FTE security engineer.

High

Existence and timeliness of an android17-security-release branch is unverified; Google's pages were unreachable.

Verify with git ls-remote from the build host; until then assume pinned tag plus vendor patches is all the patching.

High

WebView-hosting apps, intents, document viewers and CaptivePortalLogin can reintroduce browsing on a no-browser OS.

Exclude Browser2 and HTMLViewer; review CaptivePortalLogin, DocumentsUI, QuickSearchBox, Settings link-outs; choose a WebView source.

Stage 1: Pin android-17.0.0_r1 in a forked Zune manifest; compose a Zune product from base_* makefiles without Browser2/HTMLViewer; point the supervision config at Zune's parental-controls agent ... Full scope: detail sheet.

Zune's OS image is designed as a default-deny allowlist, not subtracted from stock AOSP.

R03 unverified

Clean minimal AOSP product configuration

DECISION

Compose Zune products from base_system.mk plus an explicit allowlist instead of stock phone products; enforce with a CI diff of the built image; overlays via RRO only.

WHY IT MATTERS

A no-browser kids OS cannot let upstream components arrive silently; Android 17 already adds a Web App installer module. An allowlist makes drift a build failure.

KEY POINTS

- Stock products cannot be subtracted: inherit-product leaves markers; no PRODUCT_PACKAGES_REMOVE exists. Only skipping the inherit, Soong overrides and built-in switches work. R03 F2
- Settings and the android package are prevent-disable; SystemUI throws if config_recentsComponent is empty. 3-button navigation lets a non-Quickstep Zune launcher work. R03 F5
- Many Mainline APEXes (wifi, bt, conscrypt, adservices) contribute boot jars and cannot be pruned by name; neuter via flags, feature masks, RRO. R03 F4, F7
- AOSP's prebuilt WebView is not updated regularly (Chromium's own integrator guide); shipping WebView means a signed update path forever. R03 F6
- Neutering without deleting: unavailable-feature masks (Print, Credentials, NFC), static RRO overlays, config knobs, flags like RELEASE_WEBAPP_MODULE=false. R03 F3, F7

KEEP / REMOVE / REPLACE (SELECTED)

Component	Verdict	Why
SystemUI, Settings, SettingsProvider	KEEP (hard dependency)	Prevent-disable; hide Settings from launcher, gate behind parent PIN
Launcher, Setup, Camera, Gallery, Music, Updater	REPLACE with Zune apps	ZuneHome, ZuneSetup, ZuneCamera, ZunePhotos; ZuneUpdater uses silent A/B update_engine
Browser2, QuickSearchBox, HTMLViewer, BookmarkProvider	REMOVE	Browser surface
CaptivePortalLogin	REMOVE in v1 (Q3 open)	WebView mini-browser; breaks hotel and school Wi-Fi sign-in
WebView engine	KEEP engine only (Q2 open)	Only if an app needs it; curated YouTube is the likely need
Mainline APEXes: Bluetooth, NFC, UWB, Wi-Fi	KEEP, neuter	No UI, features masked, DISALLOW restrictions; remove Tag
Telecom, TeleService, TelephonyProvider, CarrierConfig	KEEP (hard on a radio device)	Pixel always has a radio; SIM optional

0

Draft makefiles build-tested
sourced
Report states nothing was build-tested; all makefiles are dr

300+

Modules in base_system.mk
sourced
Includes the whole Mainline set; read in GrapheneOS platform

5

Module release flags on in 17.0 (cp2a) sourced
TELEPHONY, TELECOM_MAINLINE, WEBAPP, NPUMANAGER, CONSCRYPT_N

Superseded: R03 Q1 and its 'DECIDE (Q1)' row (Dialer, Messaging, CellBroadcast, EmergencyInfo, ImsServiceEntitlement, ONS, Stk) are resolved by D19: no cellular voice or SMS.

High

Nothing is build-verified: feature masks on a Pixel vendor image, release-flag overrides and PackageManager removal are unproven.

Build on Cuttlefish first; test on userdebug with privapp enforcement on; run on-device smoke tests.

High

WebView: AOSP prebuilt is stale and Chromium ships about every 4 weeks; Zune must run a signed update path forever.

Name an owner for monthly WebView updates (Vanadium, self-build or prebuilt); decide Q2.

Medium

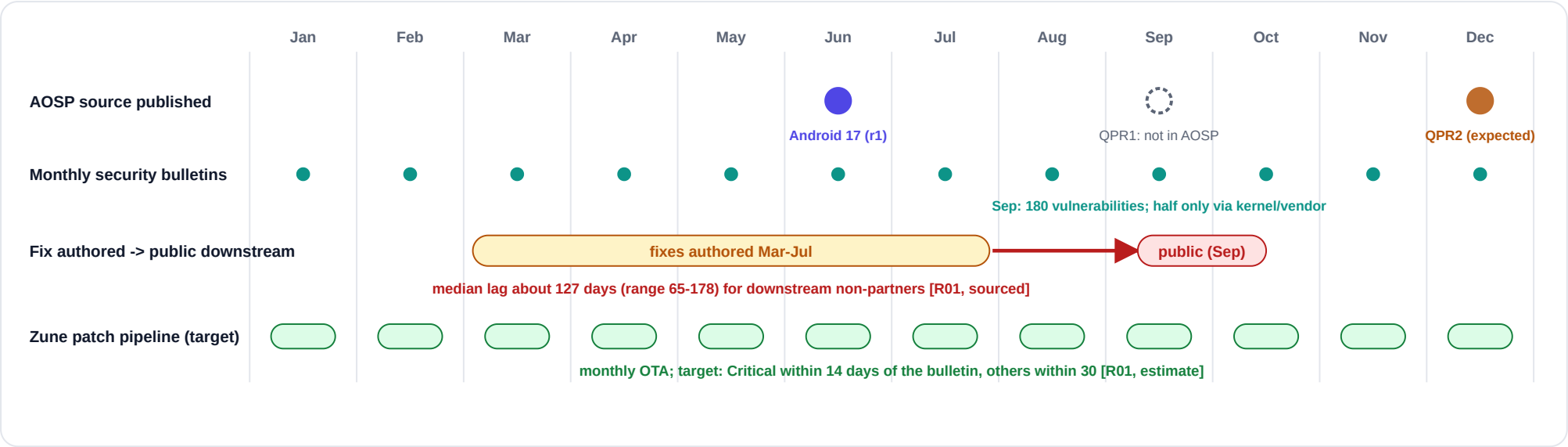
Silent capability growth: Android 17 adds WebApp, NpuManager, ImsStack, Login, PersonalContext, ContactsPicker projects versus Android 16.

Allowlist test over installed-files.txt plus release-flag overrides (RELEASE_WEBAPP_MODULE=false); CI diff every rebase.

Stage 1: Compose zune_system and zune_product from base_system.mk plus an explicit allowlist; develop on Cuttlefish and target Pixel 10a/9a via a fork of GrapheneOS adevtool; replace launcher, setup and updater with in-tree privileged Zune apps ... Full scope: detail sheet.

Security patching is the central business risk

A child-safety product needs monthly fixes; public AOSP now ships twice a year



WHY IT IS HARD (R01)

- Base is AOSP tag android-17.0.0_r1, build CP2A.260605.016, API 37, security patch level 2026-06-05; GrapheneOS and LineageOS 24.0 pin it. R01 F1
- Observed: AOSP source lands about Q2 and Q4. GrapheneOS had no AOSP rebase Jan-May 2026; QPR1 (Sep 2026) is not in AOSP. R01 F2
- A 49-commit batch including security fixes, authored Mar-Jul 2026, landed in GrapheneOS on 2026-09-09; median lag 127 days. R01 F2
- Android 15 security tags had 2-3 month gaps in 2025; mirror shows no Android 16 security branch five months after release. R01 F2
- AOSP carries a native supervision framework (SupervisionManager, Settings Supervision screen); three config hooks are empty for Zune's agent to fill. R01 F6

HOW WE SHIP FIXES (R10)

- OTA: fork GrapheneOS's 1,377-line MIT Updater; static CDN files with ed25519-signed metadata. Rejected hawkBit and Mender: neither drives update_engine. R10 F7, F8
- Keys: AOSP key set plus one RSA-4096 AVB key per model, generated air-gapped; two-person offline signing; CI holds only the metadata key. R10 Design, F1-F3
- AVB key rotation is effectively a recall: custom keys are set only while unlocked, and lock transitions wipe data and rollback indexes. R10 F5
- Roll forward only: update_engine's SPL guard needs green boot state; Zune's custom-key devices are yellow, so Zune enforces SPL monotonicity itself. R10 F6
- Patching is a named 0.5-1 FTE job: monthly SPL train, 72-hour hotfix lane. No GMS means Zune owns WebView and Mainline freshness. R10 Patch train, F10

2 / year

AOSP source drops (Q2, Q4)
sourced
QPR1 (Sep) is not in AOSP; QPR2 expected ~Dec

~127 days

median fix lag, author to downstream **sourced**
range 65-178; 49-commit batch, Sep 2026

14 / 30 days

our patch SLA (critical / other)
estimate
design target, not yet achievable

1 FTE+

platform/security engineer plus kernel support **estimate**
founder staffing decision

Zune forks GrapheneOS's update and signing plumbing; keys never touch the build host.

R10 unverified

Release engineering: signing, OTA, verified boot, anti-theft, build infra

DECISION

Recommended: fork GrapheneOS's 1,377-line MIT Updater into ZuneUpdater (update_engine plus static CDN files); sign offline with Zune-owned keys, one AVB key per hardware model; roll forward only.

WHY IT MATTERS

Signing-key custody and update delivery decide whether a bad build or leaked key can hurt children's phones, and how costly recovery is. Patch cadence is a standing people cost.

KEY POINTS

- OTA: fork GrapheneOS's 1,377-line MIT Updater; static CDN files with ed25519-signed metadata. Rejected hawkBit and Mender: neither drives update_engine. R10 F7, F8
- Keys: AOSP key set plus one RSA-4096 AVB key per model, generated air-gapped; two-person offline signing; CI holds only the metadata key. R10 Design, F1-F3
- AVB key rotation is effectively a recall: custom keys are set only while unlocked, and lock transitions wipe data and rollback indexes. R10 F5
- Roll forward only: update_engine's SPL guard needs green boot state; Zune's custom-key devices are yellow, so Zune enforces SPL monotonicity itself. R10 F6
- Patching is a named 0.5-1 FTE job: monthly SPL train, 72-hour hotfix lane. No GMS means Zune owns WebView and Mainline freshness. R10 Patch train, F10

KEY COMPROMISE AND ROTATION

Key	Compromise effect	Rotation
AVB key (one per model)	Malicious images; needs physical write access	Recall: unlock, wipe, re-flash
releasekey (OTA)	Denial of service; remote takeover only with AVB too	New otacerts.zip, signed by old key, via OTA
platform and other app keys	System-privileged app delivered by APK update	Effectively permanent; HSM, minimal use

1,377 lines, MIT

GrapheneOS Updater size and license
sourced
Read in
GrapheneOS/platform_packages_apps_Updater
branch 17

43 tags, 2026-01-08 to 2026-09-25

GrapheneOS release cadence
sourced
git ls-remote on platform_manifest;
about one per 6 days

about 125 days

Non-partner patch lag
reported
About 40 framework fixes, taken
from report 01

Superseded: US-launch framing (A1 replaced by D18, India): COPPA/state-breach counsel duties and decision 9 'confirm US-only launch'. Telephony (D19): G5 call and SMS-vault checks (emergency path stays per A9).

High

Security-branch access (android17-security-release) is unverified; non-partner patch lag was about 125 days.

Authorise partner-access outreach to Google/ODM; cross-check against GrapheneOS and LineageOS branches.

High

AVB key leak or rotation forces a per-phone unlock, wipe and re-flash.

One AVB key per model caps blast radius; strictest custody; offline host, two-person signing.

Medium

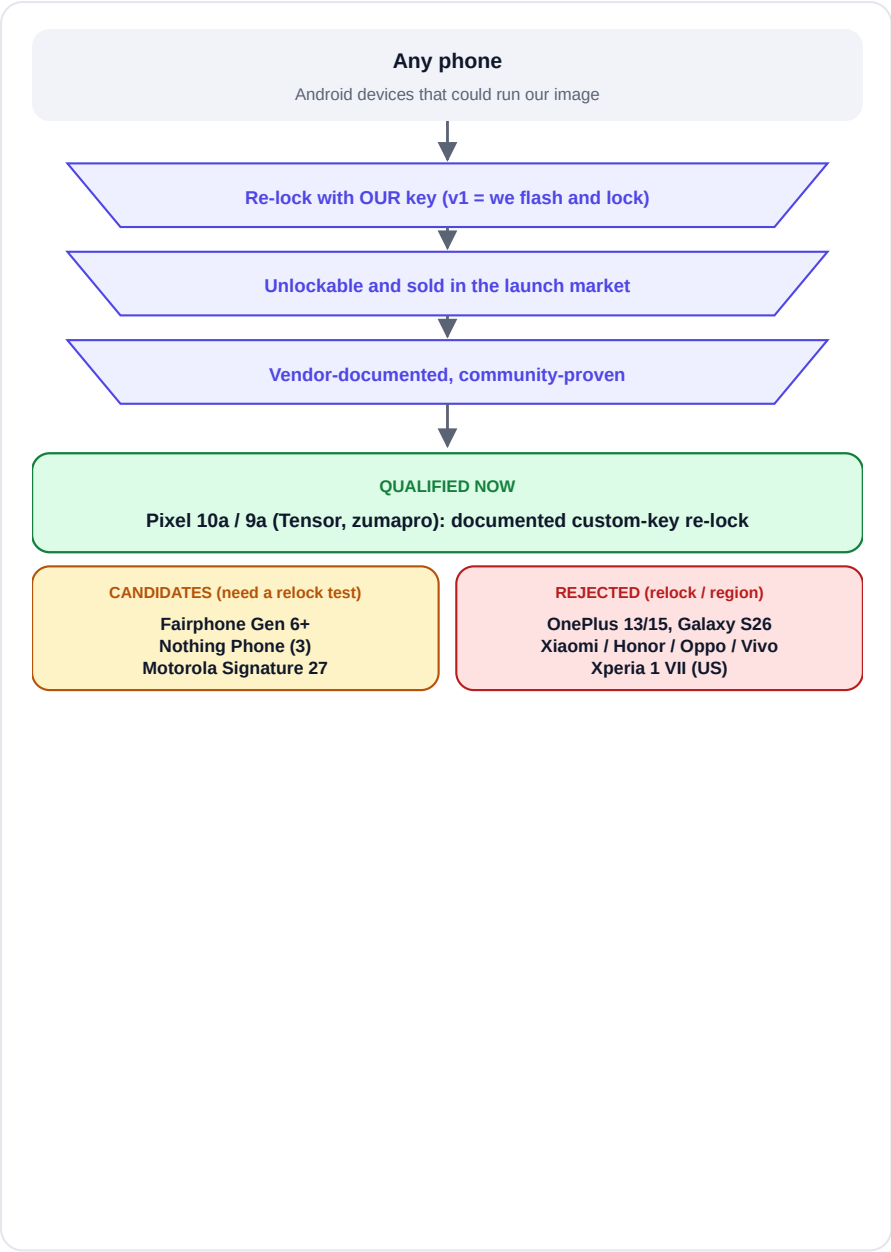
OEM firmware (bootloader, modem, vendor) needs redistribution rights to ship in Zune's OTA, or that layer goes stale.

OEM patch record is a qualification criterion; support promise ends at earlier of OEM firmware end and 5 years.

Stage 1: Full-OTA-only ZuneUpdater on a static CDN with ed25519-signed metadata and rings; offline two-person signing; per-model AVB keys; station-set Activation Lock secret; monthly SPL train with 72-hour hotfix lane; crash-summary telemetry.
Full scope: detail sheet.

Start on the one phone that can be re-locked with our own key

Device strategy: Pixel 10a/9a now; Snapdragon and own hardware later



CANDIDATES CONSIDERED (R15; US PRICES; INDIA AVAILABILITY IN R20)

Device	SoC / RAM	US price	Relock evidence	Verdict
Fairphone Gen 6+	7s Gen 4 / 12 GB	\$649	Family works (FP3/4/5) by user report; Gen 6+	Candidate; was the launch bet
Fairphone 6	7s Gen 3 / 8 GB	Import; US price unsupported	Same family evidence; LineageOS 23.2 tree	Dev reference; different SoC from Gen 6+
Nothing Phone (3)	8s Gen 4 / 12 GB	\$799	User-reported works	Best high-end Snapdragon candidate
Nothing Phone (4a) Pro	7 Gen 4 / 8-12 GB	\$499-599	(4a) reported; Pro not listed	Price-floor candidate
Motorola Signature 27	8 Elite Extreme Gen 6 / 16 GB	Unannounced	GrapheneOS-confirmed; not shipped	Watch; qualify 2027
Sony Xperia 1 VII	8 Elite / 12 GB	\$1.3-1.5k	1 V/1 VI work; US unlock reportedly blocked	Reference BSP only, not launchable
OnePlus 13/15; Galaxy S26	8 Elite family	\$899-999+	Keys rejected; no OEM unlock	Reject
Pixel 10a (reference)	Tensor G4 / 8 GB	\$499	Vendor-documented; GrapheneOS-proven	v1 device (D17)

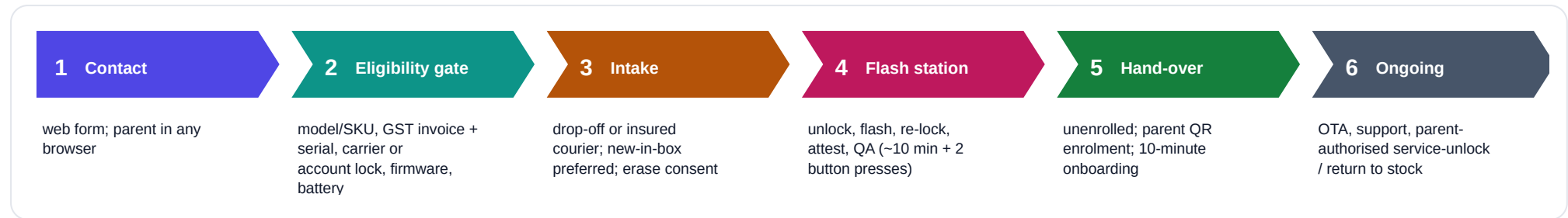
DEVICE SUPPORT CONTRACT (R02)

Tier	Requirement
MUST	AVB 2.0 with custom-key relock; OEM unlock can be disabled
MUST	A/B partitions; stable AIDL HALs and VINTF
MUST	Hardware-backed keystore; fastboot-flashable; 4 GB+ RAM
SHOULD	5+ years of vendor firmware; GKI kernel
SHOULD	Complete Camera2 HAL; MTE
Outcome	All Pixel 8-10 models qualify (GrapheneOS); a-series chosen on price

Relock evidence for non-Pixel phones is user reports only. Tamper strength on Snapdragon is also bounded by OEM bootloader exploits and leaked programmers (R15).

v1 delivery: we flash and hand back the customer's own phone

Software-led: no inventory, no hardware sales (D15, D16); customer self-install comes later



WHAT MAKES IT WORK (R18)

- Unlock and lock each need a physical on-device confirmation and wipe; 'two presses' is Pixel-only (Fairphone adds unlock_critical). Cannot be scripted. R18 F1
- A carrier id on Pixel persist can block bootloader unlock; SIM-unlock differs from OEM unlock, so test the toggle per unit. R18 F2
- Pixel bootloader updates raise an anti-rollback counter; newer-than-bundle phones are blocked pre-unlock. Pixel 10a reportedly excluded from May 2026 increment. R18 F9
- Rotating the AVB key is a recall, not an OTA: each phone needs service-unlock, confirmation and wipe. Keep per-model keys. R18 F7
- With OEM unlocking off, a locked phone cannot be re-flashed over USB; a parent-authorised service-unlock must be built before second batch (G1). R18 F8, \$4.5
- Attestation proves lock state and key hash, but needs Google's remote key provisioning, is relayable, and omits the OEM-unlock toggle. R18 \$4.2 step 5

SELLING THE IMAGE (R17)

- Relock with Zune's AVB key is mandatory for every 'Zune Ready' device; no unlocked tier; verified by attestation at the station. R17 \$1, \$4.1
- Custom-key relock is vendor-documented only for Pixel; Fairphone 3/4/5, Nothing, Sony, Motorola g-series are user-reported; Fairphone Gen 6/6+ untested. R17 F3
- Google's Pixel image terms bar modification or redistribution except per an enclosed, unread licence; commercial flashing is unresolved (gate G5). R17 F1, \$9 claim 1
- v1 posture: fetch OEM firmware at flash time, flash unmodified, pin digests in Zune's vbmeta; its 'strongest' legal rating is inferred, not evidenced. R17 \$3 option b

5-20 min; about 10 unattended plus 2 presses

Flash time per phone **reported**
Search summaries

50-60 min hands-on; 90-120 min wall

Hands-on time per phone **estimate**
Unmeasured; measure in first batch

8-10 phones/day; 15-20 once QA scripted

Throughput, one technician, 4-up
estimate
1,000 devices about 8 weeks with 3 technicians

\$3-5k per 4-phone cell

Station cell BOM **estimate**
Bare-metal Ubuntu, certified cables, charging rack

~\$122 / ~\$103 / ~\$85 at 10 / 100 / 1,000

Fully loaded cost per device
estimate
USD; India unit cost about INR 1,850 (R20)

2.1-5k devices

Self-install break-even **estimate**
Corrected from 2.5-5k; \$150-250k build and \$50-70 saving unsourced

Every Zune phone relocks with our key; OEM firmware redistribution rights remain unresolved.

R17 partly verified

Selling the Image: Installer, Qualification, Updates, Licensing

DECISION

Web installer (WebUSB, fastboot.js) runs first as Zune's provisioning station; customer self-install later. Relock with our AVB key is mandatory; no OTA with OEM firmware before written OEM terms.

WHY IT MATTERS

Relocking resists a child with no computer, but firmware updates must then ship in our own OTA, so OEM redistribution rights become a gating legal dependency.

KEY POINTS

- Relock with Zune's AVB key is mandatory for every 'Zune Ready' device; no unlocked tier; verified by attestation at the station. R17 §1, §4.1
- Custom-key relock is vendor-documented only for Pixel; Fairphone 3/4/5, Nothing, Sony, Motorola g-series are user-reported; Fairphone Gen 6/6+ untested. R17 F3
- Google's Pixel image terms bar modification or redistribution except per an enclosed, unread licence; commercial flashing is unresolved (gate G5). R17 F1, §9 claim 1
- v1 posture: fetch OEM firmware at flash time, flash unmodified, pin digests in Zune's vbmeta; its 'strongest' legal rating is inferred, not evidenced. R17 §3 option b
- Installer follows GrapheneOS's WebUSB fastboot.js (MIT) flow: 21-codename allowlist, unlock, flash, avb_custom_key, lock; Zune adds stricter pre-flight. R17 F5

UNLOCK AND CUSTOM-KEY RELOCK BY OEM (COMMUNITY DATA, 2026-09)

OEM	Unlock	Custom-key relock	Stance
Pixel	OEM toggle, then fastboot flashing unlock; carrier variants locked	Documented	Reference platform; D17 starts here
Fairphone	Per-device website code; two-step unlock	Reported FP3/4/5; Gen 6/6+ unverified	Was commercial candidate; D17 makes it later second source
Nothing (3)/(3a)	Plain fastboot	Reported; Nothing (2a) fails	Stage 2 candidate
Sony Xperia 1 V/VI	Portal plus IMEI code; US/carrier units may never unlock	Reported	Reference BSP only
Motorola	Portal code; g-series	Reported for g-series	Watch
Samsung, OnePlus, Xiaomi, ASUS	Samsung dropped unlock (One UI 8); OnePlus Deep Testing; ASUS tool ended	Newer OnePlus and Xiaomi ignore custom keys	Exclude

21 codenames (incl. tegu = Pixel 9a, stallion = 10a)

GrapheneOS installer device allowlist sourced
Read in web-install.js

374 of 743; 188 ship firmware (184 overlap)

LineageOS wiki entries needing OEM firmware first sourced
Recounted at lineage_wiki@902bd57, 2026-09-29

5 (FCM 7, 8, 202404, 202504, 202604)

Android 17 vendor interface levels accepted sourced
Read in GrapheneOS and LineageOS forks, not Google's tree

Superseded: US-only launch, US unlocked SKUs and USD pricing (\$59 + \$12.99/mo) (D18); VoLTE, E911, WEA, SMS/MMS qualification and cellular guardian-call lapse rules (D19); Fairphone Gen 6+ as commercial launch device (D17) ...

High
Google's firmware licence unanswered, Fairphone terms unsigned; R20 treats about 100 devices as the tolerated ceiling.
Re-flash service in v1, then pause; counsel and OEM outreach in parallel (gate G5).

High
Attestation depends on Google's remote key provisioning or a Zune proxy; Google's terms for non-certified OSes unverified.
Set the RKP host property or run a proxy; confirm Google's terms before launch.

High
Pixel QPR1 firmware ships before AOSP source; auto-updated customer phones fail the anti-rollback pre-flight.
Track Google's latest factory image; budget a vendor-on-r1 backport for every monthly update.

Stage 1: Station-mode web installer for one Pixel platform (9a/10a) with strict pre-flight, relock and attestation check; manual qualification against the Device Support Contract; Zune OTA server (staged rollout, health gating) ... Full scope: detail sheet.

India's child-data law has no clear exemption for parent-visible monitoring, Zune's core feature.

R19

partly verified

India: Law and Regulation

DECISION

Run an invite-only, parent-contracted pilot through an Indian entity with India-region hosting, Rule 10 parent verification at the flash visit, purpose-limited monitoring and 112 plus SOS, after Indian counsel sign-off.

WHY IT MATTERS

Rs 200 crore is the maximum penalty for breaching DPDP s.9 (children), and parent message review is the feature at legal risk. Indian counsel's opinion gates the pilot.

KEY POINTS

- DPDP s.9(3) bans tracking or behavioural monitoring of children; the Fourth Schedule has no parental-control or messaging exemption. R19 §2.1
- Defensible reading: the parent is part of the Data Principal (s.2(j)(i)) and consents, so Zune is a conduit. Unsettled; counsel gate G1. R19 §1, §10 Q1
- Child means under 18. Rule 10 verifies parents via held or provided details or a DigiLocker-type token; cards are not named. R19 §2.1
- Rule 12 never lifts s.9(2), which bars processing likely to harm a child's well-being; it bites on the AI assistant and messenger. R19 §2.1
- Dates: s.9 and Rule 10 bind 13 May 2027; MeitY floated 13 Nov 2026 (ungazetted). Behave as if in force now. R19 §2.1

RISK-RANKED OBLIGATIONS

Obligation	Applies	Severity	Design implication
DPDP s.9(3)	May 2027 (maybe Nov 2026)	Critical, Rs 200 crore	Strict-reading design plus counsel opinion
Rule 10 verification	Same	Critical	DigiLocker token or staff-checked ID; consent ledger
Under-18 account rule (proposed)	Unknown	High	Closed-contact positioning; MeitY outreach
POCSO ss.19-21; 3 h/2 h takedown	Now	High	Runbook, evidence vault, 24x7 on-call
Security, breach, CERT-In	Now / May 2027	High, Rs 250 crore	Encryption, logs, breach runbooks
Telecom Act ss.3, 42; TIUE; interception	Now, unclear	High	Counsel view; verified-parent traceability
2026 DoT authorisation and user-ID rules	In force Jun-Aug 2026	Unknown until read	Counsel reads before G1; may reach messenger, calls, phone custody

Rs 200 crore

Maximum penalty, DPDP s.9
sourced

Security failures Rs 250 crore; breach-notification failures

Under 18

Child threshold sourced
Act s.2(f); age bands do not reduce duties

13 May 2027

s.9 and Rule 10 effective
sourced
One source says 14 May; 16 Dec 2025 corrigendum unread

Superseded: Tiered-visibility default (excerpts plus metadata at 13+) conflicts with 'parent-visible' in D7/D11 and is not adopted until the founder decides; Hindi parent notices need an explicit exception to D20.

High
s.9(3) has no guidance or Board ruling; parent message review may count as prohibited monitoring.
Purpose-limited monitoring; written counsel opinion; seek s.9(4), s.9(5) or s.17(5) exemption dialogue.

High
Reported under-18 social-media rule and Karnataka under-16 plan could capture a 1:1 child messenger.
Closed-contact positioning: no discovery, handles, feeds or groups; MeitY outreach after counsel.

High
Number-free child messenger is policy-adverse given DoT SIM-binding direction (Nov 2025).
Invite-only, parent-linked identity; each child traces to a verified adult.

Stage 1: Invite-only ~200-family pilot: Indian private limited; AWS Mumbai (Hyderabad DR); Indian NTP and 180-day logs; one verified adult per family (DigiLocker token or staff-checked ID) plus mobile OTP; consent ledger and both-family consent per contact pair ... Full scope: detail sheet.

200 Indian families are reachable; the INR 43-48k Pixel is the binding constraint.

R20

partly verified

India: Market, Connectivity, Language and Pricing

DECISION

Default: launch in Bengaluru, in person, four phases to ~200 families; Wi-Fi first, optional nano-SIM; telephony deny-all except 112; English UI; Pixel 9a/10a only; pricing unvalidated.

WHY IT MATTERS

The funnel is the real risk: families must own or buy a Pixel at INR 43-48k, and unit economics at INR 399/month are thin once cloud cost is counted.

KEY POINTS

- Pixel 10a: INR 49,999 launch, 47,999 now. The 39,999 sale price needs a bank offer plus exchange; budget INR 43-48k per unit. R20 §2.1
- Keep stock Telecom, IMS and CarrierSettings (Jio is VoLTE-only; 112 needs IMS) but deny everything except 112. R20 §1.3, §4.2
- Platform emergency numbers come from the modem, so '112 always works' is unproven; enforce 112-only in Zune's screen and field-test per carrier. R20 §2.2
- Child's SIM is an ordinary KYC'd SIM in the parent's name; moved to another phone, it escapes all OS controls. R20 §4.2
- Language (D20): English UI; assistant understands Hinglish but replies in simple English; a 300-prompt Indian-English/Hinglish safety eval gates release. R20 §4.4

STAGED PLAN TO ~200 FAMILIES

Stage	Families (cumulative)	How	Gate to next
C0	12 (12)	Staff and friends, own Pixels, in person, free	Carrier matrix passes; no open child-safety P1; counsel signs terms
C1	30 (42)	Bengaluru school-parent community; founding price	95%+ first-pass flash; 85%+ daily use; NPS 40+
C2	70 (112)	Bengaluru plus courier to Hyderabad, Pune	OTA 98% in 72 h; written Google firmware answer; 90%+ payments
C3	90 (200)	Three cities	75%+ eight-week retention; activation fee covers cost

INR 49,999 launch; INR 47,999 now

Pixel 10a price in India reported
Flipkart Big Billion Days 39,999 needs bank offer plus excha

INR 43-48k

Planning cost per phone estimate
Refurbished Pixel 9a about INR 29.8-37.2k; 8a about 25.9k (C

INR 2,499 activation + INR 399/month or 3,999/year

Proposed price (unvalidated) estimate
GST-inclusive; USD/INR about 95.97 (RBI reference 2026-09-28

Superseded: Nothing overturned by a D-number (written against D17-D21). Its public priced funnel and INR 199 founding price yield to R19's free invite-only first phase until counsel gates clear ...

High

Funnel: INR 35-48k phone for a child; Pixel is small in India (about 2% of premium, weak source).
Waitlist page to test demand; refurbished after eligibility test; Pixel 8a only if price-sensitive.

High

Carrier behaviour on a custom OS (Jio IMS, SIM-less 112, Vi, BSNL) is untested.
Gate G0 carrier matrix before any family; promise no SIM-less 112 until tested.

High

Google firmware redistribution unanswered; blocks beyond about 100 devices.
Written Google answer gates phase C2; Indian counsel and Google outreach now.

Stage 1: Four staged phases from Bengaluru. Pre-flash checks: GST invoice, IMEI via Sanchar Saathi KYM (SMS 14422), reject grey and US-SKU units, OEM-unlock and bootloader reads, battery health, nano-SIM tray. Full scope: detail sheet.

Compliance: India law now governs the launch; the US map is a later-market reference

R11 superseded

Compliance, Legal and Regulatory Map (US-framed)

DECISION

R11 recommended a US-only launch with COPPA, California AB 1043 and SB 1119 compliance, EU/UK deferred, and no hardware sales. D18 (India first) supersedes the market; no-hardware, open-source and trademark items survive.

WHY IT MATTERS

Do not rely on any US date, fine or consent method here. What survives is the strictest-tier design posture, open-source and trademark hygiene, and the never-sell-handsets stance.

KEY POINTS

- Proposed: US-only launch for ages 6-12, with FTC-approved COPPA safe-harbor membership before general availability. Dropped: D18 makes India first. R11 §1
- Proposed: California AB 1043 age-bracket API (four brackets) in the first image. India has no OS age-signal law (R19); dropped for now. R11 F2; R19 §6
- Proposed: parents reading kid SMS under an interception-law counsel opinion. D19 removes SMS; messenger and AI-chat reading remain a legal question (R19). R11 §4.1 row 2
- Proposed: defer EU/UK (9-18 months extra, estimate) and avoid India; D18 reverses the India call. R19 calls it wrong. R11 §1, §4.3; R19 §6
- Survives: never sell or stock handsets (D15/D16), keeping manufacturer-type duties (CPSIA, FCC/CE, CRA) off Zune. R11 §1, row 14

about \$53,088 per violation

Amended COPPA penalty
reported
US only; secondary source; no longer the launch regime

\$2,500 negligent / \$7,500 intentional per child

AB 1043 penalty **sourced**
California AG only; read in legislature data mirror

2027-01-01

AB 1043 operative date **sourced**
Earlier-set-up devices by 2027-07-01; not applicable in Indi

RISK-RANKED OBLIGATIONS AND THEIR STATUS TODAY

Obligation	Severity	Design implication	Status under D18-D21
Amended COPPA (US under 13)	Critical	Parental consent at enrolment; retention schedule; security program	Replaced by DPDP s.9 and Rule 10 (R19)
Interception law (parents read kid messages)	High	Child-visible 'parents can see this' signal; written parental consent	Maps to Telecom Act s.42, IT Act s.69 (R19)
AI-minor laws (CA SB 1119, SB 243)	High	Tutor not companion; crisis routing; parent alerts; incident log	Design survives; California is not first market
California AB 1043 age-bracket API	High (hard date)	Birth-date step at setup; four-bracket API; no sharing	Does not apply in India (R19)
CSAM and NCII (2258A, REPORT Act, TAKE IT DOWN)	High	Photos stay on device; NCMEC registration; 48-hour takedown path	Becomes POCSO reporting to Indian police (R19)
OEM and Google firmware licence	High	No OTA firmware until a written answer	Survives (R17 gate G5)
YouTube terms and made-for-kids	High (business)	Offline licensed tier first; made-for-kids only; kill switch	Survives

Superseded: US-only launch and 'avoid India' sequencing (D18); US beta of up to 100 devices (D18, D21); kid-SMS reading and cellular E911/Wi-Fi-calling items (D19); US consent methods and AB 1043, CPSIA and FCC rows (R19 section 6).

High
Nearly every legal date and fine is secondary or memory; primary texts unread.
Counsel re-reads primary texts before any reliance; treat US dates as unverified.

High
Card or KBA parental consent (R11 row 1) is not a named Rule 10 method in India.
Use DigiLocker token or staff-inspected government ID plus consent ledger (R19).

High
Interception exposure when parents read children's messages persists under India's Telecom Act and IT Act.
Child-visible indicator, written parental consent, Indian counsel opinion before first external family (R19 G1).

Stage 1: Superseded US sequence: private beta with up to 100 provisioned US devices, counsel engaged, COPPA consent live. Reusable build-in checklist: portal consent ledger, child-visible 'parents can see this' indicators, AI safety kit, trust-and-safety kit ... Full scope: detail sheet.

Own hardware is gated, not scheduled; first device is a rebadged ODM phone

R21unverified

Own-hardware roadmap for India

DECISION

Recommended: run the ~200-user Pixel phase first (D21); send paper RFQs now, commit money only at gates H1-H4. First own device: rebadged ODM 4G phone, voice off except 112.

WHY IT MATTERS

Own hardware could widen the market below the INR 40k+ Pixel price and remove every unlock path, but adds patching, memory-cost and key-custody risk.

KEY POINTS

- Case is price and tamper-proofing, not margin: Pixel 10a is INR 47,999; an own device at INR 15-18k widens the funnel. R21 s1
- First device: rebadged ODM-platform 4G LTE phone on Snapdragon 4 Gen 2 or 6-series; precedents HMD Fusion X1 and Light Phone III. R21 s1, 2.1
- Patching is the central risk: entry SoCs lack Google long-support, so we own kernel, firmware and HAL patching; promise 3 years. R21 s1, 2.1
- Memory shock: mobile DRAM roughly doubled in Q2 2026 and is 43-45% of a low-end BoM; Nothing cancelled CMF Phone 3 Pro. R21 2.5
- No Indian incentive fits a startup: PLI ended 31 Mar 2026; MPMS needs INR 1,000 crore turnover (Indian brands), 10,000 crore otherwise. R21 2.4

UNIT ECONOMICS BY LIFETIME VOLUME (USD, ALL ESTIMATES)

Lifetime units	Path	Fixed cost	Team (3 yrs)	Peak working capital	Contribution	Net
1,000	(a) pilot lot	0.25M	0.30M	0.12M	0.07M	-0.49M
10,000	(a) lots of 2-3k	0.25M	0.30M	0.3-1.2M	0.65M	About +0.1M (breakeven ~8.5k)
100,000	(b) ODM design	1.5M	1.2M	3-5M	7M	+4.3M (breakeven ~39k)

INR 47,999 (~39,999 on sale)

Pixel 10a India price reported

From report 20; USD 1 about INR 96.

USD 115 (range 95-150)

Assumed landed cost per own phone estimate

Author assumption.

INR 16,999 / ~INR 2,070 (USD 21)

Assumed MRP / hardware contribution estimate

After 6% warranty reserve and 3% payments/logistics; GST cre

High

BSP lag and patch ownership: entry SoCs have no Google long-support; 3-year promise needs unverified ODM SLAs.

Secure Hardware Contract: patch SLA (critical fix to OTA within 30 days), 4-year BSP, source escrow.

High

Memory cost and allocation: first PO may price 25%+ above today's quotes or go unfilled; small brands get no priority.

Gate H3 requires memory allocation secured before first PO; Pixel path stays the base.

High

Key custody: a fused key cannot be rotated; loss or leak is a fleet recall.

HSM with two-person custody before any ODM contact; ODM gets only the public hash.

Stage 1: Not v1 (D21). Now, with no hardware spend: device-agnostic vendor/zune layer; HSM-held hardware root key with two-person custody before any ODM contact; OTA that updates modem, bootloader and vendor partitions; Secure Hardware Contract as the RFQ spec ... Full scope: detail sheet.

Confidential. Figures are labelled sourced / reported / estimate; see Evidence status.

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Unit economics: inputs, not a forecast

Prices and costs from the reports; most are estimates and none is validated with customers

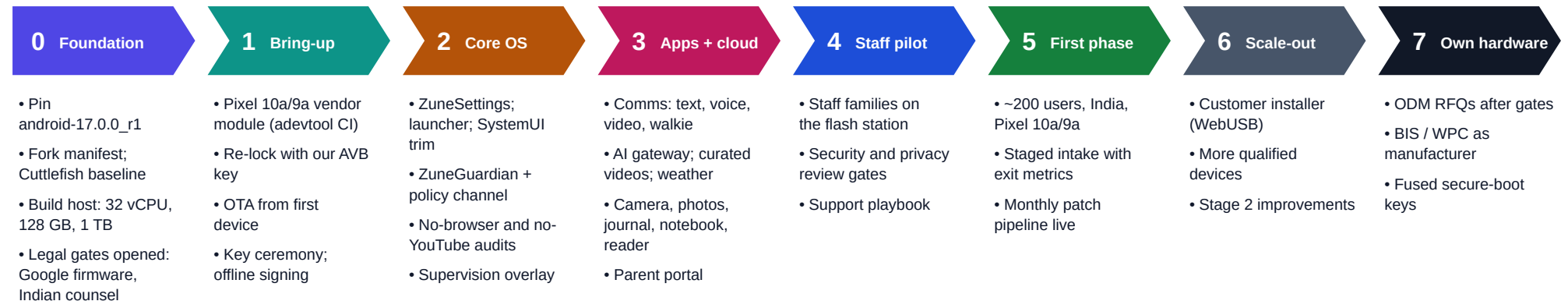
Src	Figure	Value	Basis	Note
R17	v1 station cost per unit	\$25-40 (labour only)	estimate	R18 puts fully loaded cost at \$85-125
R17	Self-install support load	20-30% need help; about \$4-6 blended	estimate	Author's inference
R17	New-platform qualification	4-6 engineer-weeks plus 3 units (about \$1.2k)	estimate	Sibling model 1-2 weeks; 2-3 days per monthly release
R17	US software-only price anchors	Bark \$14/mo or \$99/yr; Pinwheel \$15/mo	reported	Search summaries; vendor pages blocked; India anchors in R20
R18	Station cell BOM	\$3-5k per 4-phone cell	estimate	Bare-metal Ubuntu, certified cables, charging rack
R18	Fully loaded cost per device	~\$122 / ~\$103 / ~\$85 at 10 / 100 / 1,000	estimate	USD; India unit cost about INR 1,850 (R20)
R18	Bare-install price anchors	\$85 mail-in; \$60 remote-guided	reported	GrapheneOS installers; price install only, not QA or enrolment
R18	Fixed costs	Counsel \$10-25k; HSM/KMS \$1-5k; insurance \$1-3k/yr	estimate	Excludes payment fees and cloud cost
R20	Pixel 10a price in India	INR 49,999 launch; INR 47,999 now	reported	Flipkart Big Billion Days 39,999 needs bank offer plus exchange; about
R20	Planning cost per phone	INR 43-48k	estimate	Refurbished Pixel 9a about INR 29.8-37.2k; 8a about 25.9k (Cashify)
R20	Proposed price (unvalidated)	INR 2,499 activation + INR 399/month or 3,999/year	estimate	GST-inclusive; USD/INR about 95.97 (RBI reference 2026-09-28)
R20	In-person unit cost	about INR 1,850	estimate	Courier adds INR 800-3,000 round trip; omits payment fees and cloud co
R20	Cloud cost per child	INR 180-200/month	estimate	INR 199 founding price is below this
R20	Monthly net at 200 families, full price	about INR 26-30k	estimate	After GST, cloud cost and about 2% gateway fees; original INR 68k was
R20	Open-Meteo Standard plan	USD 29/month, 1M calls	reported	Commercial licence; free tier is non-commercial
R21	Pixel 10a India price	INR 47,999 (~39,999 on sale)	reported	From report 20; USD 1 about INR 96.
R21	Assumed landed cost per own phone	USD 115 (range 95-150)	estimate	Author assumption.
R21	Assumed MRP / hardware contribution	INR 16,999 / ~INR 2,070 (USD 21)	estimate	After 6% warranty reserve and 3% payments/logistics; GST creditable.
R21	Net subscription per month	INR 140 (INR 399 gross)	estimate	After cloud ~INR 190 and payment fee 8; 2.5 paying years = USD 44.
R21	Breakeven volume	~8.5k (rebadge) / ~39k (ODM design)	estimate	Vendor ODM guide claims ~22k units; low reliability.
R21	Mobile DRAM shock	~2x in Q2 2026; sub-INR 10k phones +32%	reported	LPDDR4X 4 GB +75%; budget shipments -65% in H1 2026.
R21	BIS CRS (IS 13252)	INR 53,000 fee + lab INR 10-80k	reported	30-45 working days; valid 2 years; licence in factory's name.

Src	Figure	Value	Basis	Note
R07	COGS per typical child (240 turns/month)	\$1.42	estimate	Sonnet 5.5 plus guards; \$1.10 with 25% FAQ cache hits; \$0.78 with Haik
R07	Claude Sonnet 5.5 price per MTok	\$2 in / \$10 out	sourced	Cache read \$0.20; read on Anthropic pricing page
R07	Claude Haiku 4.5 price per MTok	\$1 in / \$5 out	sourced	Cache read \$0.10
R07	Cloud voice cost avoided by on-device voice	\$2-4 per child-month (TTS)	estimate	From memory prices; cloud STT would add \$0.1-0.2
R13	LiveKit Cloud Ship plan	\$50/mo incl. 150k participant-min; \$0.0005 overage	reported	Aggregator pages only; livekit.com blocked; \$0.12/GB egress
R13	LiveKit Cloud Scale plan	\$500/mo incl. 1.5M min; \$0.0004 overage	reported	\$0.10/GB; region pinning, SOC 2 report and BAA reportedly Scale and ab
R13	Video cost at 10k kids	~\$0.06 per kid per month	estimate	Assumes 10% make one 10-min call a day; excludes PTT and included band
R13	Video cost at 100k kids	~\$5k/month on Scale; self-host saves ~60-80%	estimate	Revisit at \$3-5k per month
R14	Open-Meteo plan call budgets	Standard 1M, Professional 5M, Enterprise 50M+ per month	sourced	Read from pricing page source; call weighting (variables x models) unc
R14	Open-Meteo first two plan prices	about \$29 and \$99 a month	reported	Search summary only; pricing table unreadable.
R14	Open-Meteo free tier	<10,000 calls/day, 300k/month, non-commercial only	sourced	Apps with subscriptions count as commercial.
R14	OpenWeather alternative cost	about EUR 1.8k (10k) / 8.8k (100k) a month	reported	Search summaries disagree; about 10x Open-Meteo at scale.
R14	Beta cost	tens to low hundreds of dollars a month	estimate	Depends on plan tier; assumptions inferred.
R10	Running infra cost	\$0.5-0.9k/month	estimate	2 bare-metal hosts plus ephemeral runners; from memory, prices not che
R10	One-time infra, lab and signing hardware	\$10-18k	estimate	Device lab \$4-8k; signing hardware about \$2.5k; build hosts about \$6k
R02	Pixel 10a US price / support end	\$499 / Mar 2033	reported	Launched 2026-03-05; secondary sources only. R21 cites India price INR
R02	Pixel 9a US price	\$499 launch; ~\$300-430 now	reported	Support to Apr 2032.
R02	Incumbent kids' phones	\$100-240 device + \$15-20/mo	reported	Pinwheel Plus 6 \$219 + \$14.99/mo; all sources blocked, treat as soft.
R02	ODM vendor quotes (low reliability)	FOB \$48-115; MOQ 3-5k; NRE \$5-65k	reported	Vendor marketing: 'typical ODM launch \$215k'.
R02	Own-phone cost (author estimate)	\$0.5-1.5M, 12-24 months	estimate	Certified cellular kids' phone; R21 revises.
R02	ODM gate (author proposal)	25k+ units/yr, \$1.5M+ committed	estimate	R21 calls it too coarse; a rebadge breaks even near 8-10k units.
R02	Flash/QA/ship per unit (estimate)	~\$35	estimate	10-20 port station at ~10 min/unit. The \$450 landed figure assumed we

Read with care. The report skeptics flagged that the proposed price and cost figures conflict in places (cloud cost per child can erase subscription margin; US-dollar and rupee plans differ; public-funnel pricing conflicts with the invite-only, free first phase proposed on legal grounds). A financial model should follow a priced waitlist test.

Roadmap: gates before dates

Sequenced by dependency; durations come after staffing and funding decisions



STAGE 1: make every feature exist, end to end, in its simplest form

STAGE 2: make it better

GATES

- **G0 Legal:** written answer or counsel opinion on Google firmware; Indian counsel on DPDP and OTT rules; no paid launch before it.
- **G1 Device:** re-lock verified on a sacrificial unit of every qualified model; OEM terms signed.
- **G2 Security:** monthly patch pipeline, key custody (offline HSM), OTA tested on real devices.
- **G3 Pilot exit:** install success, support load, child-safety incidents and bypass attempts within thresholds.

EXTERNAL DEPENDENCIES

- **Google:** Pixel firmware licence; partner security-patch access (unknown for a startup)
- **Indian counsel:** DPDP consent architecture; OTT messaging; handset rules
- **LLM vendor:** terms for products used by children; India availability
- **YouTube:** API compliance review for a closed, curated client
- **Hosting:** India-region cloud; LiveKit (self-host or cloud)

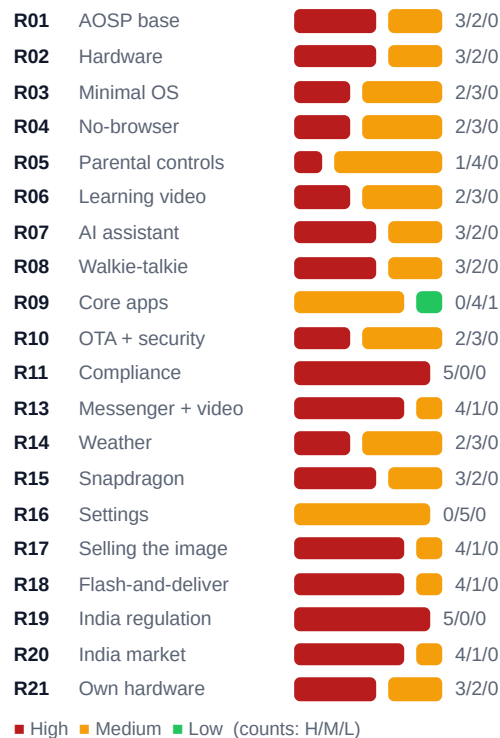
No calendar dates are stated because no staffing or funding assumptions exist yet. Phases derive from reports R01, R02, R15, R16, R17, R18, R21.

Confidential. Figures are labelled sourced / reported / estimate; see Evidence status.

Risk register: the top risks

Top high-severity risks per topic; all 100 risks (55 high) are on the detail sheets

RISK CONCENTRATION



Src	Topic	Risk	Mitigation
R01	AOSP base	Public AOSP patches can trail authoring by 2.5-6 months and arrive batched; Mainline and WebView updates are Zune's to ship.	Monthly patch-ingest pipeline; OTA SLA: Critical within 14 days of bulletin, others 30; about 1 FTE security engineer.
R02	Hardware	Legal basis for shipping Google's Pixel binaries commercially is unresolved; Google's licence text was never read.	Get counsel and a Google contact first; keep stock-firmware-plus-policy as fallback.
R03	Minimal OS	Nothing is build-verified: feature masks on a Pixel vendor image, release-flag overrides and PackageInstaller removal are unproven.	Build on Cuttlefish first; test on userdebug with privapp enforcement on; run on-device smoke tests.
R04	No-browser	Chromium/WebView is the largest attack surface and imposes a permanent update cadence; GPL-2.0 patch obligations need legal review.	Commit to an update SLA at Chromium cadence; choose Vanadium prebuilt or own build; legal review.
R05	Parental controls	Server holds children's location and transcripts; a breach is existential. 24/7 SOS staffing is a dependency.	Per-family KMS envelope encryption with crypto-shredding, retention limits, dual-approved break-glass staff access, hash-chained audit log.
R06	Learning video	YouTube can cut referrer access for our origin at any time; only recourse is an appeals form.	Remote kill switch; Videos works on day one from Tier 1 and Tier 3 content.
R07	AI assistant	Anthropic may decline, add conditions, audit or suspend; single-vendor dependency.	Provider abstraction; OpenAI or self-hosted fallback; written confirmation, ZDR and DPA before launch.
R08	Walkie-talkie	Android 17 native audio-playback enforcement is unread; incoming audio could be silently dropped on device.	Test on device or Cuttlefish with 'adb shell cmd audio set-enable-hardening enable' before other media work.
R10	OTA + security	Security-branch access (android17-security-release) is unverified; non-partner patch lag was about 125 days.	Authorise partner-access outreach to Google/ODM; cross-check against GrapheneOS and LineageOS branches.
R11	Compliance	Nearly every legal date and fine is secondary or memory; primary texts unread.	Counsel re-reads primary texts before any reliance; treat US dates as unverified.

Risk register (continued)

Top high-severity risks per topic; all 100 risks (55 high) are on the detail sheets

Src	Topic	Risk	Mitigation
R13	Messenger + video	Android 17 may silently drop incoming audio and block camera/mic from a boot-started service; inferred, not tested.	Keep room client in the visible call app; test with 'adb shell cmd audio set-enable-hardening enable' first.
R14	Weather	Alert design (NWS, WEA) assumes a US launch; this report has nothing for India's alerts.	Reconcile through reports 19 and 20; NWS is US-only.
R15	Snapdragon	OEM firmware redistribution licence unknown; only Sony's Open Devices exists and its licence was never read.	Get written OEM permission first; D15/D16 may shrink exposure because OEM firmware is already on the customer's phone.
R17	Selling the image	Google's firmware licence unanswered, Fairphone terms unsigned; R20 treats about 100 devices as the tolerated ceiling.	Re-flash service in v1, then pause; counsel and OEM outreach in parallel (gate G5).
R18	Flash-and-deliver	Customer phones in unfixable states: carrier id, anti-rollback, FRP, account locks.	Hard eligibility gate; customers self-test with screenshots; new-in-box phones preferred.
R19	India regulation	s.9(3) has no guidance or Board ruling; parent message review may count as prohibited monitoring.	Purpose-limited monitoring; written counsel opinion; seek s.9(4), s.9(5) or s.17(5) exemption dialogue.
R20	India market	Funnel: INR 35-48k phone for a child; Pixel is small in India (about 2% of premium, weak source).	Waitlist page to test demand; refurbished after eligibility test; Pixel 8a only if price-sensitive.
R21	Own hardware	BSP lag and patch ownership: entry SoCs have no Google long-support; 3-year promise needs unverified ODM SLAs.	Secure Hardware Contract: patch SLA (critical fix to OTA within 30 days), 4-year BSP, source escrow.

What is verified, and what is not

Every report went through research; skeptic passes re-check the load-bearing claims. Do not take flagged claims as fact.

0
verified

9
partly
verified

9
unverified

3
superseded

How to read this. Evidence came from GrapheneOS and LineageOS source on GitHub plus web search: Google's own AOSP tree, Google licence pages and most Indian legal texts were unreachable from the research environment. A follow-up session with AOSP access will re-verify. Skeptic passes for several reports are still to be re-run.

Src	Report	Status	Note	Flags
R01	AOSP base	partly verified	Passed skeptic pass; core evidence from GrapheneOS/LineageOS manifests because Google's source.android.com and googlesource were ...	9
R02	Hardware	partly verified	Second pass confirmed GrapheneOS/LineageOS facts from GitHub and corrected two ...	8
R03	Minimal OS	unverified	No skeptic second pass; claims were read in GrapheneOS/LineageOS source files, but nothing was build-tested.	10
R04	No-browser	unverified	No skeptic pass in file. Most claims read in GrapheneOS 17 source, but Google's AOSP tree was blocked and key designs are ...	7
R05	Parental controls	unverified	No second-pass skeptic section exists in this report. Many claims were read in primary files, but GrapheneOS trees ...	8
R06	Learning video	unverified	No skeptic pass. YouTube policy and most licence claims are search summaries (primary pages blocked) ...	6
R07	AI assistant	unverified	No skeptic pass. Anthropic terms, ZDR, prices, LiteRT-LM and speech-engine facts read first-hand ...	7
R08	Walkie-talkie	unverified	No second-pass skeptic review exists. Android 17 facts read in primary sources ...	8
R09	Core apps	unverified	No skeptic pass. Library, licence and AOSP-tree claims were read in primary files ...	7
R10	OTA + security	unverified	No second-pass skeptic section. Evidence is mostly primary files, but from GrapheneOS/LineageOS trees not Google's ...	8
R11	Compliance	superseded	No skeptic pass; nearly all dates secondary or memory. Core US-only recommendation overturned by D18 ...	6

Src	Report	Status	Note	Flags
R12	Telephony (dropped)	superseded	Core recommendation (carrier calls/SMS with framework allowlist) overturned by D19.	8
R13	Messenger + video	partly verified	Skeptic pass: 15 confirmed, 11 corrected, 1 refuted, 5 unverified of 32 rows.	9
R14	Weather	partly verified	Skeptic pass done: Open-Meteo, Google terms and GrapheneOS source read in primary files ...	8
R15	Snapdragon	superseded	Launch call overturned by D17; skeptic pass found no Snapdragon relock evidence beyond user reports ...	8
R16	Settings	partly verified	Skeptic pass done with corrections, but AOSP primary sources unreachable; checked against GrapheneOS/LineageOS trees ...	7
R17	Selling the image	partly verified	Skeptic pass with corrections; flashing mechanics reproduced from GrapheneOS/LineageOS files ...	7
R18	Flash-and-deliver	partly verified	Pixel flash and relock mechanics reproduced from real GrapheneOS/LineageOS files ...	8
R19	India regulation	partly verified	Skeptic pass re-read DPDP Act, Rules, Telecom Act, POCSO judgment via GitHub mirrors; other Indian sources blocked (secondary) ...	7
R20	India market	partly verified	Skeptic pass corrected many items; market and legal facts mostly search summaries ...	7
R21	Own hardware	unverified	No skeptic pass; evidence is mostly WebSearch summaries because fetches were blocked.	8

Decisions and help we need

The open items that change the plan most; each has a recommended default in the research

Area	Decision / ask	Why it matters	Owner
Legal	Written answer or counsel opinion on Google's licence to redistribute Pixel firmware commercially	Blocks any paid launch	Counsel + Google outreach
Legal	Indian counsel: DPDP s.9(3) monitoring, messenger classification, parent access vs interception law	Gate before any external family	Indian counsel; Indian private limited
Product	Target age range of children (assumed ~6-13; India treats under-18 as a child)	Drives safety design, AI, content	Founder
Product	Emergency calling: keep a 112-only path? (default yes; to field-test on data-only SIMs)	Safety + regulatory	Founder + counsel
Security	Fund monthly patching: about 1 platform/security engineer plus kernel support; seek partner access	Central business risk	Founder / hiring
Security	WebView supply: Vanadium prebuilt vs own Chromium build, with a Chromium-cadence update SLA	Largest attack surface	Tech lead
Content	Videos vs decisions D2/D3: research recommends a licensed offline base plus an optional made-for-kids-only YouTube tier behind a kill switch (ask YouTube in writing; drop at 8 weeks if refused)	Conflicts with a YouTube-only curation promise	Founder + legal
Commercial	Price: test INR price on a waitlist; ~Rs 1,250/month is ~15x JioShield (Rs 1,000/yr); first phase invite-only and free until Indian payments are ready	Willingness to pay is the binding constraint	Founder
AI	Obtain the LLM vendor's written approval for a child-facing product; India-region inference	Gate one for the assistant	Founder
Commercial	Pricing: invite-only and free first phase until Indian payments are ready; test INR price on a waitlist	Unit economics unvalidated	Founder
Operations	First city (Bengaluru proposed), staff the flash station, spare-phone reserve (4-6) as an exception to no inventory	Pilot logistics	Founder
Hardware	Own-hardware gates H1-H4; paper RFQs under NDA; key custody in an HSM; promise 3 years of patches	Later phase	Founder
Brand	Name clearance: 'Zune' is a codename with Microsoft trademark history	Before any public use	Founder + counsel

To be completed by the founder

Facts the research cannot supply. Nothing on this page is estimated.

Team and roles

Founder background; who owns platform/security, backend, apps, ops; key hires needed (security engineer + kernel support is in the plan).

Traction and validation

Waitlist size, parent interviews, pilot commitments, partners, letters of intent.

Funding ask and use of funds

Round size and terms. Cost categories supported by the research: build infrastructure (about \$10-18k one-time, \$0.5-0.9k/month, estimate), legal (Indian counsel, Google), device lab, flash station, cloud (AI, comms), security staffing.

Financial model

Revenue, margin and runway. The research gives unvalidated pricing and unit-cost inputs only; a model should be built after pricing is tested on a waitlist.

Decision log

Dated founder decisions; later decisions supersede earlier ones. Canonical source: docs/REQUIREMENTS.md

#	Decision	Date
D1	No web browser on the device.	2026-10-02
D2	No way for the end user to reach YouTube (no app, site, link-out, or deep link). Videos appear only inside the Zune Videos section.	2026-10-02
D3	Videos are curated YouTube content, shown only if relevant to the child's age group or to a topic the child asked about (research). Fully controlled experience.	2026-10-02
D4	SUPERSEDED by D19 (2026-10-02). (was: Cellular voice calls and SMS are IN scope.)	2026-10-02
D5	SUPERSEDED by D19 (2026-10-02). (was: Parent controls an allowlist of numbers that can call the device (inbound) and which numbers the child can call/communicate with (outbound).)	2026-10-02
D6	SUPERSEDED by D19 (2026-10-02). (was: SMS is not readable on the device ; it is readable in the parental-control interface.)	2026-10-02
D7	WhatsApp-like in-product messenger between kids using Zune devices. 1:1 only (no groups). Text + emoji only. Under the parental contact controls (approved contacts only). (Was "same as calls/SMS"; see D19.)	2026-10-02
D8	Video calling only with approved participants: other Zune kids, or parents via the parental-control interface. The parent-side calling interface is deferred ("later").	2026-10-02
D9	Weather app that also teaches children about weather.	2026-10-02
D10	Walkie-talkie (push-to-talk).	2026-10-02
D11	AI assistant with a ChatGPT-style conversation UI; kids can ask questions and send images .	2026-10-02
D12	Parental controls are browser-based (parent portal).	2026-10-02
D13	The product is limited to a small, curated set of supported devices . High-end Snapdragon -class devices are acceptable ("very high grade"); the device list is not limited to Pixels. Supersedes the Pixel-first recommendation in `docs/research/02-hardware-target.md` until topic 15 reports.	2026-10-02
D14	The device has a Settings app with the minimum settings: Wi-Fi and mobile data stay fully standard ("as it is") , plus only what is fairly required. Nothing unnecessary .	2026-10-02
D15	Business model: sell the OS image (software); customers bring a qualified phone from a curated supported-device list and install it themselves. We do not sell or inventory hardware.	2026-10-02
D16	Version 1 delivery is company-provisioned . The first batch of users contact us; we flash the image on their qualified phone and give it to them . Customer self-install (the BYO installer of D15) comes in a later version. (Phones are assumed to be customer-supplied, consistent with D15; correct this if we will supply phones.)	2026-10-02
D17	Start on a Pixel . The founder accepted beginning with a Pixel (Pixel 10a / 9a class, "a good performing phone") as the first supported device. A Snapdragon phone is not required for v1 ; Snapdragon candidates (Fairphone Gen 6+, Nothing Phone (3)) become a later second-source / premium tier, subject to relock bring-up and written OEM terms (report 15). Supersedes D13's Snapdragon preference for v1.	2026-10-02
D18	India is the first launch market . Supersedes assumption A1 (US). Implications to research: India's child-privacy law (DPDP Act and Rules; a "child" is under 18), telecom/messaging rules (DoT, TRAI), Indian carriers and VoLTE, emergency numbers and cell-broadcast alerts, Indic languages, rupee pricing and recurring-payment rules, Pixel supply and warranty in India.	2026-10-02
D19	No cellular voice calls and no SMS . All communication is internet-based and WhatsApp-style: in-product text messages, voice calls and video calls , 1:1 only (no groups), approved contacts only, parent-controlled. Supersedes D4, D5 and D6 . The device needs only data connectivity (Wi-Fi and mobile data, D14).	2026-10-02
D20	English only for version 1: no Hindi or other Indian languages (Stage 2 at the earliest).	2026-10-02
D21	First phase: about 200 users on Pixel phones; eventually the company builds its own hardware . Own hardware is a later roadmap phase, not v1. The 10/25/100 cohort sizes in report 18 are replaced by a staged ~200-user first phase.	2026-10-02

AOSP base release, branches, cadence and build host (1/3)

R01

partly verified

Detail sheet: Build on pinned AOSP Android 17; security-patch lag is the central risk.

DECISION

Base on AOSP tag android-17.0.0_r1 (CP2A.260605.016), zero-fork-first with our own device/zune and vendor/zune trees; plan one rebase onto the Q4-2026 AOSP drop before beta freeze.

Passed skeptic pass; core evidence from GrapheneOS/LineageOS manifests because Google's source.android.com and googlesource were unreachable; several corrections applied.

WHY IT MATTERS

No GMS means no Play system updates, and Zune should plan for no partner early access. Mainline and WebView fixes ship only through Zune's own OTA.

KEY POINTS

- Base is AOSP tag android-17.0.0_r1, build CP2A.260605.016, API 37, security patch level 2026-06-05; GrapheneOS and LineageOS 24.0 pin it. R01 F1
- Observed: AOSP source lands about Q2 and Q4. GrapheneOS had no AOSP rebase Jan-May 2026; QPR1 (Sep 2026) is not in AOSP. R01 F2
- A 49-commit batch including security fixes, authored Mar-Jul 2026, landed in GrapheneOS on 2026-09-09; median lag 127 days. R01 F2
- Android 15 security tags had 2-3 month gaps in 2025; mirror shows no Android 16 security branch five months after release. R01 F2
- AOSP carries a native supervision framework (SupervisionManager, Settings Supervision screen); three config hooks are empty for Zune's agent to fill. R01 F6
- Stock AOSP handheld products ship Browser2; CaptivePortalLogin and HTMLViewer are other web-rendering surfaces. Never inherit handheld_product.mk unchanged. R01 F5
- WebView source is undecided: AOSP prebuilt (freshness unchecked, no update path), own Chromium build, or licensed prebuilt. R01 F3b
- Siso is the Android 17 default build tool; kernels still build separately with Bazel/Kleaf; Cuttlefish gives hardware-free CI (needs /dev/kvm). R01 F5, F7

NUMBERS

Metric	Value	Basis
AOSP base tag	android-17.0.0_r1 (CP2A.260605.016, SDK 37) Read in GrapheneOS and LineageOS manifests; googlesource itself unreachable	sourced
Security patch level of tag	2026-06-05 Strong but indirect: inferred from first downstream diff; tag not readable	sourced
Median fix lag, author to GrapheneOS	127 days (range 65-178) 49-commit batch merged 2026-09-09; 'first public' date is inference	sourced
Android 15 security-tag gaps	2-3 months No tag Jul/Aug 2025; r10 2025-09-02, r11 2025-11-03; mirror data ends 2025-11	sourced
Repos hosted by GrapheneOS / LineageOS	12.2% / 10.1% GrapheneOS 129 of 1,057 (88 AOSP forks = 8.3%); LineageOS 108 of 1,067, not all forks	sourced
Zune AOSP fork target	under 30 repos Judgement with no supporting data; kids lockdown may need more	estimate
Proposed CI build host	32 vCPU, 128 GB RAM, 1 TB NVMe, Ubuntu 24.04 GrapheneOS states 136 GiB sync + 100 GiB build, 32 GiB RAM; a floor if Zune builds Chromium or kernels	estimate
Clean / incremental build time	1.5-3 h / 3-15 min No source states it; Google's 40-minute reference (72 cores) unverified	estimate

BASE OPTIONS CONSIDERED

Option	Verdict	Why
A. Pure AOSP android-17.0.0_r1 plus Zune trees	Chosen	Clean provenance, least diff; we own patching, WebView, OTA, kernel
B. Fork GrapheneOS	Reuse ideas, not the tree	Hardened, Pixel-focused; 129 own repos; early access partial, not transferable
C. LineageOS base	Only if hardware needs its device trees	108 own repos and app set; not locked down
D. Wait for Q4-2026 drop	Rejected	Idle about 2-3 months; rebase onto it instead
E. Track android-latest-release	Never in CI	Unreproducible; base shifts at each drop

Manifest strategy: fork the manifest (zune/manifest, pinned snapshot plus zune.xml).

ANDROID 15 PUBLIC SECURITY-TAG CADENCE

Tag	Date	Note
r3 to r9	2024-12-02 to 2025-06-02	Monthly
(none)	Jul and Aug 2025	Two bulletins with no tag
r10	2025-09-02	3-month gap
r11	2025-11-03	2-month gap

Android 16 security tags	None seen	Mirror stale after 2025-11; absence unproven
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Source: aosp-mirror tags. The mirror stopped syncing, so the Android 16 result is inconclusive.

AOSP base release, branches, cadence and build host (2/3)

R01

partly verified

Detail sheet: Build on pinned AOSP Android 17; security-patch lag is the central risk.

FOUNDER DECISIONS RAISED BY R01 AND CURRENT STATUS

Decision	Status	Note
AOSP 17 base plus Q4-2026 rebase	Base stated in REQUIREMENTS; rebase plan unconfirmed	Older vendor-BSP Android would change security posture
Security staffing and OTA budget	Open	About 1 FTE platform/security engineer plus kernel support; monthly OTA
Android partner / early access	Open	Plan for none; partner status is upside
No GMS	Assumed (A3)	Removes Play updates; drives WebView and Mainline OTA burden
Telephony and SMS	Resolved: out (D19)	Changes attack surface, certification and module list
Build infrastructure	Open	Own cloud (32 vCPU class) versus ODM-provided builds
WebView source	Open	AOSP prebuilt, own Chromium build, or licensed prebuilt
Device and vendor BSP	Pixel 10a/9a class first (D17)	Vendor kernel and interface-level fit with Android 17 not researched

RISKS

Sev.	Risk	Mitigation
High	Public AOSP patches can trail authoring by 2.5-6 months and arrive batched; Mainline and WebView updates are Zune's to ship.	Monthly patch-ingest pipeline; OTA SLA: Critical within 14 days of bulletin, others 30; about 1 FTE security engineer.
High	Existence and timeliness of an android17-security-release branch is unverified; Google's pages were unreachable.	Verify with git ls-remote from the build host; until then assume pinned tag plus vendor patches is all the patching.
High	WebView-hosting apps, intents, document viewers and CaptivePortalLogin can reintroduce browsing on a no-browser OS.	Exclude Browser2 and HTMLViewer; review CaptivePortalLogin, DocumentsUI, QuickSearchBox, Settings link-outs; choose a WebView source.
Medium	Supervision framework is new, partly flag-gated and changed between Android 16 QPR2 and 17; rebases can flip behaviour.	Pin to the base; prefer build-time resource overlay over RRO (RRO route unverified); retest every rebase.
Medium	Hardware may force a vendor BSP kernel (6.1/6.6/6.12) and HAL branch that lags AOSP 17.	Check Android 17 compatibility of the chosen device's kernel line before committing; vendor interface level not researched.

SCOPE

Stage 1: Pin android-17.0.0_r1 in a forked Zune manifest; compose a Zune product from base_* makefiles without Browser2/HTMLViewer; point the supervision config at Zune's parental-controls agent; baseline-build on Cuttlefish with CI per merge request (goldfish sdk_phone16k nightly for 16 KB pages); stand up a weekly security-branch watcher, monthly patch ingest and the OTA SLA.

Stage 2:

- Rebase onto the Q4-2026 AOSP drop (QPR2, API 37.2; about Dec 2026/Jan 2027, date unverified) and freeze before beta.
- Next rebase candidate Q2 2027 (Android 18); report marks this an inference.
- If chosen: own Chromium WebView build with its own patch pipeline and a larger build host.
- Pursue Android partner or early-access status, treated as upside rather than base case.
- Siso remote cache (self-run RBE backend) or ccache wrappers if builds are slow; ccache effect under Siso untested.

OPEN ITEMS

- Verify android17-security-release branch and android-security-17.* tags with git ls-remote from the build host.
- Decide WebView source: AOSP prebuilt, own Chromium build or licensed prebuilt; name an owner for updates.
- Approve about 1 FTE platform/security engineer plus kernel support and a monthly OTA service.
- Decide whether to pursue Android partner status; base case is no early access, as GrapheneOS itself was not accepted.
- Check Android 17 compatibility of the Pixel vendor kernel and interface level; R01 did not research it.

AOSP base release, branches, cadence and build host (3/3)

R01

partly verified

Detail sheet: Build on pinned AOSP Android 17; security-patch lag is the central risk.

DO NOT TAKE AS FACT

Claim	Why
Release date 2026-06-16	Not independently confirmed; nearest hard dates are 2026-06-18 (GrapheneOS, GSI) and 2026-06-19 (LineageOS).
Google publishes AOSP source only in Q2 and Q4; QPR2 AOSP drop around December 2026	Google's own wording unreadable (source.android.com blocked); only GrapheneOS's observed rebase pattern supports it. QPR2 is API 37.2, in beta; date unverified.
Kernel: GKI android17-6.18, EOL 2030-07-01, 5.15 supported	Unverified; AOSP 17 manifests carry kernel prebuilts for 6.1, 6.6, 6.12, 6.18 only, no 5.15.
Sept 2026 bulletin: 180 vulnerabilities (95 + 85); CalyxOS timeline; GrapheneOS '3 months' early access	Sources blocked; not independently checked.
Fixes became public only in September 2026	Inference from downstream import dates; data shows when two downstreams imported them, not when Google published.
A timely public security-only branch exists for the current Android release	Unproven; none observed for Android 17, and none for Android 16 in the stale mirror.
Clean build 1.5-3 h, incremental 3-15 min, host sizing, Google's 40-minute / 400 GB figures	Estimates; Google figures come from blocked pages. Ubuntu 24.04 is proven only as GrapheneOS's supported host.
Zune needs under 30 AOSP forks	Judgement with no data; GrapheneOS hosts 129 repos and a kids launcher/Settings/SystemUI lockdown may need more.
Browser2 is still in Android 17 handheld_product.mk; an RRO can fill the supervision config	Both inferred: Browser2 from android-16.0.0_r3 plus the manifest; no overlayable declaration found, build-time overlay is safer.

Superseded: R01 Decision 5 (telephony/SMS in or out) is answered by D19: out. R01's hardware caveats pre-date D17 (Pixel first) and D21 (about 200 Pixel users); nothing else is overturned.

Hardware target and device-support strategy ^(1/3)

R02

partly verified

Detail sheet: Pixel 10a/9a is the best-evidenced route to a relockable custom Android

DECISION

Start on Pixel 10a/9a (D17), keep the product layer device-agnostic, and defer own hardware until volume and cash gates are met; Snapdragon is a later second source.

Second pass confirmed GrapheneOS/LineageOS facts from GitHub and corrected two; AOSP source, Google licence, prices and press claims stayed unreachable.

WHY IT MATTERS

Device choice fixes the security story (relock with our own key), the patch burden, the legal exposure over Google binaries, and whether the child can ever unlock the phone.

KEY POINTS

- Google no longer publishes Pixel device trees; builds use an AOSP tag, an adevtool-generated vendor module and Google's prebuilt 6.1 kernels. R02 F1
- Bring-up is tooling, not reverse engineering, but recurs every QPR: stock images lead the AOSP tag. Budget at least one platform engineer. R02 F1 [CORRECTED]
- GrapheneOS ships Android 17 on Pixel 10a and 9a with its own verified-boot key: strong indirect evidence relock works. Own test pending. R02 F4 [CORRECTED]
- The 10a and 9a share the zumapro platform but have separate vendor modules, kernel repos and QA matrices. R02 s1 [CORRECTED]
- Pixel 11 is not ready: GrapheneOS excludes gen 11 from builds as '[temporary]', and lists no Pixel 11 as supported (2026-10-01). R02 F7
- GSI on any Treble phone is rejected: not redistributable, no community Android 17 GSI, relock likely fails on stock vendors. R02 F8
- With bootloader locked and OEM unlocking off, a bad image hard-bricks the phone; OTA needs A/B rollback tests and staged rollout first. R02 s4 rule 6 [CORRECTED]
- Reported: incumbents (Pinwheel, Bark, Gabb, Troomi) rebadge commodity phones and sell subscriptions; none builds hardware. Stock firmware plus policy is the fallback. R02 F9

NUMBERS

Metric	Value	Basis
Pixel 10a US price / support end	\$499 / Mar 2033 Launched 2026-03-05; secondary sources only. R21 cites India price INR 47,999.	reported
Pixel 9a US price	\$499 launch; ~\$300-430 now Support to Apr 2032.	reported
Incumbent kids' phones	\$100-240 device + \$15-20/mo Pinwheel Plus 6 \$219 + \$14.99/mo; all sources blocked, treat as soft.	reported
adevtool 'switch Pixels to 17 QPR1 images' (2026-09-06)	15 files, +1,229/-144 lines Measures the recurring vendor-module skew work.	sourced
ODM vendor quotes (low reliability)	FOB \$48-115; MOQ 3-5k; NRE \$5-65k Vendor marketing: 'typical ODM launch \$215k'.	reported
Own-phone cost (author estimate)	\$0.5-1.5M, 12-24 months Certified cellular kids' phone; R21 revises.	estimate
ODM gate (author proposal)	25k+ units/yr, \$1.5M+ committed R21 calls it too coarse; a rebadge breaks even near 8-10k units.	estimate
Flash/QA/ship per unit (estimate)	~\$35 10-20 port station at ~10 min/unit. The \$450 landed figure assumed we buy the phone (superseded by D15).	estimate

PIXEL LINEUP CONSIDERED (US PRICES, OCT 2026)

Model (codename)	Launch	US price	Support to	Fit
Pixel 11 family (malibu)	2026-08-20	\$899-1,899	7 yrs claimed	No: not yet supported by GrapheneOS
Pixel 10a (stallion)	2026-03-05	\$499	Mar 2033	v1 primary
Pixel 10 family (laguna)	2025-08	From \$799	~2032	Too costly for kids
Pixel 9a (tegu)	2025-04	\$499 launch; ~\$300-430 now	Apr 2032	v0/v1 sibling
Pixel 8a (akita)	2024-05	\$499 launch; refurb \$170-339	~May 2031 (inferred)	Cheap post-launch SKU

Prices are secondary-source and unverified; re-check before quoting. Pixel 10a and 9a share platform zumapro (Tensor G4).

Hardware target and device-support strategy (2/3)

R02 partly verified

Detail sheet: Pixel 10a/9a is the best-evidenced route to a relockable custom Android

BUILD PATHS CONSIDERED

Path	Time to v1	Unit economics	Main risk	Verdict
A. Pre-flashed new Pixel 10a/9a	3-5 months after image ready	~\$450 landed; \$529-599 bundle nets ~10-15%	Blob licence; Pixel openness; price	v1 path; bundle model superseded (D15/D16)
A2. Refurb Pixel 8a (Google Certified)	+1 bring-up	~\$260-340 landed; sell ~\$349	Unknown lock state and history	Post-launch SKU
B. BYO installer (Pixel only)	2 months	No hardware cost; high support cost	Bricked or carrier-locked phones	School/IT pilot only; D15: later version
B2. GSI on any Treble phone	n/a	n/a	No relock; frozen vendor; no community Android 17 GSI	Reject
C. ODM own phone	12-24 months	\$0.5-1.5M upfront; \$50-115 FOB	Cash, BSP lag, we own patches, certification	Gate at scale (D21: later)
D. Second-source OEM, pre-flashed	6-9 months	\$150-350 devices	Per-OEM unlock/relock unverified	After v1 (D17)
E. Stock firmware + managed-device policy	2-3 months	Any Samsung/Moto/BLU phone	'No browser' is policy, not image	Fallback only

Costs and times are author estimates unless noted. Path A assumes we sell the phone.

DEVICE SUPPORT CONTRACT (ALL MUSTS REQUIRED)

Tier	Requirement
MUST	AVB 2.0 with custom-key relock; OEM unlock can be disabled
MUST	A/B partitions; stable AIDL HALs and VINTF
MUST	Hardware-backed keystore; fastboot-flashable; 4 GB+ RAM
SHOULD	5+ years of vendor firmware; GKI kernel
SHOULD	Complete Camera2 HAL; MTE
Outcome	All Pixel 8-10 models qualify (GrapheneOS); a-series chosen on price

R15 later raised the RAM floor to 8 GB+ for the Snapdragon list.

RISKS

Sev.	Risk	Mitigation
High	Legal basis for shipping Google's Pixel binaries commercially is unresolved; Google's licence text was never read.	Get counsel and a Google contact first; keep stock-firmware-plus-policy as fallback.
High	Pixel openness is shrinking: no device trees, AOSP source twice a year, LineageOS has no official Android 17 builds.	Second-source OEM later (D17); no bet on Pixel 11; dedicated platform engineer for vendor-module skew.
High	Monthly security patches come via a gated Google partner path; startup access is unknown, so we may lag.	Treat as founder/legal action item; until resolved we trail GrapheneOS and Google.
Medium	Locked bootloader with OEM unlocking off: a boot-failing OTA hard-bricks the phone; nobody can unlock to recover.	A/B rollback tests, strict key discipline, staged rollout; founder decides who may re-enable OEM unlock.
Medium	Price and supply: Pixel costs more than competitors' phones; bulk terms and 10a/9a stock unknown.	Customer supplies the phone (D15), moving inventory risk off us; confirm Indian supply and warranty.

SCOPE

Stage 1: v0 now to about Dec 2026: Cuttlefish plus two unlocked Pixels (9a tegu, 10a stallion). v1: Pixel 10a (9a as sibling), Wi-Fi first with optional data SIM (D14, D19), customer-supplied phone flashed by us (D15/D16). Factory flow: unlocked stock, flash image plus avb_custom_key, fastboot flashing lock, disable OEM unlocking, QA script. 'No browser' is enforced in the image.

Stage 2:

- Keep Pixel a-series primary through about 2028; qualify one second-source OEM (Motorola moto g, Fairphone, Sony Xperia 10) behind a written Device Support Contract.
- Add a cheap refurb Pixel 8a SKU after launch.
- Own ODM kids device only past the gate: 25k+ units/yr, \$1.5M+ committed, 5-year BSP commitment, factory key provisioning, partner bulletin access (R21 revises).
- Consumer BYO installer deferred to a later version (D15/D16); report suggests a school/IT pilot first.
- CI-generated vendor modules: fork adevtool (MIT), pin a Google build per device per month, diff vendor-specs, smoke-test physical devices.

OPEN ITEMS

- Counsel and Google contact on licence to ship modified Pixel firmware commercially; licence text unread in both passes.
- Our own relock test of Pixel 10a/9a on Android 17; so far evidence is GrapheneOS's shipping, not our device.
- Access to Google's Partner Security Bulletin patches for a startup is unknown; needed for monthly patches.
- Who may ever re-enable OEM unlocking on a shipped device: parent PIN or factory RMA only? Founder decision.
- Pixel unlock status, supply, warranty and India pricing are not covered: the report uses US carrier variants and US prices.

Hardware target and device-support strategy (3/3)

R02

partly verified

Detail sheet: Pixel 10a/9a is the best-evidenced route to a relockable custom Android

DO NOT TAKE AS FACT

Claim	Why
Pixel 10a is \$499 and supported to March 2033; Pixel 11 starts at \$899	Unverified: price and press sources blocked. Consistent with GrapheneOS FAQ 7-year guarantee for gen 8+, but US prices, not India.
Pixel 11 lacks MTE and GrapheneOS could not finish its port	MTE claim unchecked (press only); second pass says the port is '[temporary]' excluded and in progress, so 'could not finish' overstates.
Pixel factory images may not be modified or redistributed; commercial licence unresolved	Google's licence text was unreadable in both passes; GrapheneOS shipping is a precedent, not a licence.
Incumbent prices and hardware (Pinwheel \$219, Troomi \$99.95 + \$19.95/mo, Gabb \$199)	All cited sites blocked; treat figures as soft.
Google is the only OEM documenting avb_custom_key; Fairphone and Moto g work	Unverified: the avbroot issue was not readable here; later reports show user reports only.
Fairphone 6 at EUR 549 with 8 years of updates; Motorola 2027 GrapheneOS partnership	Unverified; reviewer's memory suggests EUR 599. Do not quote.
ODM \$0.5-1.5M and 12-24 months; \$450 landed; 10-15% bundle margin; 5-10% return reserve	Author estimates, not sourced; vendor ODM figures are marketing and optimistic.
CPSIA, EU RED/EN 18031 and Cyber Resilience Act compliance paths	Unverified (sources blocked, CRA dates from memory); US/EU items, not India (D18).

Superseded: (1) Selling pre-flashed Pixels as a bundle (Path A: ~\$450 landed, \$529-599 bundle, \$0.5-0.9M inventory float, pre-orders) is superseded by D15/D16: customer supplies the phone, we flash it. (2) US launch assumptions (US prices, CPSIA, FCC/PTCRB, US carrier acceptance, US-or-EU question) are superseded by D18 (India first). (3) 'No BYO installer for consumers' now reads: BYO is a later version (D15/D16). (4) ODM timing and the 25k units/yr gate are replaced by D21 and the R21 gates. (5) D13's Snapdragon preference had overruled the Pixel-first call; D17 restores a Pixel start. The no voice/SMS stance matches D19.

Clean minimal AOSP product configuration (1/3)

R03

unverified

Detail sheet: Zune's OS image is designed as a default-deny allowlist, not subtracted from stock AOSP.

DECISION

Compose Zune products from `base_system.mk` plus an explicit allowlist instead of stock phone products; enforce with a CI diff of the built image; overlays via RRO only.

No skeptic second pass; claims were read in GrapheneOS/LineageOS source files, but nothing was build-tested.

WHY IT MATTERS

A no-browser kids OS cannot let upstream components arrive silently; Android 17 already adds a Web App installer module. An allowlist makes drift a build failure.

KEY POINTS

- Stock products cannot be subtracted: inherit-product leaves markers; no `PRODUCT_PACKAGES_REMOVE` exists. Only skipping the inherit, Soong overrides and built-in switches work. R03 F2
- Settings and the android package are prevent-disable; SystemUI throws if `config_recentsComponent` is empty. 3-button navigation lets a non-Quickstep Zune launcher work. R03 F5
- Many Mainline APEXes (wifi, bt, conscrypt, adservices) contribute boot jars and cannot be pruned by name; neuter via flags, feature masks, RRO. R03 F4, F7
- AOSP's prebuilt WebView is not updated regularly (Chromium's own integrator guide); shipping WebView means a signed update path forever. R03 F6
- Neutering without deleting: unavailable-feature masks (Print, Credentials, NFC), static RRO overlays, config knobs, flags like `RELEASE_WEBAPP_MODULE=false`. R03 F3, F7
- Guards: CI diff of upstream makefiles on each rebase; allowlist test over `installed-files.txt`; on-device checks: no http(s) VIEW handler, empty browser role. R03 section 5
- Only platform-privileged apps (Home, Setup, Guard, Updater) are in-tree; other apps ship as SHA256-pinned signed prebuilts from Gradle CI. R03 F8, section 5
- AOSP's Supervision stack (SupervisionManager, `config_systemSupervision`) is a candidate parental-controls hook; maturity unknown. Enforcement model still open: platform role, Device Owner or ZuneGuard. R03 F7, Q5

NUMBERS

Metric	Value	Basis
Draft makefiles build-tested	0 Report states nothing was build-tested; all makefiles are drafts	sourced
Modules in <code>base_system.mk</code>	300+ Includes the whole Mainline set; read in GrapheneOS <code>platform_build@17</code>	sourced
Module release flags on in 17.0 (cp2a)	5 TELEPHONY, TELECOM_MAINLINE, WEBAPP, NPUMANAGER, CONSCRYPT_NSC	sourced
New AOSP projects in Android 17 vs 16	6 WebApp, NpuManager, ImsStack, Login, PersonalContext, ContactsPicker	sourced
WebView version pinned in cp2a	145.0.7632.218 From GrapheneOS release config; standalone prebuilt; may differ from AOSP's own	sourced
Plumbing to copy from <code>generic_system.mk</code>	about 50 lines Report's own estimate; must be diffed on each rebase	estimate
Open founder decisions	6 of 7 Q1 (cellular voice/SMS) resolved by D19; Q2-Q7 open	sourced

Clean minimal AOSP product configuration (2/3)

R03 unverified

Detail sheet: Zune's OS image is designed as a default-deny allowlist, not subtracted from stock AOSP.

KEEP / REMOVE / REPLACE (SELECTED)

Component	Verdict	Why
SystemUI, Settings, SettingsProvider	KEEP (hard dependency)	Prevent-disable; hide Settings from launcher, gate behind parent PIN
Launcher, Setup, Camera, Gallery, Music, Updater	REPLACE with Zune apps	ZuneHome, ZuneSetup, ZuneCamera, ZunePhotos; ZuneUpdater uses silent A/B update_engine
Browser2, QuickSearchBox, HTMLViewer, BookmarkProvider	REMOVE	Browser surface
CaptivePortalLogin	REMOVE in v1 (Q3 open)	WebView mini-browser; breaks hotel and school Wi-Fi sign-in
WebView engine	KEEP engine only (Q2 open)	Only if an app needs it; curated YouTube is the likely need
Mainline APEXes: Bluetooth, NFC, UWB, Wi-Fi	KEEP, neuter	No UI, features masked, DISALLOW restrictions; remove Tag
Telecom, TeleService, TelephonyProvider, CarrierConfig	KEEP (hard on a radio device)	Pixel always has a radio; SIM optional
PackageInstaller	KEEP, neuter	DISALLOW_INSTALL restrictions; removal needs base_system.mk edit, unproven

PRODUCT COMPOSITION OPTIONS

Option	For	Against
A. Inherit stock products, subtract via overrides and RRO	No build/make fork	Needs a stub module per removal; new upstream apps appear silently
B. Fork build/make and delete lines	Proven (GrapheneOS, LineageOS, kioskhome-gsi)	Every rebase conflicts; additions still arrive silently
C. Own allowlist from base_*.mk (recommended)	Default-deny; no build/make fork; drift caught by CI diff	Copy about 50 plumbing lines; a missed item shows as build or runtime error

No stock phone-class product the report read omits handheld_system.mk, so subtraction always starts from a loaded base.

FOUNDER QUESTIONS IN R03 AND CURRENT STATUS

ID	Question	Status
Q1	Cellular voice and SMS in v1?	Resolved: no (D19)
Q2	Ship a WebView?	Open; report recommends yes with a named monthly-update owner
Q3	Support captive-portal Wi-Fi in v1?	Open; report recommends no, document hotspot workaround
Q4	Bluetooth, NFC, USB file transfer defaults	Open; proposed Bluetooth on, NFC off, USB transfer off
Q5	Parental-control enforcement model	Open; owned by the parental-controls topic
Q6	Navigation: 3-button versus gestures	Open; report recommends 3-button with no Recents
Q7	Patch budget: RRO-only or small patches	Open

RISKS

Sev.	Risk	Mitigation
High	Nothing is build-verified: feature masks on a Pixel vendor image, release-flag overrides and PackageInstaller removal are unproven.	Build on Cuttlefish first; test on userdebug with privapp enforcement on; run on-device smoke tests.
High	WebView: AOSP prebuilt is stale and Chromium ships about every 4 weeks; Zune must run a signed update path forever.	Name an owner for monthly WebView updates (Vanadium, self-build or prebuilt); decide Q2.
Medium	Silent capability growth: Android 17 adds WebApp, NpuManager, ImsStack, Login, PersonalContext, ContactsPicker projects versus Android 16.	Allowlist test over installed-files.txt plus release-flag overrides (RELEASE_WEBAPP_MODULE=false); CI diff every rebase.
Medium	Small images may break Android 17 recovery sideload (COW space exhaustion); source is a search snippet only.	Use update_engine network OTA as the path; test factory-recovery flows.
Medium	Without a dialer, R03 infers no emergency calling; A9 keeps the built-in emergency path by default.	Telecom stack stays (hard dependency); counsel consulted before any change (A9, report 19).

Clean minimal AOSP product configuration ^(3/3)

R03

unverified

Detail sheet: Zune's OS image is designed as a default-deny allowlist, not subtracted from stock AOSP.

SCOPE

Stage 1: Compose zune_system and zune_product from base_system.mk plus an explicit allowlist; develop on Cuttlefish and target Pixel 10a/9a via a fork of GrapheneOS adevtool; replace launcher, setup and updater with in-tree privileged Zune apps; RRO overlays for config; other apps as pinned signed prebuilts; CI allowlist diff on every build.

Stage 2:

- Candidate: gesture navigation, which needs a Quickstep-compatible launcher fork (Q6).
- Candidate: small patches to Settings or SystemUI beyond RRO-only (Q7).
- Candidate: captive-portal Wi-Fi support, removed in v1 with a hotspot workaround (Q3).
- Candidate: Device Owner path, which keeps ManagedProvisioning, if chosen (Q5).
- Ongoing: monthly signed WebView refresh and upstream makefile diff on each rebase.

OPEN ITEMS

- Q2: confirm shipping a WebView with a named owner for monthly signed updates (report recommends yes).
- Q5: choose enforcement model: platform Supervision role, Device Owner (keeps ManagedProvisioning) or privileged ZuneGuard.
- Reconcile emergency calling with A9: platform emergency path stays by default; counsel review before any removal.
- Build-test the drafts on Cuttlefish, then a Pixel image: feature masks, flag overrides, recents behaviour, privapp enforcement.
- Q3, Q4, Q6, Q7: captive-portal Wi-Fi, Bluetooth/NFC/USB defaults, navigation mode, RRO-only versus small patches.

DO NOT TAKE AS FACT

Claim	Why
All makefiles, overlays and product layout in the report	Drafts; nothing was build-tested.
unavailable-feature masks (Print, Credentials, NFC) work on a Pixel vendor image	Pattern read only in Cuttlefish slim; effect on Pixel unverified.
Zune can override release flags via PRODUCT_RELEASE_CONFIG_MAPS	Inferred from generic_system.mk's gms_mainline use; not tested.
Without a dialer there is no emergency calling	Inference only; A9 keeps the platform emergency path by default, pending legal review (report 19).
Android 17 breaks recovery sideload for small system images	Search snippet of an X post, not read directly.
com.android.webapp is Android 17's Web App installer, off by default	Secondary source (a GitHub project README), not AOSP.
PackageInstaller neutering via DISALLOW_INSTALL_* and CarrierDefaultApp removal	From memory, unchecked.
WebView 145.0.7632.218 and AOSP 17 hosting of external/chromium-webview	Version read from GrapheneOS release config; repo absent from GrapheneOS 17 and LineageOS 24 manifests. R01 second pass says AOSP carries it and GrapheneOS deletes it.
Android 17 introduces the Supervision stack	R01 second pass: framework predates 17 (Android 16 QPR2) and flags still change per release.
Missing privapp-permissions entry denies permissions or breaks boot	Sources conflict; must be tested on userdebug.

Superseded: R03 Q1 and its 'DECIDE (Q1)' row (Dialer, Messaging, CellBroadcast, EmergencyInfo, ImsServiceEntitlement, ONS, Stk) are resolved by D19: no cellular voice or SMS. R03's data-only recommendation matches; its no-emergency-calling inference is overridden by A9 (platform emergency path stays by default).

No-browser lockdown: threat model and enforcement (1/3)

R04

unverified

Detail sheet: No-browser is enforced in the locked image, not by deleting an app

DECISION

Compile 'no browser' into a locked, verified image: user build, Zune AVB key, WebView caller gate, no http(s) handler, INTERNET allowlist. Policy only tunes it.

No skeptic pass in file. Most claims read in GrapheneOS 17 source, but Google's AOSP tree was blocked and key designs are untested inference.

WHY IT MATTERS

The product's core promise depends on it. Real kid-control bypasses are hidden-browser bugs, so the defensible claim is narrow: no browser app, no way to type a URL.

KEY POINTS

- AOSP 17 is not browser-free: CarrierDefaultApp (disabled http/https wildcard handler, JS WebView), OsuLogin and HTMLViewer are escape hatches. R04 F1
- AOSP 17 ships no WebView implementation. Vanadium (hardened Chromium, GPL-2.0 patches) or an own Chromium build means owning a Chromium update pipeline. R04 F2
- WebView gate: system_server already sees caller UID, so a ~20-line allowlist restricts WebView to two Zune packages (feasibility inferred). R04 F3
- App-level origin allowlists leak: shouldInterceptRequest misses redirects, blob:, javascript: and assets; Network Security Config has no allowlist. R04 F4
- eBPF denies INTERNET at socket creation per app. Stage 1 grants it to about 8 packages; the EPUB reader gets none. R04 F6
- Always-on lockdown VPN ('Zune Guard') adds domain/SNI control but AOSP leaks, and ECH and QUIC blind SNI filters; deferred to Stage 2. R04 F6
- Embedded YouTube in the Videos WebView carries the one Medium residual risk: its UI is third-party code with links we cannot remove. R04 Bypass table #7
- Physical access bounds everything: a recovery wipe cannot be blocked, but leaves an inert device until a parent re-pairs, not a browser. R04 Bypass table #16

NUMBERS

Metric	Value	Basis
Bypass vectors modelled	23 Count of the report's table; likelihood and residual ratings are engineering judgement	estimate
WebView caller-gate patch	~20 lines Inferred from reading system_server code; not built or tested	estimate
Patches to AOSP (target)	under 5 Design target, not measured	estimate
Packages with INTERNET in Stage 1	~8 Design target; EPUB reader gets none	estimate
Zune packages allowed to use WebView	2 Design target	estimate
Chromium WebView version (Vanadium, LineageOS prebuilt)	154.0.8037.x Read in GrapheneOS and LineageOS WebView repos	sourced

TOP BYPASS VECTORS AND RESIDUAL RISK

Vector	Likelihood	Mitigation	Residual
http(s) link tapped in any app	High	No handler (CI gate), baked Intent Firewall rule	Low
Hidden WebView hosts: CaptivePortalLogin, OsuLogin, HTMLViewer, CarrierDefaultApp	Medium	Remove from product; WebView caller allowlist; captive_portal_mode=0	Low
Videos WebView: YouTube UI, end cards	High	One iframe, no navigation, proxy/allowlist	Medium
AI assistant as web gateway	High	No browse tools, strip URLs, server filters	Web-derived content still reaches child
ADB, Developer options, trade-in mode	High	user build, ro.adb.secure=1, trade-in flag off	Low
OEM unlock, fastboot, stock reflash	High	OEM-unlock bit off; locked bootloader, Zune AVB key	Low on Pixel; higher on BROM-exploitable SoCs
Recovery factory reset	High	Cannot block; device inert until re-paired; backend alert	Parent policy lost; no browser gained
Reflashed or rooted device	Low-Med	Server attestation of verified-boot key	Beyond image control

Ratings are engineering judgement [I]; 23 vectors in the full report.

No-browser lockdown: threat model and enforcement (2/3)

R04

unverified

Detail sheet: No-browser is enforced in the locked image, not by deleting an app

WEBVIEW STRATEGY OPTIONS

Option	Pro	Con	Verdict
A. No WebView	Zero engine surface	Breaks EPUB reader and YouTube IFrame; native Player SDK deprecated	Only if Videos goes native/licensed
B. Provider + app-level limits only	Cheapest	shouldInterceptRequest leaks	Reject as sole control
C. Provider + system_server gate + no-INTERNET EPUB + proxied Videos	Small patches; closes hidden WebViews	Still ships Chromium; patch burden	Stage 1 (recommended)
D. C plus Chromium allowlist fork	Native-code origin gate	Patches on Chromium's cadence	Stage 2 if needed

NETWORK STRATEGY OPTIONS

Option	Pro	Con	Verdict
DNS allowlist only	Easy	IP literals, ECH/HTTPS-RR, broad googlevideo.com	A layer, not a control
Per-UID INTERNET + firewall chains	Kernel-enforced	Per-app only, not per-destination	Stage 1 base
Always-on lockdown VPN (Zune Guard)	Domain/SNI/QUIC control	Battery, performance, AOSP leaks	Stage 2
Server gateway (WireGuard to Zune)	Central enforcement	Holds only if device is forced onto it	Pair with Guard

RISKS

Sev.	Risk	Mitigation
High	Chromium/WebView is the largest attack surface and imposes a permanent update cadence; GPL-2.0 patch obligations need legal review.	Commit to an update SLA at Chromium cadence; choose Vanadium prebuilt or own build; legal review.
Medium	Embedded YouTube UI exposes links we cannot remove, capping Videos containment at medium.	Single iframe with no navigation, proxy allowlist; Chromium allowlist patch or licensed/native video in Stage 2.
Medium	ECH, QUIC and HTTPS-RR erode DNS/SNI enforcement; AOSP always-on VPN lockdown has known leaks.	Treat DNS as a layer; use per-app INTERNET as base; gate content at the single-iframe Videos page.
Medium	Physical access: parent-assisted unlock, retail-unlocked devices, SoC-level flashing and recovery wipes cannot be fully stopped.	Company flashes and relocks (D16); server attestation of verified-boot key; Pixel start (D17) lowers SoC-level risk.
High	AI assistant is a web gateway: web-derived content can still reach the child even with no browse tools.	No browse tools, strip URLs, server-side filters; never claim 'no web access'.

SCOPE

Stage 1: Image: user build, ro.adb.secure=1, locked bootloader with Zune AVB key, OEM-unlock forced off, no Developer options, fw.max_users=1. Browser2, QuickSearchBox, CaptivePortalLogin, CarrierDefaultApp, HTMLViewer, CertInstaller and similar excluded; AOSP Settings not used as child UI. Under 5 patches: WebView caller allowlist, verified Intent Firewall directory, system-asserted restriction set, Settings without Developer options. Network: INTERNET for about 8 packages, strict Private DNS to a Zune resolver, Zune captive-portal URLs. Videos through a proxied single-iframe WebView; EPUB WebView has no INTERNET. CI gates every build (no http(s) VIEW handler, WebView and INTERNET allowlists, exported-activity diff).

Stage 2:

- Always-on lockdown VPN ('Zune Guard') with domain/SNI allowlist
- Chromium-level origin allowlist patch if needed
- Logging 'sink' activity so parents see attempted links
- Advanced Protection hardening: unknown-sources ban, USB data off when locked, 2G off, MTE
- Server gateway (WireGuard) paired with Guard; explore licensed/native video to drop the YouTube WebView

OPEN ITEMS

- Sales model: pre-flashed relocked only vs image-only (weaker claim); v1 partly settled by D15/D16 company-provisioned delivery.
- WebView supply: Vanadium prebuilt vs own Chromium build, with an update SLA at Chromium cadence.
- Videos architecture: accept a network-limited YouTube WebView in Stage 1; explore licensed/native video later.
- Accept captive-portal Wi-Fi (hotel, school) as unsupported in v1?
- Marketing wording ('no browser app', never 'unbypassable') and acceptance that a recovery wipe leaves an inert device.

No-browser lockdown: threat model and enforcement (3/3)

R04

unverified

Detail sheet: No-browser is enforced in the locked image, not by deleting an app

DO NOT TAKE AS FACT

Claim	Why
AOSP 17 behaviour (WebView, Intent Firewall, eBPF, adbd trade-in mode)	Read from GrapheneOS 17 branches, not Google's android-17.0.0_r1; source.android.com and googlesource were blocked.
A ~20-line WebView caller allowlist in system_server closes hidden WebViews	Feasibility is inference from reading code; no patch was built or tested on Cuttlefish.
AOSP 17 offers no WebView implementation	Absence inferred from the AOSP and LineageOS manifests, not proven.
Family Link hidden-browser bypass via Play Services; YouTube embed domain lists	Search summaries only (androidpolice, SonicWall/Meraki KBs); sources not read in full.
CaptivePortalLogin behaviour	googlesource blocked; search summary only.
Likelihood and residual ratings for all 23 bypass vectors; 'under 5 patches' budget	Engineering judgement and design targets, not measured.
DISALLOW_INSTALL_APPS effect on Zune Updater; Pixel recovery wipe under custom AVB key; supervision profile-owner behaviour	Listed by the report as unverified.

Superseded: Hardware-sales framing ('sell pre-flashed devices') is reframed by D15/D16: we sell the OS image and flash customer-supplied phones. Vector #6 'SMS unreadable (D6)' is moot under D19 (no SMS). Hardware gate for non-Pixel devices is deferred by D17 (Pixel first).

Parental-controls platform: portal, on-device enforcement, sync ^(1/2)

R05

unverified

Detail sheet: Google's parental controls aren't in AOSP, so Zune builds engine, backend and portal.

DECISION

Build ZuneGuardian, a system app that is Device Owner and holds the SYSTEM_SUPERVISION role, enforcing signed cloud policy locally. Device Owner is a one-way door awaiting founder approval.

No second-pass skeptic section exists in this report. Many claims were read in primary files, but GrapheneOS trees, not Google's android-17.0.0_r1 tag.

WHY IT MATTERS

Parent control is the core promise. With no Google services there is no Family Link, so control plane, tamper resistance and offline behaviour must be built in-house.

KEY POINTS

- AOSP ships supervision plumbing (SupervisionService, role, bound app service) with all config hooks empty; consumer features and Family Link backend are Google's. R05 F1
- SYSTEM_SUPERVISION role is system-only and exclusive; grants SUSPEND_APPS, lock-task, time, safe-boot and factory-reset controls, but not WIPE_DATA, private DNS or VPN. R05 F2
- Platform policy is thin: only per-app ALLOWED/BLOCKED/TIME_LIMIT, and TIME_LIMIT suspends immediately (open TODO). Zune must own the schedule, contact and content engine. R05 F3
- Cloud is authority; device enforces signed JWS ES256 bundles locally. Past expiry plus 14-day grace, it falls to guardians and emergency only. R05 4.4
- Sync is one persistent TLS WebSocket as a doorbell (policy.update, then HTTPS GET); polling and alarms as fallback. No FCM, no UnifiedPush. R05 4.5
- Fail closed: all apps are system apps; a persistent-data-block claim lock survives recovery wipes, leaving wiped devices inert. R05 4.2, 4.7
- Pairing: portal shows a one-time QR plus 8-character code (10 min, single use) in a parent session; the device scans it. R05 4.2
- Stack: Go, Postgres, Redis, per-family KMS envelope encryption, React PWA portal with passkeys; attestation is a signal, not the root of trust. R05 4.6

NUMBERS

Metric	Value	Basis
Infra cost per device at 1k devices	\$0.25-0.50/month \$250-500/mo total; excludes LiveKit, AI, SMS, carrier, staff; about 2x either way	estimate
Infra cost per device at 10k devices	\$0.09-0.22/month \$0.9-2.2k/mo; logs and Postgres writes dominate	estimate
Infra cost per device at 100k devices	\$0.06-0.15/month \$6-15k/mo; about 100k sockets	estimate
Backend design point	1,000 devices First phase is about 200 users (D21); report made no cost estimate for 200	estimate
Policy propagation target	p95 under 10 s Design SLO, not measured	estimate
Offline grace default	14 days After policy expiry; then guardians and emergency only; founder to confirm	estimate
Retention defaults	msgs 180 d / usage 90 d / location 30 d Design defaults, not counsel-reviewed	estimate
Pairing code	8 chars, 10 min, single use Design parameter, untested	estimate

ARCHITECTURE OPTIONS

Option	Tamper resistance	Update agility	Complexity / verdict
(a) Device Owner DPC only	Good	High (APK)	Medium; lacks role plumbing and framework policy store
(b) Platform service + hooks	Highest	OTA only	High; rebase every Q2/Q4; only for targeted patches
(c) Device Owner + role app + ZunePolicyService	High	High for logic	Medium; RECOMMENDED
(d) Android Management API / Family Link	n/a	n/a	Impossible without Google services

Device Owner can be set only during provisioning; adding it later means a wipe.

TAMPER MATRIX (SELECTED)

Attack	Mitigation
Uninstall or disable Guardian	System priv-app, Device Owner, DISALLOW_UNINSTALL_APPS, no app-list UI
Clock or time zone change	Forced auto time, no UI, trusted-time floor, rollback flagged to parents
Settings factory reset	DISALLOW_FACTORY_RESET, no UI
Recovery wipe	Persistent-data-block claim lock; inert unpaired state; portal alert on re-pair
Safe mode	DISALLOW_SAFE_BOOT; all apps are system apps
ADB or developer options	user build, no developer UI, DISALLOW_DEBUGGING_FEATURES
Reflash or bootloader	Locked bootloader, our AVB key, OEM unlock off; ABL/EDL bug risk
Network blackhole or crash	Local enforcement; grace then minimal mode; role rebind

SIM-swap and WebView link-out rows omitted: the first is superseded by D19, the second belongs to report 04.

Parental-controls platform: portal, on-device enforcement, sync (2/2)

R05

unverified

Detail sheet: Google's parental controls aren't in AOSP, so Zune builds engine, backend and portal.

BACKEND INFRA COST PER DEVICE (ESTIMATE)

Devices	Shape	Monthly	Per device
1k	2 small nodes, small HA Postgres, small Redis	\$250-500	\$0.25-0.50
10k	3 nodes, mid HA Postgres, Redis	\$0.9-2.2k	\$0.09-0.22
100k	4-6 nodes (100k sockets), large Postgres + replica, Redis cluster	\$6-15k	\$0.06-0.15

Rough, about 2x either way; excludes LiveKit, AI, SMS, carrier, staff.

RISKS

Sev.	Risk	Mitigation
Medium	GrapheneOS trees are newer than android-17.0.0_r1, and Google's source was unreachable; cited files may differ.	Re-verify every cited file on android-17.0.0_r1 once AOSP is reachable.
Medium	Role grants may miss needed restrictions (private DNS, VPN, OEM unlock, wipe); supervision API is flagged and still moving.	Device Owner fills gaps; EnforcementBackend interface isolates API; validate on Cuttlefish; CI runs every rebase.
High	Server holds children's location and transcripts; a breach is existential. 24/7 SOS staffing is a dependency.	Per-family KMS envelope encryption with crypto-shredding, retention limits, dual-approved break-glass staff access, hash-chained audit log.
Medium	Device Owner is a one-way door: it can be set only during provisioning, so adding it later means a wipe.	Decide now; cheap at provisioning; keeps private-DNS and VPN options open for Stage 2.
Medium	Socket battery cost and carrier NAT timeouts are unmeasured; RKP attestation refresh needs a configured server host.	Fallbacks: JobScheduler 15 min, alarms; station-recorded identity if RKP is unreachable.

SCOPE

Stage 1: Device Owner plus supervision role at setup; QR pairing; signed policy bundles for screen time, downtime, app rules and contact allowlists; parent-PIN approvals; remote lock; browser portal with passkeys; one active device per child.

Stage 2:

- Multiple active devices per child (schema already supports N device keys per child).
- Private DNS and VPN controls, kept open by choosing Device Owner now.
- Caregiver (grandparent) roles with time-boxed access, if not pulled into Stage 1.
- Parent-side voice and video calling in the portal (deferred by D8).
- Location sharing, default off and opt-in, if not pulled into Stage 1.

OPEN ITEMS

- Founder approval of Device Owner plus supervision role (one-way door at provisioning).
- Resale and transfer: hard claim lock with support override (recommended) versus soft lock.
- Who staffs safety alerts 24/7, and whether location sharing is in Stage 1.
- Offline grace default (14 days) and minimal-mode behaviour; caregiver roles in Stage 1 or 2.
- Parent scans at hand-over (recommended) versus station pre-binds the phone to the family at intake.

DO NOT TAKE AS FACT

Claim	Why
Android 17 supervision service, role grants and empty config hooks	Read in GrapheneOS branch 17 (newer than the tag) and AOSP16; Google's source was blocked, so nothing was read on android-17.0.0_r1.
Family Link features need a Google account; Android Management API needs Play Services	Search summaries and secondary sources only; not read at source.
Push without GMS works via persistent socket plus Doze allowlist	Headwind code was read, but socket reliability under Doze and OEM killers is inferred; battery and carrier NAT timeouts are unmeasured.
Infra cost per device (\$0.06-0.50/month)	Judgement-based estimate, about 2x either way, excludes LiveKit, AI, SMS, carrier and staff.
Retention periods, location default and trusted-time design	Design choices by the author, not counsel-reviewed and not tested.
COPPA amendment (comply by 2026-04-22) and California AB 1043 (2027-01-01) legal hooks	Secondary sources; US-law framing superseded by India launch (D18); India's DPDP position is in later reports.
Release wipe via MASTER_CLEAR; ManagedProvisioning can be skipped	Memory/inference, unchecked; needs a Cuttlefish test of Device Owner timing and UX strings.
Family Link bypass list (clock, factory reset, second account, ADB, safe mode)	Single secondary source (Bitdefender blog).

Superseded: US assumptions (A1 replaced by D18, India): AWS us-east-2 region, COPPA and California AB 1043 legal hooks, 911 emergency number, America/Chicago example, USD framing. Telephony (D19 supersedes D4-D6): dialer allowed in bedtime lock, E.164 phone-number contact keys with in/out call direction, ZunePhone and SMS vault, SMS fallback for SOS, SIM-swap row, SMS wake-up. Built-in emergency path stays by default (A9). 1k design point sits above the ~200-user first phase (D21).

Curated learning video: legality, technology, alternatives (1/3)

R06 unverified

Detail sheet: YouTube is a gated, killable add-on; Videos must work without it.

DECISION

Conditional go: offline licensed content plus paid partners are the base; YouTube is an optional made-for-kids-only tier behind a kill switch, dropped if refused or silent at 8 weeks.

No skeptic pass. YouTube policy and most licence claims are search summaries (primary pages blocked); Oak, Kolibri, Kiwix, Media3 and wrapper code read first-hand.

WHY IT MATTERS

The founder's core video promise (D2, D3) rests on a platform that can revoke access at will and whose rules collide with our containment design. The fallback protects launch.

KEY POINTS

- No third-party YouTube Kids API; the sanctioned route is the IFrame player in a WebView, which needs no Play Services. R06 §1, F5
- Four YouTube rules collide with our containment: no overlays on the player, no disabling links, no ad blocking, no offline caching. R06 §2.2
- Every embedded video must be checked as Made-For-Kids (MFK); we publish MFK-only for ages up to 12 to avoid mislabel exposure. R06 F1, §2.4
- D3 topic requests are answered from our own catalog index, not live YouTube search, which allows about 100 searches a day fleet-wide. R06 §2.3
- Tier 2: one isolated WebView page on our own HTTPS origin, one iframe, navigation blocked in our client, remote kill switch. R06 §4.1
- Tier 1 is offline and native: Media3 player with no WebView and no INTERNET permission; signed content packs; ~40 h starter pack. R06 §4.1
- Licence traps, non-commercial: Khan Academy (CC BY-NC-SA), TED/TED-Ed, PBS KIDS. Kolibri and Kiwix packs include Khan and TED. R06 §2.5
- Curation is the real cost, not hosting: ~1,500-video launch catalog needs about 45 editor-days single review, 80 with double review. R06 §4.2, §2.6

NUMBERS

Metric	Value	Basis
Live YouTube searches per day, fleet-wide	~100 search.list costs 100 units in its own 10,000/day bucket since 2026-06-01	reported
Disney settlement for mislabelled child-directed videos	\$10M Order entered 2025-12-23; law-firm summaries	reported
Launch catalog curation effort	45 / 80 editor-days ~1,500 videos at ~35 videos per editor-day, single / double review	estimate
Offline starter pack	~40 h, ~18 GB Assumes ~0.45 GB per hour at 480p	estimate
Streaming cost on Cloudflare Stream	~\$0.60 per child-month 20 min/day at 480p; listed \$1 per 1,000 delivered minutes	estimate
Streaming cost on Bunny Stream	~\$0.03-0.05 per child-month Listed ~\$0.005-0.01/GB; Europe and North America pricing	estimate
150 h catalog stored on Cloudflare	~\$45 per month Listed \$5 per 1,000 stored minutes per month	estimate
Starter pack download per device (Bunny)	~\$0.20 40 h pack downloaded once	estimate

YOUTUBE RULES THAT COLLIDE WITH THE ZUNE DESIGN

YouTube rule	Zune element hit	Verdict
No overlays in front of the player	Touch shield (report 04)	Drop the shield; block navigation in WebView client
No removing, obscuring or disabling player links	D2: no path to youtube.com	Grey area; needs written YouTube answer
No modifying or blocking ads	Network allowlist could drop ad hosts	Allowlist must include Google ad hosts
No download, cache or offline copies	Offline videos	No offline YouTube; Tier 1 covers offline
No background or audio-only player	Listen mode	Not offered
No charging users to watch embedded content	Product is sold; D3 gating	Keep YouTube free; PIN and screen-time locks grey

Derived-metrics policy for custom scores ML age-suitability pre-filter Unclear; make all labels human editorial
Rules come from search summaries of official YouTube pages, not first-hand reads.

Curated learning video: legality, technology, alternatives (2/3)

R06unverified

Detail sheet: YouTube is a gated, killable add-on; Videos must work without it.

CONTENT SOURCES: CAN WE SHIP THEM IN A PAID IMAGE?

Source	Licence found	Commercial use
Oak National Academy (UK video)	OGL v3.0, attribution	Yes; confirm paid-product terms in writing
PhET simulations	CC BY 4.0	Yes (HTML5, needs WebView)
NASA media	Generally not copyrighted; no logos or endorsement	Yes, with care
StoryWeaver (EPUB stories)	CC BY 4.0	Yes; feeds the EPUB reader
Blender open movies	CC BY 3.0	Yes
Khan Academy	CC BY-NC-SA 4.0	No without a deal
TED / TED-Ed	CC BY-NC-ND 4.0 per TED terms	No; commercial needs a licensing request
PBS KIDS / LearningMedia	Personal or educational non-commercial; no redistribution	No

Crash Course and OpenStax: treat as No (unverified). Being in a Kolibri or Kiwix pack does not clear an item.

OPTIONS CONSIDERED

Option	For	Against	Call
A. YouTube only (literal ask)	Huge catalog, no hosting	Revocable, ads, no offline, four rule collisions	No
B. Offline licensed/open only	We own the stack; clean D2	Smaller catalog; licensing work	Base
C. B + gated YouTube + paid partners	Day-one value; YouTube optional	Two player backends; curation cost	Chosen
D. NewPipe-style extraction	Best UX, no ads	Against Terms; legal and reputational risk	Reject
E. Kolibri as runtime	Mature, MIT	Python server in image; adult UI; licences still ours	Not v1

RISKS

Sev.	Risk	Mitigation
High	YouTube can cut referrer access for our origin at any time; only recourse is an appeals form.	Remote kill switch; Videos works on day one from Tier 1 and Tier 3 content.
High	YouTube may require clickable player links, incompatible with D2; ad clicks open advertiser pages we must refuse.	Ask YouTube in writing (Q2, Q3); founder decides whether D2 or YouTube Videos wins; Tier 1 unaffected.
Medium	Mislabelled child-directed videos carry enforcement risk (Disney \$10M); legal status of YouTube data collection via our embed unresolved.	MFK-only up to age 12, per-video nightly MFK check, counsel sign-off; never claim "100% safe".
Medium	MFK-only narrows content for ages 10-14; kids see YouTube's contextual ads we cannot remove.	Tier 1 and Tier 3 must cover older learners; Tier 3 partner deals shrink YouTube share over time.
Medium	Licence gaps: Crash Course, OpenStax, TED-Ed unresolved; Oak is England-aligned and its paid-product terms unconfirmed.	Treat unresolved sources as non-commercial; get Oak's paid-product terms in writing; start Tier 3 outreach in parallel.

SCOPE

Stage 1: Tier 1 offline player on Media3 with a ~40 h signed starter pack from open-licensed sources; human-labelled catalog index (age, topic) answering D3 requests; gated Tier 2 YouTube WebView page, MFK-only, behind a remote kill switch; parent block-only overrides and report button; nightly health check; 8-week gate on written YouTube answers.

Stage 2:

- Fetch further content packs over Wi-Fi beyond the ~40 h starter pack.
- Grow Tier 3 partner licences (Khan, TED, Britannica, kids catalogs) so YouTube reliance shrinks over time.
- Revisit Kolibri-style runtime only after v1; v1 borrows just its licence taxonomy.

OPEN ITEMS

- Written YouTube answers to five questions (MFK-only commercial OS, navigation refusal, overlay, derived metrics, device-partner route); gate at 8 weeks.
- If YouTube requires working in-player links, founder must choose between D2 and YouTube Videos.
- Counsel sign-off on child-privacy exposure of YouTube data via our embed; report assumed US/COPPA, now India DPDP (reports 19, 20).
- Budget for curation staff and Tier 3 licences, and who signs outreach to YouTube, Khan, TED, Britannica.
- Get Oak's terms for a paid product in writing; Indian-curriculum fit of Tier 1 sources not assessed in this report.

DO NOT TAKE AS FACT

Claim	Why
YouTube policy rules (MFK check, no overlays, no link or ad changes, no offline, Referer rule) and 'no third-party Kids API'	Read via search summaries; developers.google.com, support.google.com and youtube.com were blocked. Wording and section numbers may differ; re-read before legal reliance.
Khan, TED and PBS are non-commercial; Crash Course and OpenStax licences	Mostly secondary sources. Crash Course 'NC' is from memory; OpenStax sources conflict (CC BY-NC-SA vs CC BY); TED-Ed licence unresolved.
Hosting costs (\$0.60 and \$0.03-0.05 per child-month, \$45 per month, \$0.20 per device)	Estimates from third-party price pages and an assumed 0.45 GB per hour at 480p; Europe/North America pricing, not India delivery.
Curation effort of 45-80 editor-days and the staffing model	Engineering judgement at an assumed 35 videos per editor-day; no source.
Disney \$10M order and amended COPPA compliance date 2026-04-22	Law-firm summaries only and US-specific; India-first (D18) changes the applicable law.
YouTube may refuse stale Chromium WebView builds	The report itself marks this unverified engineering judgement.

Curated learning video: legality, technology, alternatives (3/3)

R06

unverified

Detail sheet: YouTube is a gated, killable add-on; Videos must work without it.

Superseded: US launch assumption (A1) is replaced by D18 (India first): decision 3 'US (Common Core/NGSS) or UK (Oak)?' is overtaken; COPPA/FTC exposure and US-standards schema fields (CCSS, NGSS, UK NC) must be reconciled to India DPDP (reports 19, 20); hosting prices are US/Europe, not India. D20 (English only) fits the report. Core tiered recommendation is not overturned.

Kid-safe AI assistant: architecture, models, safety, policy (1/2)

R07unverified

Detail sheet: Cloud AI tutor, not companion; Anthropic's written approval for children is gate one.

DECISION

Stage 1: cloud assistant via Zune AI Gateway (Claude Sonnet 5.5, Haiku 4.5 guards, zero data retention), on-device voice, tutor-not-companion design. On-device LLM only as Stage 2 offline fallback.

No skeptic pass. Anthropic terms, ZDR, prices, LiteRT-LM and speech-engine facts read first-hand; OpenAI, Google, Mistral and all regulation items are search summaries.

WHY IT MATTERS

The AI assistant (D11) carries per-child running cost, child-safety exposure and single-vendor dependence. Anthropic's approval and a release-blocking safety eval gate launch; image input is the highest-risk part.

KEY POINTS

- Anthropic allows minors' API products with age gating, moderation, monitoring and AI disclosure; under-13s not spelled out, so get written confirmation. R07 F1
- Zero data retention (ZDR) is available on request for the Messages API; flagged content may still be kept up to 2 years. R07 F2
- Gemini API is out: terms bar clients directed at or likely accessed by under-18s. OpenAI is the backup, needing ZDR first. R07 F3
- Gateway pipeline: PII scrub and Haiku input screen, Sonnet 5.5 call grounded in curated packs, output screen before display or speech. R07 §4
- Voice is on-device from day one: push-to-talk, sherpa-onnx, Moonshine or Whisper STT, Kokoro or Matcha TTS; AOSP ships no usable speech stack. R07 F6
- Product is a tutor, not a companion: four modes, three age bands, no persona, no personal-fact memory, no web, no image generation. R07 §1, §4
- Image input (D11) is highest-risk; recommended v1 scope is objects, drawings and worksheets, with on-device people/NSFW gate. R07 §4
- Safety gating: scripted T0 responses for self-harm, abuse, grooming; release-blocking eval (~600 benign per band, 800 adversarial); kill switch. R07 §4

NUMBERS

Metric	Value	Basis
COGS per typical child (240 turns/month)	\$1.42 Sonnet 5.5 plus guards; \$1.10 with 25% FAQ cache hits; \$0.78 with Haiku as main model	estimate
Claude Sonnet 5.5 price per MTok	\$2 in / \$10 out Cache read \$0.20; read on Anthropic pricing page	sourced
Claude Haiku 4.5 price per MTok	\$1 in / \$5 out Cache read \$0.10	sourced
US-pinned inference surcharge	1.1x Only Claude 4.6+; Haiku 4.5 cannot be geo-pinned on first-party API	sourced
Retention of flagged content under ZDR	up to 2 years Must be disclosed in our privacy notice	sourced
Cloud voice cost avoided by on-device voice	\$2-4 per child-month (TTS) From memory prices; cloud STT would add \$0.1-0.2	estimate
On-device LLM RAM floor (Gemma 4)	8 GB (E2B) / 12 GB (E4B) Google allowlist 2026-10-01; Gemma3-1B needs 6 GB	sourced
Daily turn caps by age band (5-7 / 8-10 / 11-13)	10 / 25 / 30 Proposed defaults, not measured	estimate

AI VENDOR OPTIONS

Vendor	Finding	Call
Anthropic (Claude)	Minors allowed with safeguards; ZDR on request; under-13 not explicit	Primary; written confirmation first
OpenAI	Under-18 guidance; no under-13 personal data without ZDR	Backup
Google Gemini API	Terms bar clients directed at or likely accessed by under-18s	No-go without written exception
Mistral	Consumer 13+; API business-only from 2026-08-05	Unverified
Self-hosted (Qwen3, gpt-oss-safeguard)	Apache 2.0 weights and guard models	Stage 2-3

OpenAI, Google and Mistral rows come from search summaries; vendor pages were blocked.

COST PER ACTIVE CHILD PER MONTH (USD, ESTIMATE)

Usage	Turns/month	Sonnet 5.5 + guards	With 25% FAQ cache	Haiku 4.5 main
Light	100	\$0.59	\$0.46	\$0.32
Typical	240	\$1.42	\$1.10	\$0.78
Heavy (daily cap)	600	\$3.55	\$2.75	\$1.95

Assumes 2,000 cached and 1,160 uncached input tokens, 160 output tokens, two Haiku screens per turn, 8% image turns. US-pinned adds 1.1x. Excludes Trust & Safety staffing.

Kid-safe AI assistant: architecture, models, safety, policy (2/2)

R07 unverified

Detail sheet: Cloud AI tutor, not companion; Anthropic's written approval for children is gate one.

AGE BANDS AND LIMITS

Band	Modes	Limits
5-7	Ask Why, Story Maker, voice-first	No free chat, no photos by default; 1-3 sentences; 10 turns/day
8-10	Adds Homework Helper (hints only), Look at This	2-5 sentences, grade 3-5; 25 turns/day
11-13	Adds deeper explainers, writing feedback	Grade 6-8; 30 turns/day; sensitive topics get curated answer plus adult redirect

Target age range is unconfirmed (A8); India's DPDP Act treats everyone under 18 as a child.

RISKS

Sev.	Risk	Mitigation
High	Anthropic may decline, add conditions, audit or suspend; single-vendor dependency.	Provider abstraction; OpenAI or self-hosted fallback; written confirmation, ZDR and DPA before launch.
High	Child jailbreaks, multi-turn drift; self-harm false negatives are the worst case, false positives frustrate families.	Layered guards, scripted T0 responses, release-blocking red-team evals, kill switch, human Trust & Safety queue.
High	Image input: PII, self-generated sexual imagery, CSAM reporting duty; abuse disclosures where parent is the risk.	Objects/homework only in v1, on-device gate, EXIF stripped, ephemeral; abuse policy written by counsel and child-safety expert.
Medium	Child-voice STT accuracy, battery, thermals and on-device latency are unmeasured.	Measure kid-speech WER on our own consented set; gate launch on it.
Medium	Transcripts are a breach honeypot; Anthropic keeps flagged data up to 2 years even under ZDR; cost spikes from abuse.	Per-family KMS, 30-day default retention, per-device token caps and circuit breaker; disclose 2-year flagged retention.

SCOPE

Stage 1: Cloud gateway (Go) with Claude Sonnet 5.5 main model and Haiku 4.5 input/output guards under ZDR; on-device push-to-talk voice; four modes (Ask Why, Homework Helper, Story Maker, Look at This) across three age bands; limited image input; parent alerts and transcripts; release-blocking eval harness; kill switch.

Stage 2:

- On-device LLM (Gemma 4 E2B/E4B via LiteRT-LM) as offline fallback on 8 GB+ devices, answering only from local curated packs.
- Self-hosted open-weight guards first (Qwen3Guard-Stream, gpt-oss-safeguard), then self-hosted models (Qwen3, gpt-oss) at scale (Stage 2-3).
- Expose on-device TTS as a system TextToSpeechService (or GrapheneOS Speech Services, MIT) for the EPUB reader and Weather.

OPEN ITEMS

- Anthropic written confirmation for under-13 use, ZDR, DPA and child-safety system prompt; direct API vs Bedrock; OpenAI named fallback.
- Minimum age for assistant, and parent visibility for 11-13: full transcripts versus summaries plus alerts; target age range (A8) unconfirmed.
- Abuse-disclosure policy (who is told when a parent may be involved) and Trust & Safety staffing, possibly 24x7.
- Pricing: report assumes AI bundled in a subscription, 30-turn daily cap, COGS ceiling; REQUIREMENTS has no recurring-price decision.
- Reconcile to India (D18): DPDP consent, data location for US-pinned inference, local crisis resources, retention (reports 19, 20).

DO NOT TAKE AS FACT

Claim	Why
Anthropic permits products for children under 13	Guidelines cover minors generally and do not address under-13s; the report makes written confirmation go/no-go gate one.
OpenAI ZDR requirement, Gemini under-18 bar, Mistral terms	openai.com and ai.google.dev were blocked; search summaries only; Vertex AI coverage unverified; Mistral unverified.
US regulation items (COPPA 2026-04-22, FTC 6(b), CA SB 243, GUARD Act, EU AI Act)	Secondary summaries; GUARD Act status after May 2026 unknown; COPPA audio exception and SB 243 scope unverified; none is India law.
COGS of \$1.42 per typical child-month	Engineering estimate from listed prices and an assumed token mix and 240 turns; no measured usage.
On-device speed (52 tok/s Galaxy S26 Ultra; 2-5 tok/s mid-range CPU) and kid-speech accuracy	Speed comes from a blog; battery and thermals not found; child-voice WER unmeasured.
AOSP has no usable speech stack	Inferred from GrapheneOS 17 and LineageOS 24.0 manifests, not AOSP source; labelled engineering judgement.
Crisis lines (988, Childhelp), cloud TTS/STT prices, Kokoro and Gemma 4 weight licences	From memory or unread; marked 'verify' in the report. US lines do not apply to India-first.

Superseded: US launch assumption (A1) replaced by D18 (India first): COPPA, FTC 6(b), CA SB 243, WA/OR laws, GUARD Act, CA AB 1043, US-only inference, US crisis lines (988, Childhelp), NCMEC reporting and USD framing need reconciling to India DPDP (child = under 18; reports 19, 20). Device tier table (Nothing Phone (3), Fairphone Gen 6+, Nothing 4a Pro, Signature 27) superseded by D17 (start on Pixel 10a/9a); Pixel fit with the 8 GB on-device floor is not assessed. Spanish Stage 2 is overtaken by D20 (English only). Core architecture not overturned.

Walkie-talkie (push-to-talk) and safe kid-to-kid voice (1/3)

R08unverified

Detail sheet: Walkie-talkie designed as our own Opus relay, avoiding a third-party voice processor; untested.

DECISION

Build push-to-talk as Opus frames over Zune's own authenticated WebSocket relay, cloud-only, with parent-visible metadata only in v1; LiveKit SFU is Plan B.

No second-pass skeptic review exists. Android 17 facts read in primary sources; law, vendor terms, prices and costs are memory, secondary or inferred.

WHY IT MATTERS

Push-to-talk carries children's voice, so transport choice sets privacy exposure (no third-party voice processor), per-kid cost and what parents can see. The report is an unbuilt design.

KEY POINTS

- Transport: Opus frames over Zune's own authenticated WebSocket instead of a WebRTC SFU; half-duplex 1:1 speech needs no SFU, ICE or UDP. R08 S1
- Audio: Opus VOIP wideband at 16 kbps, 20 ms frames bundled in threes, 120-200 ms jitter buffer; about 300-480 ms mouth-to-ear (unmeasured). R08 S4 Audio
- Android 17 forces a split: transmit starts from the visible Walkie activity; incoming playback runs in a privileged system service with MODIFY_AUDIO_SETTINGS_PRIVILEGED. R08 F1, F2
- No Google services: Nearby Connections is unusable; Wi-Fi Aware and BLE 'near me' mode deferred to Stage 2; v1 is cloud-only. R08 F4, F5
- Waking a no-GMS device needs a persistent socket plus battery exemption; Doze allows one exact alarm per 9 minutes. R08 F3
- Safety limits: one talker per edge, 30 s cap, 60 transmissions/hour per edge, kid mute control, 'Tell a grown-up' alert to guardians. R08 S4 Trust and safety
- Parent visibility in v1 is metadata only; 72-hour recorded review waits for counsel. No grooming detection claimed for voice. R08 F10, S1
- SOS: 3-second hold alerts guardians via Web Push with last location (if enabled); UI states Walkie cannot call emergency services. R08 S4 SOS

NUMBERS

Metric	Value	Basis
Speech bitrate	16 kbps Opus wideband libwebrtc source recommends 16-20 kbps; Android 10+ has a MediaCodec Opus encoder	sourced
Mouth-to-ear latency, own relay	~300-480 ms (SFU ~185-335 ms) Unmeasured; judged acceptable for half-duplex	estimate
Bandwidth per direction, continuous talk	29.6 kbit/s (13.3 MB/h) WebSocket/TLS/TCP, 60 ms bundles; WebRTC at 20 ms packets is 36 kbit/s	estimate
Data per kid per month	~53 MB per direction At 8 talk-minutes per day	estimate
Relay egress at 10k kids	~540 GB/month, \$27-49 Egress price \$0.05-0.09/GB from memory	estimate
LiveKit Cloud, same load	\$5.7k (Scale) to \$7.2k (Ship) per month Join-on-press; built on aggregator prices carried over from R13	estimate
ASR moderation, 100% coverage	\$0.08-0.16 per kid per month About 3.6 eight-thread nodes at an assumed \$0.30 per node-hour	estimate
Roblox v2 voice classifier (English)	63.9% precision, 58.2% recall at 1% FPR Read in repo README; none of its six classes is grooming; faster-whisper about 15x realtime	sourced

TRANSPORT OPTIONS FOR PUSH-TO-TALK

Option	Strength	Weakness	Verdict
WebRTC + SFU (LiveKit, Pion, mediasoup, Janus)	Mature jitter and NAT handling; shares one SDK with video	Join latency; third party holds child audio; needs UDP or TURN fallback	Plan B
Store-and-forward clips over HTTPS	Cheapest; works offline; moderate before delivery	Not live; every clip is stored child voice	Stage 2, 'missed' only
Opus over Zune WebSocket	No setup on warm socket; per-frame policy kill; no vendor processor	We write jitter buffer, floor control, relay; TCP stutter on bad cellular	Recommended
Mumble/Murmur	Proven PTT, BSD-3, Opus	Built for open servers; no schedules or push	Fallback if schedule slips
Matrix / Element Call	E2EE, federation	Group-shaped; AGPL or commercial; E2EE conflicts with parent review	Reject

Own-relay build effort about 1.5-2k lines (estimate).

Walkie-talkie (push-to-talk) and safe kid-to-kid voice (2/3)

R08

unverified

Detail sheet: Walkie-talkie designed as our own Opus relay, avoiding a third-party voice processor; untested.

COST AT 10K KIDS, 8 TALK-MINUTES PER DAY (ALL ESTIMATES)

Item	Monthly	Basis
Own relay egress (~540 GB)	\$27-49	Inferred; \$0.05-0.09/GB from memory
LiveKit Cloud, join-on-press	\$5.7k (Scale) or \$7.2k (Ship)	Sessions at 3x talk time; secondary prices
Self-hosted LiveKit, 2 nodes plus bandwidth	\$300-600 plus ops	Inferred
Relay peak (3% concurrent)	One 4-vCPU Go node	About 300 streams, 10k msgs/s; needs load test
ASR moderation, opted-in clips	\$0.08-0.16 per kid	Self-hosted faster-whisper; assumed \$0.30/node-hour

Modelled at 10k kids; D21 first phase is about 200 users and was not modelled.

ANDROID 17 CONSTRAINTS AND DESIGN ANSWERS

Constraint	Evidence	Zune design answer
Background audio needs a visible activity or non-short foreground service	Android docs; GrapheneOS source (Java path only)	Playback in privileged system service; transmit from visible activity
Microphone foreground service cannot start from background	Android docs	Hold-to-talk on visible Walkie screen only
Doze: one exact alarm per 9 min; exemption keeps network	Android docs	Persistent socket, battery exemption, partial wake lock
Nearby Connections is GMS-only	google/nearby source; microG implements only Exposure Notification	Cloud-only v1; Wi-Fi Aware or BLE near-me in Stage 2
No push-to-talk keycode in Android 17 KeyEvent	GrapheneOS source read	On-screen hold-to-talk; headset-hook PTT in Stage 2
Emergency call disconnects self-managed VoIP calls	Telecomm source read	Walkie never claims emergency calling

RISKS

Sev.	Risk	Mitigation
Medium	TCP head-of-line stutter on weak cellular could make walkie audio glitchy.	Test on target carriers; if more than a few percent glitch, switch to LiveKit Plan B (Opus-only, SPEECH preset).
High	Android 17 native audio-playback enforcement is unread; incoming audio could be silently dropped on device.	Test on device or Cuttlefish with 'adb shell cmd audio set-enable-hardening enable' before other media work.
Medium	Persistent-socket battery and carrier NAT behaviour unmeasured; report assumed Snapdragon devices, D17 starts on Pixel.	Measure on Pixel 10a/9a first; ntfy reports 0-1% battery in 17 h on one phone (n=1).
High	Law and vendor terms unverified, and the report's US framing (COPPA, FCC, all-party consent) no longer matches the India launch.	Counsel review under India's DPDP Act (D18) before any child voice is stored; keep v1 metadata-only.

High	A hostile adult with a guardian account is the main residual threat; stored child voice would become the top breach target.	Verifiable guardian consent (R13); no stored audio in v1; never market voice moderation.
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SCOPE

Stage 1: MVP: 1:1 kid-to-kid between approved contacts; hold-to-talk, 30 s cap, floor control with busy/unavailable tones; Opus relay on the existing socket; no ring-through, no stored audio; metadata log, rate limits, 'Tell a grown-up', SOS alert; server-side schedules abort open streams. Ship gates: verifiable parental consent (COPPA term; India basis per D18), dual-guardian consent, named T&S owner before the first cross-family link.

Stage 2:

- Review mode: 72-hour recorded clips with dual-guardian opt-in, only after counsel and a T&S/reporting runbook
- Voice messages for unavailable contacts (store-and-forward clips), if founder agrees
- Local 'near me' mode: BLE L2CAP + Opus baseline; Wi-Fi Aware where FEATURE_WIFI_AWARE is true
- Headset-hook and volume-key push-to-talk
- Self-hosted faster-whisper flags on opted-in clips (assistive, never a gate)

OPEN ITEMS

- Transport: own WebSocket relay (default) or LiveKit PTT for one shared SDK; founder decision pending.
- Offline use: cloud-only v1 (default) or 'works with no internet' required at launch.
- Parent visibility: metadata-only v1 (default), then 72 h dual-guardian review after counsel.
- Incoming audio: auto-play when allowed (default) or tap-to-hear always.
- Fund a named trust-and-safety owner before the first cross-family link (R13 raises the same gate).

Walkie-talkie (push-to-talk) and safe kid-to-kid voice (3/3)

R08

unverified

Detail sheet: Walkie-talkie designed as our own Opus relay, avoiding a third-party voice processor; untested.

DO NOT TAKE AS FACT

Claim	Why
Relay latency 300-480 ms versus SFU 185-335 ms	Inferred arithmetic, unmeasured; device audio latency varies and has no runtime API.
LiveKit Cloud cost \$5.7k-\$7.2k per month at 10k kids	Built on aggregator prices via R13 (official page blocked) and an assumed usage model (sessions at 3x talk time).
Own relay costs tens of dollars per month at 10k kids	Egress price from memory; peak load and one-node sizing are unmeasured and need a load test.
Voice moderation costs \$0.08-0.16 per kid per month	Assumes \$0.30 per node-hour and 100% coverage; extrapolated from one CPU benchmark in a README.
COPPA, 18 U.S.C. 2258A, all-party consent states, FCC interconnected-VoIP rules	Memory or secondary only (sources blocked), unverified, and US-specific; superseded by the India launch (D18).
Privileged apps are exempt from Android 17 background-audio blocking	Only the Java focus path was read; native audioserver playback enforcement is unread.
Nearby Connections is closed on Android	Inferred from missing platform code and GMS intents in the repo, not from a Google statement.
Relay is about 1.5-2k lines and 'Zello-style' works well	Estimate and memory; no prototype built. No verified child-directed terms for LiveKit, Agora, Twilio, Daily or Vonage.

Superseded: Telephony paragraph (carrier VoLTE voice/SMS, native 911 dialer) and SOS one-tap cellular call to a guardian: superseded by D19 (optional portal SMS to a guardian number is server-side; confirm). US assumptions (A1: COPPA, 2258A, FCC E911, one US region): superseded by D18 India. Snapdragon targeting: D17 (Pixel first). 10k-kid sizing: D21 (~200 users first). A9 default keeps only the platform emergency path.

First-party app suite: stack, in-tree vs APK, per-app design (1/3)

R09 unverified

Detail sheet: Kid apps built in-house on Compose: no store, no GMS, roughly 80 person-weeks (estimate)

DECISION

Recommended: at most six OS-coupled components in-tree; every other app in a Gradle monorepo, separately signed, silently updated. Compose UI, Radium EPUB reader without INTERNET access.

No skeptic pass. Library, licence and AOSP-tree claims were read in primary files; effort figures, book-source licences and priv-app behaviour are inference or memory.

WHY IT MATTERS

Sets build cost, update path and attack surface for every on-device app. No store UI and few INTERNET holders leave fewer ways to break out or leak.

KEY POINTS

- Two tiers: at most six OS-coupled components built in-tree with the platform key; every other app is Gradle-built, separately signed, silently updated. R09 S1 pt1, F3
- One stack: Kotlin, Compose, in-house ZuneKit design system with three age-band profiles, Hilt, Room; minSdk and targetSdk both 37. R09 S1 pt2, S4.2
- Launcher written from scratch in Compose (5 pw). Launcher3 forks need a rebase every Q2/Q4 drop; Lawnchair is based on Android 16. R09 S3, F7
- EPUB: Radium Kotlin Toolkit 3.4.0 (BSD-3, WebView-based, reflowable and fixed-layout). KORader, crengine, FBReader, Libreria rejected as copyleft or proprietary. R09 F6-F7, S3
- Radium's EPUB navigator is a WebView that intercepts every navigation and serves resources in-process, making a no-INTERNET Reader plausible (inference). R09 F6, S4.3
- Only Assistant, Walkie, Videos, Setup, Updater, ZuneComms and Guardian hold INTERNET; WebView exists only in Reader and Videos. R09 S1 pt6
- Google services replaced: Sentry SDK for crash reports, sherpa-onnx for TTS and speech (AOSP has none), LocationManager for location, bundled OFL fonts. R09 S4.2, F1
- An egress lint fails CI on android.webkit.*, http(s) ACTION_VIEW or INTERNET outside the allowlist; Cuttlefish boots the image in CI. R09 S4.2, S4.4

NUMBERS

Metric	Value	Basis
Stage-1 app effort	61.5 person-weeks Per-app estimates summed; excludes Messenger, calls, Weather, Settings (costed in other reports).	estimate
With test, accessibility, usability	about 80 pw 5 Android engineers for 16-20 weeks beside the platform track.	estimate
Radium Kotlin Toolkit	3.4.0 (2026-09-11), BSD-3, minSdk 24 Read in LICENSE, CHANGELOG, README.	sourced
Jetpack versions	Ink 1.0.0, CameraX 1.6.2, Media3 1.11.1, Room 2.8.5 Official release pages via a summarising fetch tool.	reported
Compose launcher	5 pw vs forking Launcher3 Fork rejected: five variants, rebase every quarterly drop.	estimate
Kid usability gate	about 80% unaided completion; 6-8 children per age band per wave Inference; not validated.	estimate
In-house device rack	6-10 real phones Firebase Test Lab cannot flash the Zune OS.	estimate
v1.1 backlog	about 20 pw Music, planner, dictionary, games, family album, Journal backup.	estimate

STAGE-1 APP SUITE AND EFFORT

App	Tier	Stage-1 MVP	pw
Core, ZuneKit, CI pipeline	n/a	Policy client, attested HTTP client, egress lint, design system	8
ZuneHome + ZuneSetup	A	Approved-app grid, bedtime screen; QR pairing, age band, kid name	9
Camera + Photos	B	CameraX photo/video, no geotags; MediaStore grid, explicit send-to-parent	8
Journal	B	Text, drawing, voice note; Room + SQLCipher; private, child-shares or parent-visible	6
Notebook	B	Typed + Jetpack Ink handwriting, checklists, PDF export to parent only	6
Reader	B	Radium EPUB, dyslexia fonts, read-aloud, bookmarks; no INTERNET	6
Assistant UI + Walkie UI	B	Chat bubbles, image attach (EXIF stripped); one hold-to-talk button	9
Videos UI + Clock + Calculator + Recorder	B	Offline Media3 player, IFrame wrapper, kill switch; utilities	9.5

Total 61.5 pw (estimate). Tier A = platform-signed in-tree; Tier B = Gradle, own key, android_app_import.

First-party app suite: stack, in-tree vs APK, per-app design (2/3)

R09unverified

Detail sheet: Kid apps built in-house on Compose: no store, no GMS, roughly 80 person-weeks (estimate)

REPLACING GOOGLE MOBILE SERVICES DEPENDENCIES

GMS-bound item	Replacement
Firebase Crashlytics, Analytics, FCM, Remote Config	Sentry SDK (MIT), scrubbed; push over ZuneComms socket; signed-policy config
Play Services Location	Platform LocationManager (GNSS only)
ML Kit	Bundled LiteRT models; none in v1
Maps SDK, Credential Manager, Play Integrity	MapLibre later; none; key attestation via Guardian
Downloadable fonts	Bundled OFL fonts (Lexend, Andika, OpenDyslexic)
TextToSpeech / SpeechRecognizer	sherpa-onnx engine (report 07)

WorkManager and CameraX core have no play-services dependency (read in androidx build files).

EPUB ENGINE OPTIONS

Engine	Licence	Fixed layout	Verdict
Radium Kotlin	BSD-3	Yes	Choose; harden the WebView
KOReader / crengine	AGPL-3.0 / GPL-2.0	No (inference)	Reject: copyleft
FBReader, Libreria, Moon+	GPL, AGPL parts, proprietary	n/a	Reject; FBReader/Moon+ licences unverified
Own native renderer	ours	Hard	About 12 pw; only if WebView is banned

Licences for KOReader, CoolReader and Libreria read in repo files; FBReader and Moon+ from memory.

RISKS

Sev.	Risk	Mitigation
Medium	Radium inherits the WebView attack surface; 3.x navigators have churned between versions.	Pin versions, budget 1 pw per upgrade, intercept all navigation, no INTERNET permission in Reader.
Medium	Re-flash or repair under D16 wipes a child's private Journal.	Stage 1: parent-triggered encrypted export; automatic backup deferred to v1.1.
Medium	Public-domain classics carry dated content; editorial review is a recurring cost.	Editorial screening; a content editor is required headcount, not optional.
Low	Camera quality and vendor extensions vary by device HAL (report assumed Snapdragon).	Test per device; extensions optional (GrapheneOS Camera pattern); D17 starts on Pixel only.
Medium	A missing privapp-permissions entry can stop boot (memory, unverified); per-app SELinux domains unchecked.	CI boots Cuttlefish on every change and runs priv-app, policy and egress tests.

SCOPE

Stage 1: Core libraries and CI pipeline, then ZuneHome, ZuneSetup, Camera, Photos, Journal, Notebook, Reader, Assistant UI, Walkie UI, Videos UI, Clock, Calculator and Recorder, all on-device in simplest form. Waves 0-3 run to about week 24; Messenger, calls, Weather and Settings sit in other reports.

Stage 2:

- v1.1 backlog (about 20 pw): offline Music, Homework planner, Dictionary, 3-5 curated offline games, Family album.
- Journal automatic backup (Stage 1 has only a parent-triggered encrypted export).
- Reader v1.1: tap-to-define via offline dictionary, private highlights, reading-time stats to the parent portal.
- Commercial titles via EDRLab LCP; licence and fees unverified.
- Later, unscheduled: where's-mom location, ScratchJr-style coding, handwriting recognition, TalkBack bundling, on-device LLM.

OPEN ITEMS

- Journal default: private-to-child with opt-in sharing (recommended) or parent-visible.
- Target age range (A8 unconfirmed); it drives handwriting, target sizes and voice-first design.
- Accept WebView in exactly two apps: Reader and Videos (the report's recommendation).
- Reader launch content: public-domain and CC only (recommended), or fund publisher deals and LCP now; recheck licences for India.
- Headcount: a kid-UX designer and a content editor are required, not optional; crash-report host (managed Sentry US vs self-hosted) undecided.

First-party app suite: stack, in-tree vs APK, per-app design (3/3)

R09unverified

Detail sheet: Kid apps built in-house on Compose: no store, no GMS, roughly 80 person-weeks (estimate)

DO NOT TAKE AS FACT

Claim	Why
About 80 person-weeks, 61.5 pw Stage-1, 5 engineers for 16-20 weeks	Engineering estimates, not measured; wave plan also runs to week 24, so schedule and staffing figures are not reconciled.
Book-source licences (Gutenberg, StoryWeaver, African Storybook, OpenStax, Internet Archive)	Memory only; book-source sites were egress-blocked and the search budget ran out. Standard Ebooks is 'believed' US public domain.
No-INTERNET Reader works with Radium; files served via signature-protected provider	Inference from reading the navigator source; not built or tested.
Priv-app, SELinux and dexpreopt behaviour (untrusted_app domain, boot failure on missing entry, preprocessed + dexpreopt)	Memory or unchecked; report lists them as unverified.
Camera2 upkeep in AOSP 17; FBReader and Moon+ licences; LCP fees	Not verified. Android 17 facts come from GrapheneOS 17 and LineageOS lineage-24.0 trees because googlesource was blocked.
Child-data minimisation (30-day retention, no analytics SDK, crash payload limits); touch targets 64/56/48 dp; 80% usability gate	Engineering inference or suggestions to validate; counsel to confirm. COPPA framing is stale under D18 (India).
Library versions (Ink 1.0.0, CameraX 1.6.2, Media3 1.11.1, Room 2.8.5)	Read through a summarising fetch tool on official pages, not raw files.

Superseded: Spanish in v1.1 (decision 6) no longer fits India-first (D18); D20 sets English only for v1. 'US only' public-domain book screening and COPPA/US-hosted Sentry framing assume a US launch (A1 superseded by D18; DPDP, child = under 18). Snapdragon camera-HAL risk assumes D13 (D17: Pixel first). Phone app costed in R12 is dropped by D19 (no cellular calls/SMS). Core stack and delivery recommendations stand.

Release engineering: signing, OTA, verified boot, anti-theft, build infra (1/3)

R10unverified

Detail sheet: Zune forks GrapheneOS's update and signing plumbing; keys never touch the build host.

DECISION

Recommended: fork GrapheneOS's 1,377-line MIT Updater into ZuneUpdater (update_engine plus static CDN files); sign offline with Zune-owned keys, one AVB key per hardware model; roll forward only.

No second-pass skeptic section. Evidence is mostly primary files, but from GrapheneOS/LineageOS trees not Google's; prices, EU/UK rules and security-branch status from memory.

WHY IT MATTERS

Signing-key custody and update delivery decide whether a bad build or leaked key can hurt children's phones, and how costly recovery is. Patch cadence is a standing people cost.

KEY POINTS

- OTA: fork GrapheneOS's 1,377-line MIT Updater; static CDN files with ed25519-signed metadata. Rejected hawkBit and Mender: neither drives update_engine. R10 F7, F8
- Keys: AOSP key set plus one RSA-4096 AVB key per model, generated air-gapped; two-person offline signing; CI holds only the metadata key. R10 Design, F1-F3
- AVB key rotation is effectively a recall: custom keys are set only while unlocked, and lock transitions wipe data and rollback indexes. R10 F5
- Roll forward only: update_engine's SPL guard needs green boot state; Zune's custom-key devices are yellow, so Zune enforces SPL monotonicity itself. R10 F6
- Patching is a named 0.5-1 FTE job: monthly SPL train, 72-hour hotfix lane. No GMS means Zune owns WebView and Mainline freshness. R10 Patch train, F10
- Benchmark: GrapheneOS, a partner, cut 43 release tags in 260 days; non-partner patch lag was about 125 days (report 01). R10 F9
- Zune Activation Lock: Android 17 has FRP-secret primitives and no UI; the station sets a secret, escrowed server-side under per-family KMS. R10 F11
- Attestation is a signal, not a gate (expects our AVB key hash, locked, yellow). Whether Google's RKP serves non-GMS OSeS is unverified. R10 F12

NUMBERS

Metric	Value	Basis
GrapheneOS Updater size and license	1,377 lines, MIT Read in GrapheneOS/platform_packages_apps_Updater branch 17	source
GrapheneOS release cadence	43 tags, 2026-01-08 to 2026-09-25 git ls-remote on platform_manifest; about one per 6 days	source
Non-partner patch lag	about 125 days About 40 framework fixes, taken from report 01	reported
Running infra cost	\$0.5-0.9k/month 2 bare-metal hosts plus ephemeral runners; from memory, prices not checked	estimate
One-time infra, lab and signing hardware	\$10-18k Device lab \$4-8k; signing hardware about \$2.5k; build hosts about \$6k if bought	estimate
OTA payload size	full 1.5-2.5 GB; monthly delta 0.1-0.4 GB Lineage README example is 1.93 GB; 5,000 devices would move 0.5-2 TB/month	estimate
Fleet patch-currency target	90% on latest SPL within 10 days Design target, not measured	estimate
Support promise	contractual 3 yrs, target 5 Per-model end date: earlier of OEM firmware end and 5 years; founder to confirm	estimate

KEY COMPROMISE AND ROTATION

Key	Compromise effect	Rotation
AVB key (one per model)	Malicious images; needs physical write access	Recall: unlock, wipe, re-flash
releasekey (OTA)	Denial of service; remote takeover only with AVB too	New otacerts.zip, signed by old key, via OTA
platform and other app keys	System-privileged app delivered by APK update	Effectively permanent; HSM, minimal use

AOSP test keys are public and must never ship; about 44 APEX modules also need offline signing.

Release engineering: signing, OTA, verified boot, anti-theft, build infra (2/3)

R10unverified

Detail sheet: Zune forks GrapheneOS's update and signing plumbing; keys never touch the build host.

OTA CLIENT OPTIONS

Option	Fit	Verdict
(a) Fork GrapheneOS Updater	MIT; already drives update_engine, deltas, constraints, downgrade checks	Chosen: smallest option
(b) LineageOS Updater	Apache-2.0; JSON list with sha256/size	Not chosen
(c) hawkBit + hara-ddiclient	EPL-2.0 server; JVM client; does not drive update_engine	Adds server plus glue; revisit past about 50k devices
(d) Mender	Targets embedded Linux/Yocto; does not drive update_engine	Adds server plus glue; not chosen
(e) AOSP SystemUpdaterSample	Demo only	Not chosen

RELEASE GATES (CALL AND SMS CHECKS OMITTED)

Gate	Check
G1	Same container on two hosts; diff target-files
G2	SPDX SBOM, NOTICE; kernel and Vanadium GPL sources published
G3	Cuttlefish boot smoke, no-browser bypass suite, policy tests, N-1 to N update
G4	ro.debuggable=0, secure adb, no permissive SELinux, no test keys, SPL monotonic
G5	Real devices, 3 per model: full and incremental OTA, slot fallback, attestation, FRP
G6	Two-person offline sign-off, then signature and AVB validation scripts

Staging and rings follow: lab 48 h, 5%, 25%, 100% over about 7 days.

RISKS

Sev.	Risk	Mitigation
High	Security-branch access (android17-security-release) is unverified; non-partner patch lag was about 125 days.	Authorise partner-access outreach to Google/ODM; cross-check against GrapheneOS and LineageOS branches.
High	AVB key leak or rotation forces a per-phone unlock, wipe and re-flash.	One AVB key per model caps blast radius; strictest custody; offline host, two-person signing.
Medium	OEM firmware (bootloader, modem, vendor) needs redistribution rights to ship in Zune's OTA, or that layer goes stale.	OEM patch record is a qualification criterion; support promise ends at earlier of OEM firmware end and 5 years.
Medium	Attestation/RKP without GMS and FRP code in Google's own tree are unverified; bootloader behaviour varies per model.	Test each model on a sacrificial unit; fall back to station-recorded identity if RKP fails.
Medium	Operating load: monthly WebView build, Virtual A/B free space on full phones, battery cost of the persistent channel.	Dedicated WebView runner; free-space precheck before updates; charging and unmetered Wi-Fi constraints.

SCOPE

- Stage 1:** Full-OTA-only ZuneUpdater on a static CDN with ed25519-signed metadata and rings; offline two-person signing; per-model AVB keys; station-set Activation Lock secret; monthly SPL train with 72-hour hotfix lane; crash-summary telemetry.
- Stage 2:**
- Chain every partition to delegated keys so root vbmeta stays cold and a hot key rotates by OTA (test per bootloader).
 - APEX/Mainline side-channel updates between full OTAs.
 - HSM-backed signing via avbtool --signing_helper and --payload_signer at general availability (offline encrypted files suffice for the beta).
 - Trial fullmte.mk on MTE-capable devices; defer hardened_malloc (needs a bionic fork).
 - Revisit a hawkBit-style rollout server past about 50k devices.
- OPEN ITEMS**
- Support promise: contractual 3 years, target 5, per-model end date? (report default: yes).
 - Name three key custodians, approve about \$2.5k signing hardware, and fund a 0.5-1 FTE release/security owner.
 - Approve forking GrapheneOS's MIT Updater and static-CDN design; choose bare-metal or cloud build hosts.
 - Forced-update policy (14 days normal, 72 h critical); default screen lock; whether lost mode includes location; crash-report consent.
 - Reconcile to current decisions: India (DPDP) replaces US/COPPA assumptions; key-custody threshold (100-device beta) versus the ~200-user first phase.

Release engineering: signing, OTA, verified boot, anti-theft, build infra (3/3)

R10

unverified

Detail sheet: Zune forks GrapheneOS's update and signing plumbing; keys never touch the build host.

DO NOT TAKE AS FACT

Claim	Why
All prices and infra costs (\$0.5-0.9k/month, \$10-18k once, signing hardware about \$2.5k, CDN under \$20/month)	From memory; price pages were blocked and search quota ran out. Verify before quoting.
Android 17 FRP-secret primitives; remote wipe avoiding the secret-clearing Settings path	Read in GrapheneOS tree, not Google's; the wipe-path point is inference and needs checking in Google's tree.
Locked custom-key (yellow) devices bypass update_engine's SPL guard	Guard reading is primary, but the yellow-state consequence is inference; must be tested per model.
Cloud KMS can sign AVB via a digest-stripping signing helper	Helper trick is the author's inference from avbtool code; not tested.
Non-partner patch lag about 125 days; security-branch access	Taken from report 01, not re-measured here; AOSP security-branch status unverified.
OTA sizes and bandwidth (full 1.5-2.5 GB, 0.5-2 TB/month at 5,000 devices)	Inference from one Lineage README example; R2 pricing from memory; sizing predates the ~200-user phase.
Google's RKP backend serves a non-GMS OS	Explicitly unverified; rkpd is off without a hostname property whose value is unknown.
Chromium guide quote on AOSP prebuilt WebView staleness; GrapheneOS FAQ statements	Read from scratchpad page copies; Chromium URL not re-verified; EU CRA/UK PSTI dates from memory.

Superseded: US-launch framing (A1 replaced by D18, India): COPPA/state-breach counsel duties and decision 9 'confirm US-only launch'. Telephony (D19): G5 call and SMS-vault checks (emergency path stays per A9). Scale (D21): beta limit of 100 devices and 5,000-device bandwidth sizing predate the ~200-user first phase. Pixel-first (D17) narrows the 3-model device lab.

Compliance, Legal and Regulatory Map (US-framed) (1/2)

R11 superseded

Detail sheet: Report assumed a US launch; India-first decision (D18) replaces its legal map.

DECISION

R11 recommended a US-only launch with COPPA, California AB 1043 and SB 1119 compliance, EU/UK deferred, and no hardware sales. D18 (India first) supersedes the market; no-hardware, open-source and trademark items survive.

No skeptic pass; nearly all dates secondary or memory. Core US-only recommendation overturned by D18; parts (open-source, trademark, no-hardware) still stand.

WHY IT MATTERS

Do not rely on any US date, fine or consent method here. What survives is the strictest-tier design posture, open-source and trademark hygiene, and the never-sell-handsets stance.

KEY POINTS

- Proposed: US-only launch for ages 6-12, with FTC-approved COPPA safe-harbor membership before general availability. Dropped: D18 makes India first. R11 §1
- Proposed: California AB 1043 age-bracket API (four brackets) in the first image. India has no OS age-signal law (R19); dropped for now. R11 F2; R19 §6
- Proposed: parents reading kid SMS under an interception-law counsel opinion. D19 removes SMS; messenger and AI-chat reading remain a legal question (R19). R11 §4.1 row 2
- Proposed: defer EU/UK (9-18 months extra, estimate) and avoid India; D18 reverses the India call. R19 calls it wrong. R11 §1, §4.3; R19 §6
- Survives: never sell or stock handsets (D15/D16), keeping manufacturer-type duties (CPSIA, FCC/CE, CRA) off Zune. R11 §1, row 14
- Survives: AI assistant as tutor, not companion; crisis routing, parent alerts, audit-ready logs. Indian helplines replace 988 (R19). R11 F5, row 3
- Survives: open-source hygiene: publish GPL-2.0 kernel source or offer; no Piper (GPL-3), KOREader/MuPDF or libsignal (AGPL) in image. R11 F12, row 10
- Survives: trademark clearance: 'Zune' is a codename (Microsoft history); say 'built on the Android Open Source Project', no Android logo. R11 F12, row 12; A5

NUMBERS

Metric	Value	Basis
Amended COPPA penalty	about \$53,088 per violation US only; secondary source; no longer the launch regime	reported
AB 1043 penalty	\$2,500 negligent / \$7,500 intentional per child California AG only; read in legislature data mirror	sourced
AB 1043 operative date	2027-01-01 Earlier-set-up devices by 2027-07-01; not applicable in India (R19)	sourced
COPPA counsel and safe harbor	\$20-50k counsel; \$5-25k/yr safe harbor; 8-12 weeks Author's order-of-magnitude, USD	estimate
AI audit (CA SB 1119)	\$30-100k by mid-2027 Only if California is served	estimate
SOC 2 programme	\$30-80k; 6-12 months Type I then Type II	estimate
Trademark clearance	\$3-10k; 2-3 weeks Clear or replace 'Zune' before marketing	estimate
EU/UK extra time	9-18 months CRA Class I, RED, EAA, DSA, AI Act stack	estimate

RISK-RANKED OBLIGATIONS AND THEIR STATUS TODAY

Obligation	Severity	Design implication	Status under D18-D21
Amended COPPA (US under 13)	Critical	Parental consent at enrolment; retention schedule; security program	Replaced by DPDP s.9 and Rule 10 (R19)
Interception law (parents read kid messages)	High	Child-visible 'parents can see this' signal; written parental consent	Maps to Telecom Act s.42, IT Act s.69 (R19)
AI-minor laws (CA SB 1119, SB 243)	High	Tutor not companion; crisis routing; parent alerts; incident log	Design survives; California is not first market
California AB 1043 age-bracket API	High (hard date)	Birth-date step at setup; four-bracket API; no sharing	Does not apply in India (R19)
CSAM and NCII (2258A, REPORT Act, TAKE IT DOWN)	High	Photos stay on device; NCMEC registration; 48-hour takedown path	Becomes POC SO reporting to Indian police (R19)
OEM and Google firmware licence	High	No OTA firmware until a written answer	Survives (R17 gate G5)
YouTube terms and made-for-kids	High (business)	Offline licensed tier first; made-for-kids only; kill switch	Survives
CPSIA, FCC/CE (only if handsets sold)	Medium	Stay an installation service; never sell handsets	Not triggered until own hardware (D21); India: BIS/WPC/TEC (R19)

R11 matrix has 17 rows; costs and lead times are the author's estimates.

Compliance, Legal and Regulatory Map (US-framed) (2/2)

R11superseded

Detail sheet: Report assumed a US launch; India-first decision (D18) replaces its legal map.

OPTIONS CONSIDERED

Option	Verdict	Reason
A. Comply with AB 1043 and COPPA now, US-wide	Adopt (market superseded)	Cheap; satisfies later Colorado, Brazil, Australia rules
B. License OS permissively for AB 1856 carve-out	Reject	Blobs, apps and EULA make qualification doubtful
C. Exclude California	Reject	Largest market; others copy
D. US, UK and EU together	Defer	CRA Class I, EN 18031, EAA, DSA, AI Act, GDPR-K
E. Sell phones	Reject (D15/D16)	CPSIA, FCC/CE, CRA manufacturer duties, inventory

REGULATORY DATES AS READ BY R11

Date	Event	Market
2026-04-22	Amended COPPA Rule compliance (live)	US
2026-09-11	CRA Art. 14 vulnerability reporting live (24 h)	EU
2027-01-01	AB 1043 operative; Vermont design code; WA chatbot law	US
2027-05-13	DPDP children's rules (confirmed by R19)	India
2027-07-01	AB 1043 deadline for older devices; CA SB 1119 binds	US
2027-12-11	CRA full application	EU

Only AB 1043 and CRAAnnex III were read at primary source; every other date is secondary or memory.

RISKS

Sev.	Risk	Mitigation
High	Nearly every legal date and fine is secondary or memory; primary texts unread.	Counsel re-reads primary texts before any reliance; treat US dates as unverified.
High	Card or KBA parental consent (R11 row 1) is not a named Rule 10 method in India.	Use DigiLocker token or staff-inspected government ID plus consent ledger (R19).
High	Interception exposure when parents read children's messages persists under India's Telecom Act and IT Act.	Child-visible indicator, written parental consent, Indian counsel opinion before first external family (R19 G1).
High	Google/OEM firmware redistribution licence unanswered; blocks any OTA carrying OEM firmware.	No OTA with OEM firmware until a written answer (R17 gate G5).
High	YouTube terms and made-for-kids rules endanger the Videos tier; FTC fined Disney \$10M over mislabelled videos.	Offline licensed tier first; made-for-kids only; kill switch (R06).

SCOPE

- Stage 1:** Superseded US sequence: private beta with up to 100 provisioned US devices, counsel engaged, COPPA consent live. Reusable build-in checklist: portal consent ledger, child-visible 'parents can see this' indicators, AI safety kit, trust-and-safety kit, no analytics SDKs or ad IDs, signed builds with SBOM.
- Stage 2:**
- US GA after safe-harbor certification, AB 1043 shipped, SB 1119 readiness (superseded market)
 - UK, then EU once CRA and RED routes are costed
 - SOC 2 Type I then Type II (6-12 months)
 - Independent AI audit if California is served (about mid-2027)
 - Canada and Australia later

OPEN ITEMS

- Target age band: R11 suggested 6-12; A8 unconfirmed; DPDP treats everyone under 18 as a child.
- May photos or AI image uploads leave the device? R11 recommended no cloud photo storage.
- Clear or replace the 'Zune' name before any public use or marketing (estimate \$3-10k, 2-3 weeks).
- Codec patents (AVC, HEVC, AAC, MPEG-2): cost unknown until licensors quote; disable unneeded software codecs.
- Budget for counsel retainer, safe-harbor program, SOC 2 and insurance is undecided.

DO NOT TAKE AS FACT

Claim	Why
AB 1856 (signed 2026-09-10) removes open-source-licensed OS distributors from AB 1043	Secondary only; enrolled text unread (a digest of the 2026-08-21 Senate version was read).
Amended COPPA details and the \$53,088 per-violation penalty	Secondary: compiled GitHub research files; Federal Register and FTC were blocked.
CA SB 1119 binds broad chatbot operators from July 2027 with audits and a private right of action	Secondary news story; final definitions unread.
FTC pattern: Disney \$10M, Apitor \$500k suspended, Genshin Impact \$20M	Secondary summaries; not read in FTC orders.
AVC/HEVC/AAC patent pools bill device makers; CRA Class I needs a notified body	Training memory, unchecked.
All costs and lead times (9-18 months EU, \$20-50k counsel, audits, SOC 2)	Author's judgement, order of magnitude only.

Superseded: US-only launch and 'avoid India' sequencing (D18); US beta of up to 100 devices (D18, D21); kid-SMS reading and cellular E911/Wi-Fi-calling items (D19); US consent methods and AB 1043, CPSIA and FCC rows (R19 section 6).

Cellular calls and SMS with parent allowlists (superseded) (1/3)

R12 superseded

Detail sheet: Cellular calls and SMS were designed, then dropped; communication is now internet-only.

DECISION

Dropped. Founder removed cellular voice and SMS (D19): no dialer, no SMS vault. All communication is internet-based. Report kept only for reusable ideas.

Core recommendation (carrier calls/SMS with framework allowlist) overturned by D19. Second pass refuted 2 claims, corrected 8; AOSP source unreachable, GrapheneOS 17 branches read.

WHY IT MATTERS

Dropping carrier telephony removes a heavy engineering and legal burden (carrier deals, E911, SMS interception law, Telecom fork); the report's bypass lessons may still apply to the emergency path.

KEY POINTS

- Proposed: framework-level, deny-by-default call/SMS allowlist (policy service in Telecom/Telephony) plus a no-UI SMS vault; role-based screening fails open after 5 s. R12 S1, 2.2
- Cost seen: Telecom and Telephony ship inside the com.android.telephonycore APEX, so the fork meant a re-signed APEX rebased every AOSP drop. R12 S1 [CORRECTED], 2.1
- Dropped by founder, twice: 'No cell calling, only WhatsApp-type calling' (D19, replacing D4-D6). Comms are now in-app 1:1 text, voice, video. REQUIREMENTS D19, A9
- Dropped with it: US carrier plan (BYO SIM, T-Mobile MVNE), Pixel IMS/VoLTE carrier qualification, and US wiretap/911/WEA analysis (D18 India). R12 2.6-2.8, S6
- Survives: Android 17 self-managed VoIP calls skip Telecom's filter graph; since D19's calls are in-app VoIP, gate server-side plus on-device. R12 2.3; R13 F3
- Survives: signed, deny-by-default policy snapshot (ECDSA P-256, expiry, fail closed after 14 days unsynced), approvals queue and immutable audit log. R12 S4
- Survives: two 2026 Telecom intent-layer bypasses (one is CVE-2026-28615) show patch lag matters if telephony stays for A9's emergency path. R12 2.2, 7.11

NUMBERS

Metric	Value	Basis
Call-screening timeout before call rings anyway	5,000 ms (fail-open) Read in Telecomm@17 source (GrapheneOS branch)	sourced
AOSP emergency-callback filter-skip window	5 min (stock blocker lifts 2 h) R12's proposed 30 min was shown to be a bypass	sourced
Intent-layer Telecom bypasses fixed in 2026	2 (2026-05-29 and 2026-06-17) Commits read in git log; CVE-2026-28615 id from search summaries	sourced
Planned framework fork size	1-2k lines Second pass: no evidence; omits APEX plumbing, vault and tests	estimate
Wholesale cost per line (voice, SMS, modest data)	\$6-12 per month Unverified placeholder; no primary price found	estimate
US kids-phone incumbents, retail	\$20-89 per month Search summaries; US market; one Bark quote conflicts	reported

R12 PROPOSAL AND ITS FATE

Component	What R12 proposed	Fate
ZunePhone (dialer)	Contacts-only call list, large emergency button	Dropped (D19); A9 default keeps platform emergency path
ZuneSmsVault	Default SMS role; encrypted queue; upload to portal; wipe	Dropped (D19 supersedes D6): no SMS on device
Telecom/Telephony fork + ZunePolicyService	Signed deny-by-default allowlist inside framework, rebased each AOSP drop	Cellular purpose dropped; signed-policy idea survives
Carrier path	US only: VoLTE, BYO SIM beta, T-Mobile MVNE line at GA	Dropped (D18 India first; D19 data only)
IMS, eSIM, carrier data	Google stock blobs; licence question	Voice need dropped; emergency path and eSIM handling still open (A9, A10)
US legal analysis	Wiretap, 47 CFR 9.10, WEA, CPNI, COPPA	Superseded by D18; India analysis belongs to reports 19-20
Portal: call log, SMS inbox, approvals, audit	Per-contact rules; read-only SMS threads; immutable audit log	SMS inbox dropped; approvals and audit log reusable

Cellular calls and SMS with parent allowlists (superseded) (2/3)

R12superseded

Detail sheet: Cellular calls and SMS were designed, then dropped; communication is now internet-only.

BYPASS VECTORS R12 MAPPED, AND STATUS UNDER D19

Vector	R12 control	Under D19
Intent-layer bypass (ACTION_CALL trampoline)	Enforce below intent layer; no third-party apps	Moot without dialer; patch-lag lesson holds
MMI/USSD and call-forward codes (*21*, *67)	Reject any dial string with *, # or letters	Moot without a dialer
Emergency-callback window	AOSP flags only (5 min); alert portal	Moot with data-only SIM (A10)
In-app VoIP calls (self-managed)	Own device-side check; filter graph is skipped	Now covers every call: gate server-side too
USB modem, DIAG, AT ports	Lock USB config; verify per device	Check per device: could dial or text outside Android
Caller-ID and SMS-sender spoofing	Reject failed STIR/SHAKEN; flag unauthenticated SMS	Moot: no phone-number identities or SMS

US KIDS-PHONE INCUMBENTS (REPORTED, UNVERIFIED)

Incumbent	Host network	Retail per month
Gabb	Verizon MVNO	\$20-25
Troomi	AT&T (older sources say T-Mobile)	\$27-50
Bark	T-Mobile MVNO (announced 2022-12-02)	\$39-89; one source says from \$49
Pinwheel	Carrier-agnostic; own wireless service too	Not stated

Search-summary sources; prices not re-verified. Not an India comparison.

RISKS

Sev.	Risk	Mitigation
Medium	Telecom patch lag: android-17.0.0_r1 may predate CVE-2026-28615; matters while telephony stays for A9 emergency calls.	Base on a branch with post-June-2026 Telecomm fixes; R12 default: 30-day maximum latency for critical fixes.
High	Emergency path retained by default (A9); removing it is a safety and legal question; R12's E911 evidence is US-only and unverified.	Consult counsel before any override; India analysis sits in report 19.
High	In-app VoIP calls bypass Telecom's filter graph, so a client-side allowlist alone is not enforcement.	Server-side token gating plus on-device check against the signed policy (R13); fail closed.
Medium	Modem exposure via USB (DIAG/AT ports) can dial or text outside Android on some OEM configs (flagged for Qualcomm).	Lock USB configuration (charging or MTP only); confirm per qualified device; pre-beta fuzzing list.
Medium	eSIM provisioning without Google services for a data-only SIM inherits a Pixel stock-binary licensing question.	Check whether baseline eSIM use suffices; GrapheneOS says installed profiles work without the proprietary component.

SCOPE

Stage 1: None. Cellular calls and SMS are out of scope (D19). Stage 1 communication is in-app text, voice and video (R13 backbone); A9 default keeps only the platform emergency-call path.

Stage 2:

- Parent-side voice/video calling (deferred, D8)
- Optional end-to-end encryption to a parent-held key (R12 and R13 both list it)
- AI risk flags for bullying and grooming (R12 for SMS; R13 for messenger)
- Calendar-style schedules, time budgets and multi-guardian roles

OPEN ITEMS

- A9: keep the platform emergency-call path (112 in India) or remove it; founder must say so, counsel first (report 19).
- How non-emergency cellular calls and SMS are technically disabled on the image is not specified in the reports reviewed.
- A10: confirm data-only SIM/eSIM works and Wi-Fi-only use works; eSIM provisioning without Google services remains a licensing question.
- Patch source and maximum latency for Telecom/Telephony security fixes; R12 default is 30 days for critical fixes, undecided.
- In-app calls skip Telecom's filter graph: specify the on-device policy check and one signed policy format shared with R13.

Cellular calls and SMS with parent allowlists (superseded) (3/3)

R12

superseded

Detail sheet: Cellular calls and SMS were designed, then dropped; communication is now internet-only.

DO NOT TAKE AS FACT

Claim	Why
Fork of about 1-2k lines is acceptable	Second pass found no evidence; omits APEX plumbing, vault and tests. Moot after D19.
Wholesale \$6-12 per line per month; incumbent retail prices	Placeholders; no primary price found; one search result conflicts with the Bark low end; US market only.
AOSP 17 removed the Messaging app	UNVERIFIED: compared a GrapheneOS tree to AOSP 16; AOSP's own 17 manifest was unreadable.
911 reaches carriers even from non-initialized handsets	UNVERIFIED (2015 sunset proposal, no final order found); legal pin-cites came from search summaries; all US law.
Pixel 10a has a physical SIM tray	Probable only: search summaries agree but direct spec pages were blocked.
Soracom Air RTC Gateway, Gigs, Telnyx, 1NCE as carrier options	Leads only; availability, pricing and capabilities unverified. Moot after D19.
Vendor IMS fills the STIR/SHAKEN verdict per carrier	Unverified; Android code exposes it only from IMS call profiles.
All Android 17 findings come from AOSP-equivalent code	android.googlesource.com was blocked; code was read from GrapheneOS 17 branches, which carry extra patches.

Superseded: Entire design (ZunePhone, ZuneSmsVault, Telecom/Telephony fork for allowlists; D4-D6) superseded by D19. US carrier plan (BYO SIM, T-Mobile MVNE), US law analysis: D18 India. Pixel and Snapdragon qualification premises: D17 and D19. Cellular Stage-2 items (network-side allowlist, MMS capture, Wi-Fi calling, text-911) dropped. A9 default keeps only the emergency path.

Kid messenger, 1:1 video calling and shared comms backbone (1/3)

R13partly verified

Detail sheet: Kid messaging and video are designed around parent-approved contacts and parent-readable threads.

DECISION

Build one first-party comms service (Go, WebSocket, Postgres) with a parent-approved contact graph; server-readable text-and-emoji messenger and LiveKit 1:1 video; no recording, no E2EE in Stage 1.

Skeptic pass: 15 confirmed, 11 corrected, 1 refuted, 5 unverified of 32 rows. Android 17 read from GrapheneOS branch; AOSP blocked; prices and law secondary.

WHY IT MATTERS

Cross-family contact is where child-safety risk concentrates; the report calls one breach of children's chats existential, so trust model and moderation limits decide viability.

KEY POINTS

- One first-party comms service (Go, one WebSocket per device, Postgres, Redis); Matrix and Signal rejected, XMPP kept as the 'buy' fallback. R13 S1, 3.1
- Parent-approved contact graph governs everything; cross-family links need both guardians via invite code. No phone matching, address-book upload or kid-searchable directory. R13 S1, 4.1
- Messages are server-readable with per-family envelope encryption; both families' guardians read threads, kid is told. No E2EE in Stage 1. R13 S1, 3.2
- Text-only, 500 characters, no links, images or GIFs; age-banded emoji allowlist; server rules block URLs, numbers, handles and flag bullying or secrecy. R13 4.3
- Video on LiveKit SFU (Apache-2.0) at 360p/24 fps; server can end calls; locked-down tokens; no recording or covert listening in Stage 1. R13 F5, 4.4
- Android 17 catch: privileged background start gives no while-in-use capability, so camera, mic and audio live in the visible call app. R13 F3, 4.2
- No-GMS push: persistent socket in a privileged service, Doze-allowlisted; battery and NAT unmeasured, ntfy evidence argues against 30 s heartbeats. R13 F3, 4.5
- Known gaps: live video unmoderated; rules-only text filters miss coded language; social engineering of parents is the main cross-family threat. R13 S7 R2, R11

NUMBERS

Metric	Value	Basis
LiveKit Cloud Ship plan	\$50/mo incl. 150k participant-min; \$0.0005 overage Aggregator pages only; livekit.com blocked; \$0.12/GB egress	reported
LiveKit Cloud Scale plan	\$500/mo incl. 1.5M min; \$0.0004 overage \$0.10/GB; region pinning, SOC 2 report and BAA reportedly Scale and above	reported
Video bitrate cap	360p / 24 fps, ~0.64 Mbps incl. audio Design cap; hardware codec behaviour to be benched per device	estimate
Video cost at 10k kids	~\$0.06 per kid per month Assumes 10% make one 10-min call a day; excludes PTT and included bandwidth	estimate
Video cost at 100k kids	~\$5k/month on Scale; self-host saves ~60-80% Revisit at \$3-5k per month	estimate
Message load at 100k kids	~115 msg/s avg, ~1.2k/s peak, ~1 GB/day Assumes 100 messages per kid per day; 3-4 nodes, one Postgres primary	estimate
Custom comms build, 1:1 text path only	~3k lines of Go Second pass: excludes policy engine, tokens, moderation, portal APIs, encryption	estimate
Carrier NAT idle-drop window	25-60 s or more ntfy-android PR 113 user reports; vendor blog puts TURN use at 15-30% of WebRTC sessions	reported

MESSENGER STACK OPTIONS

Option	For	Against	Verdict
Matrix (Synapse/Conduit + rust-sdk)	Receipts, offline, multi-device built in	Room-based authz; AGPL or paid licence; heavy to strip	Reject for Stage 1
XMPP (Prosody + Smack)	Receipts, archive, stream management; permissive licences	Parent graph and moderation still custom; second runtime	Best 'buy' fallback
Signal protocol	Strong E2EE	libsignal is AGPL-3.0; blocks parent review and moderation	Reject
MQTT	Tiny keepalive	No store, ACLs or history	Not needed
Custom WebSocket + Postgres	Exact semantics; in-line moderation; one runtime	We own reliability: idempotency, ordering, resync	Recommended

Synapse AGPLv3 or paid licence and libsignal AGPL verified in repo files.

Kid messenger, 1:1 video calling and shared comms backbone (2/3)

R13

partly verified

Detail sheet: Kid messaging and video are designed around parent-approved contacts and parent-readable threads.

PARENT VISIBILITY AND ENCRYPTION MODES

Mode	Parent review	Server moderation / abuse reporting	Breach impact
A. Server-readable, per-family envelope encryption (recommended)	Full, instant	Full	DB-only leak is ciphertext; app/KMS compromise exposes plaintext
B. E2EE, parent browser as extra recipient (passkey PRF)	Only if parent key present; history lost on key loss	Client-side only	Smallest
C. Client-side escrow to parent key	Same as B	Weak	Medium
D. Metadata plus flagged excerpts	Partial	Needs plaintext anyway	Smaller

Defaults: 180-day content retention, 13-month metadata. Mode D proposed as a per-age Stage 2 option.

CHANNEL CONTROLS AND KNOWN GAPS

Channel	Controls designed	Known gap
Text messenger	Server rules block URLs, numbers, handles; flag bullying, secrecy	Coded words and spelled-out numbers beat rules
1:1 video	Approved edges only; locked tokens; portal End call; no recording	Unmoderated; kid can show handle or address on camera
Walkie-talkie (R13 design)	Live-only, no stored voice, metadata to parent	Fails without data; transport choice open (R08 relay vs LiveKit)
Cross-family link	Both guardians approve via invite; revoke ends live sessions	Guardian B's judgement; hostile adult needs only image plus portal account
Screenshots	FLAG_SECURE on call and chat windows	Cannot stop a second camera pointed at the screen

RISKS

Sev.	Risk	Mitigation
Medium	Persistent-connection battery and carrier NAT timeouts are unmeasured; ntfy evidence says short heartbeats keep the modem awake.	Start probing at minutes, not 30 s; measure on Pixel 10a/9a first (D17); fall back to longer-interval design.
High	Android 17 may silently drop incoming audio and block camera/mic from a boot-started service; inferred, not tested.	Keep room client in the visible call app; test with 'adb shell cmd audio set-enable-hardening enable' first.
High	Server-readable store of children's chats is a high-value target; the report calls one breach existential.	Per-family envelope encryption with KMS keys, break-glass staff access, audit logs.
High	Safety gate is Guardian B's judgement; after self-install ships (D15), a hostile adult needs only the image and an account.	Verified-guardian step before issuing or accepting invites; v1 intake (D16) is an extra gate; T&S reviews rapid inviters.

High	Trust-and-safety and abuse-reporting readiness is a staffing dependency (R13 cites US NCMEC); rules-only moderation misses subtle grooming.	Named T&S owner, runbook and 24 h SLA before first cross-family link; self-hosted ML in Stage 2.
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SCOPE

Stage 1: In-app 1:1 text+emoji messenger and 1:1 video between approved contacts; contact graph with dual-guardian approval; server-side moderation and flags; one device WebSocket with signed policy; portal thread viewer, approve/revoke, End call. D19 also puts in-app voice calls in Stage 1, but R13 does not design them separately.

Stage 2:

- Kid-initiated QR bump (still needs both guardians); typing and read receipts as parent toggles
- Per-age visibility: metadata plus flagged excerpts for ages 11-13
- Self-hosted ML moderation and grooming scoring; on-device nudity detection that pauses video
- Parent-side video/voice with consented listen-in (deferred by D8); voice notes and photos with scanning
- Optional parent-key E2EE tier; self-hosted LiveKit past about \$3-5k per month

OPEN ITEMS

- Which app owns the single device socket: ZuneGuardian (report 05) or ZuneComms (R13)? R13 text contradicts itself.
- D19 adds in-app voice calls; R13 designs video and push-to-talk, not a separate voice-call flow. Scope it on the backbone.
- Guardian verification and parental-consent method: R13 defaults to card transaction (US/COPPA-derived); India (DPDP) basis for counsel to confirm.
- Walkie-talkie transport: R13's LiveKit room versus R08's own Opus relay; founder decision pending.
- Thread visibility: both families read everything (default) or metadata-only for older kids; age bands assume 5-13, A8 unconfirmed.

Kid messenger, 1:1 video calling and shared comms backbone (3/3)

R13

partly verified

Detail sheet: Kid messaging and video are designed around parent-approved contacts and parent-readable threads.

DO NOT TAKE AS FACT

Claim	Why
LiveKit Cloud prices (Ship \$50/150k min; Scale \$500/1.5M min)	Aggregator pages only; official livekit.com blocked. Region pinning and SOC 2 reportedly need Scale or higher.
LiveKit Cloud is an acceptable processor of children's audio and video	No verified LiveKit position on child-directed services; DPA and child-data terms unverified.
Video costs about \$0.06 per kid per month; 100k kids about \$5k per month	Inferred from assumed usage (10% of kids, 10-minute call); excludes PTT minutes and included bandwidth.
Comms service is about 3k lines of Go	Covers only the 1:1 text path; excludes policy engine, token orchestration, moderation, portal APIs, encryption.
Privileged apps are exempt from Android 17 audio blocking	Only the Java-side exemption was read; native playback enforcement is unread; AOSP source was blocked.
Battery target of 3% per day idle	Unsupported; the cited ntty PR argues short heartbeats are expensive. Unmeasured on any device.
Tensor G4 codec behaviour; Android 17 ships without Emoji 17.0 glyphs	Both UNVERIFIED (cited page was about Pixel 10; emoji claim from a search summary only).
COPPA, 2258A 'no duty to monitor', TAKE IT DOWN Act, California AB 1043, 11 all-party states	Secondary or memory; counsel must confirm; all US law, superseded by the India launch (D18).
US region plus a DPA for hosting	Assumes US launch (A1); India first (D18) changes region, latency and data-residency, which R13 does not analyse.

Superseded: US-only launch (A1) and US legal framing (COPPA, 2258A/NCMEC, TAKE IT DOWN, CA AB 1043): superseded by D18 (India, DPDP, under-18). Section 6 SIM/data-plan and PSTN-endpoint/SMS-forward contact-graph items: superseded by D19, D14, D15. Snapdragon bench edit: overtaken by D17 (Pixel first). 10k/100k-kid sizing: D21 (~200 users first).

Weather app that teaches children about weather (1/3)

R14

partly verified

Detail sheet: Weather app teaches with authored lessons and parent-set places; alerts are US-only so far

DECISION

Backend-proxied weather via Open-Meteo paid plan (MET Norway fallback); parent-set places; authored explainers and safety cards; no LLM in app. NWS/WEA alerts are US-only: re-plan for India.

Skeptic pass done: Open-Meteo, Google terms and GrapheneOS source read in primary files; NWS, FCC, device specs are search summaries; US assumptions now stale.

WHY IT MATTERS

Weather carries a vendor-terms dependency, the child-location question and safety alerts, where a wrong design is legal or cost risk. Beta cost is small; scale needs Enterprise or self-hosting.

KEY POINTS

- Open-Meteo's free tier is non-commercial only: Zune needs a paid plan (Standard 1M, Professional 5M, Enterprise 50M+ calls/month) or self-hosting. R14 S2.1, claim 1
- Google Weather API excluded: terms bar apps whose primary purpose is weather information; corrected to a 'primary purpose' judgement. R14 S2.1, claim 2
- Devices call only Zune's backend; vendors see one server identity and ~5 km grid coordinates, never device or account IDs. R14 S1, S4.1
- Stage-1 location is parent-set in the portal. GrapheneOS's network location calls Apple's undocumented endpoint; Mozilla Location Service is dead since 2024. R14 S2.2, claim 5
- Education is authored, not LLM-generated: 20 explainers x 3 age bands plus 6 safety cards, triggered on device by live conditions. R14 S4.4
- AQI is a separate endpoint, doubling requests to about 2.6M (10k devices) and 12.6M (100k); 100k needs Enterprise or self-hosting. R14 S2.1 [CORRECTED]
- Cell-broadcast alerts: link_method=none is not enough; child can toggle alerts off. Fix: DISALLOW_CONFIG_CELL_BROADCASTS plus RRO flags; geofencing fails open. R14 S2.3 [CORRECTED]
- Corrected teaching rules: stickiness keys on dew point, not relative humidity; sun-protection chip at UV 3, not 6 (threshold needs meteorologist confirmation). R14 S4.4 [CORRECTED]

NUMBERS

Metric	Value	Basis
Open-Meteo plan call budgets	Standard 1M, Professional 5M, Enterprise 50M+ per month Read from pricing page source; call weighting (variables x models) unconfirmed.	sourced
Open-Meteo first two plan prices	about \$29 and \$99 a month Search summary only; pricing table unreadable.	reported
Corrected request volume	about 2.6M (10k devices), 12.6M (100k) a month Place-cell and refresh assumptions are inference; AQI separate endpoint is verified.	estimate
Open-Meteo free tier	<10,000 calls/day, 300k/month, non-commercial only Apps with subscriptions count as commercial.	sourced
Self-hosted Open-Meteo node	8 GB RAM minimum (16 recommended), 100 GB NVMe AGPL-3.0 code; cost 'low hundreds of dollars a month' is inferred.	sourced
OpenWeather alternative cost	about EUR 1.8k (10k) / 8.8k (100k) a month Search summaries disagree; about 10x Open-Meteo at scale.	reported
Beta cost	tens to low hundreds of dollars a month Depends on plan tier; assumptions inferred.	estimate
Authored content, report scope	20 explainers x 3 bands + 6 safety cards Report's own lean cut: 8 x 3 bands, 6 cards, 2 visuals.	estimate

WEATHER DATA SOURCES FOR A COMMERCIAL CHILD PRODUCT

Source	Commercial terms	Cost (beta / scale)	Verdict
Open-Meteo	Global; paid plan grants commercial licence; data CC BY 4.0	Standard ~\$29, Professional ~\$99 (reported); 100k: Enterprise or self-host	Primary
MET Norway Locationforecast	Global, CC BY 4.0, free incl. commercial; unique User-Agent mandatory	\$0 at fallback volumes	Fallback (terms from search summary)
NWS api.weather.gov	US only; public domain with conditions; limits undocumented	\$0 at any scale	US alerts only; stale under D18
OpenWeather One Call 3.0	Global; OdbL-based; on-screen attribution	About EUR 1.8k (10k) / 8.8k (100k) a month	Reject (cost)
Tomorrow.io	Free plan bars commercial use for evaluation accounts	From about \$50 (reported)	Reject
WeatherAPI.com	Free tier reportedly allows commercial use (unverified)	\$7/3M, \$25/5M, \$65/10M calls (reported)	Plan B; read ToS first
Google Weather API	Terms bar use where weather is the primary purpose	About \$0.15/1,000 after 10k free	Excluded (judgement)

Only Open-Meteo, NWS and Google terms were read in primary sources; others are search summaries.

Weather app that teaches children about weather (2/3)

R14partly verified

Detail sheet: Weather app teaches with authored lessons and parent-set places; alerts are US-only so far

SECOND-PASS CORRECTIONS THAT CHANGE THE DESIGN

Original claim	Correction
Google API unusable: bars weather apps	Narrower 'primary purpose' clause; exclusion prudent but a judgement
About 1.3M / 6.3M raw calls a month	About 2.6M / 12.6M: AQI is a separate endpoint
All Flash Flood and Severe Thunderstorm warnings fire WEA	Only top damage tiers do; key on CAP WEAHandling, not event names
link_method=none keeps alerts always on	Child can switch alerts off; needs DISALLOW_CONFIG_CELL_BROADCASTS plus RRO flags
Humidity 90% means sticky	Misconception; key stickiness on dew point
UV 6 or higher triggers sun-safety card	Use UV 3, escalate at 6; threshold from memory
Bundle endpoint serves any place_id	Authorise place_id per family, else cost-amplification route
Stage 1: 60 variants, 4 visuals, alert engine	Stage-1.5; lean cut is 8 explainers x 3 bands, 2 visuals

Corrections from the independent second pass on 2026-10-02.

EDUCATION BY AGE BAND

Band	Style	Example hook (corrected)	US standards
5-7	Icons, 1-2 sentences, no formulas	Cloudy today. Clouds are tiny water drops floating up high.	K-ESS2-1, K-ESS3-2
8-10	Numbers, cause and effect, predict-then-check	Dew point about 55-65 F feels sticky; above 65 F oppressive	3-ESS2-1/2, 3-ESS3-1, 5-ESS2-1
11-13	Mechanisms, data reading, weather vs climate	Pressure fell 4 hPa in 3 hours: a low is arriving	MS-ESS2-4/5/6

NGSS is a US standard (reconcile to India, D18); not all codes individually opened. Age bands differ across reports.

RISKS

Sev.	Risk	Mitigation
Medium	Open-Meteo can change terms effective immediately and end a subscription; exact prices unverified.	Vendor adapter, MET Norway fallback, stale-while-error, self-host path (AGPL, 8-16 GB RAM).
High	Alert design (NWS, WEA) assumes a US launch; this report has nothing for India's alerts.	Reconcile through reports 19 and 20; NWS is US-only.
High	Child can disable cell-broadcast alerts; carrier acceptance on a non-stock OS and non-tappable links untested.	DISALLOW_CONFIG_CELL_BROADCASTS plus RRO flags; test on Android 17; ask carrier at certification.
Medium	Authored safety content needs a science-education reviewer and a meteorologist; editorial burden recurs.	Fund both reviews before beta; consider the lean 8-explainer cut.
Medium	Legal reading of coarse location plus device identity is unresolved; the report	Parent-set places avoid child geolocation; counsel review under India's DPDP (reports 19, 20).

Sev.	Risk	Mitigation
	framed it as COPPA.	

SCOPE

Stage 1: Report's scope: current + 48 h + 7 days forecast, NWS alert banner and parent notice, 20 explainers x 3 bands, 6 safety cards, 4 Compose visuals, launcher tile, Journal note, parent place/units/alert settings, MET Norway fallback, CellBroadcast overlay. Its own check calls this Stage-1.5; lean cut: 8 explainers x 3 bands, 6 cards, 2 visuals. US-specific alert items need re-planning for India.

Stage 2:

- Predict-then-check and cloud-match games; graphs from the child's journal; GLOBE Observer cloud citizen science (child terms to check).
- Multi-place plus optional coarse GNSS 'I am here' (one-shot, rounded on device, parent opt-in).
- Capability-gated barometer and light-sensor experiments; pollen and moon phase; read-aloud once TTS exists.
- Self-hosted Open-Meteo node; per-country alerts (ECCC, DWD, Meteoalarm), licences to verify; tuned Assistant 'why' Q&A.
- Climate stories for ages 11-13 with a parent toggle (Stage 1 stays factual, local-weather framed).

OPEN ITEMS

- Weather alerts for India: NWS and WEA are US-only; whether cell-broadcast stays under D19 and data-only SIM (A10) is undecided.
- Approve Open-Meteo commercial subscription for beta, with self-host evaluation before about 5-10k devices (report default: yes).
- Fund a science-education reviewer and a meteorologist safety-card review before beta (report default: yes).
- Canonical age-band enum: reports use 5-7/8-10/11-13, 6-8, 5-9/10-13; REQUIREMENTS A8 age range unconfirmed.
- Parent-side alert delivery undefined: browser portal (D12) means Web Push, email or SMS, which no report has specified.

Weather app that teaches children about weather (3/3)

R14

partly verified

Detail sheet: Weather app teaches with authored lessons and parent-set places; alerts are US-only so far

DO NOT TAKE AS FACT

Claim	Why
Open-Meteo prices of about \$29 and \$99 a month	Search summary only; the pricing table was unreadable.
Call-volume and cost model (2.6M / 12.6M requests, 10k / 100k devices)	Inferred place-cell and refresh assumptions; whether plan budgets weight calls by variables is unconfirmed.
NWS terms, rate limits, WEA coverage and FCC opt-out rules	weather.gov and federalregister.gov were blocked; search summaries only; WEAHandling not checked in a live payload.
Geofencing fails open, so missing location cannot suppress alerts	Read in a LineageOS Android 15-era tree, not Android 17; device test still needed.
Competitor prices and terms (OpenWeather, Tomorrow.io, WeatherAPI.com, Visual Crossing) and MET Norway terms	Search summaries only; none load-bearing but unchecked.
Barometer on Pixel 9a/10a; Fairphone 6; none on Nothing Phone (3)	Spec aggregators, not OEM sheets; Snapdragon device discussion is stale under D17.
link_method=none passes carrier conformance (47 CFR 10.441)	Rule text unreadable; UNVERIFIED, ask the carrier at WEA certification.
Open-Meteo AQI outside Europe, UV 3 threshold, NGSS codes, climate-anxiety survey figure	AQI is a 45 km model, not observations; UV threshold and some NGSS codes from memory; survey not verified.

Superseded: US launch assumptions (A1 superseded by D18 India): NWS alerts, WEA design, US-English/imperial default (decision 8), COPPA framing, NGSS standards, ZIP. Spanish as first Stage-2 locale conflicts with D20 (English only). Snapdragon/Fairphone sensor discussion superseded by D17 (Pixel 10a/9a first). Keep-cell-broadcast decision leaned on R12 (superseded by D19) and needs re-confirmation. Core design (backend proxy, Open-Meteo, parent-set places, authored lessons) stands.

Curated Snapdragon device set and BSP strategy ^(1/3)

R15 superseded

Detail sheet: Snapdragon is a later tier: relock evidence is user reports only

DECISION

Snapdragon is a later second-source/premium tier (D17), not v1. Qualify candidates on the Device Support Contract, a passing relock test and written OEM terms; Pixel stays v1. Launch call overturned by D17; skeptic pass found no Snapdragon relock evidence beyond user reports; shortlist and DSC survive as the later second-source tier.

WHY IT MATTERS

Explains why most 'very high grade' Snapdragon phones are excluded or unproven on tamper-proofing, and what must be proven before a second-source or premium tier.

KEY POINTS

- Gate on a Device Support Contract, not the SoC: custom-key relock, OEM-unlock disable, A/B, AIDL HALs, 8 GB+ RAM, written OEM permission. ^{R15 F1}
- No Snapdragon phone has passed the contract; relock evidence is user reports (avbroot #299), none for Fairphone 6/6+ or Nothing (4a) Pro. ^{R15 F2 [CORRECTED]}
- Only the Pixel 10a has vendor-documented custom-key relock. Rejected: OnePlus 13/15 (bootloaders reject keys), Galaxy S25/S26 (no OEM unlock), Xiaomi class. ^{R15 F2, F3}
- Shortlist as candidates pending bring-up: Fairphone Gen 6+ (\$649), Nothing Phone (3) (\$799), Nothing (4a) Pro (\$499-599), Motorola Signature 27 (unshipped). ^{R15 F3}
- Route: our AOSP 17 system on OEM vendor/firmware partitions, re-signed with our AVB key; adevtool is Pixel-only, so Snapdragon needs per-OEM tooling. ^{R15 F4, s4}
- Android 17 accepts Android 13-17 vendors (FCM 7 to 202604, GKI 6.1-6.18), so older vendors can host our system; boot test still needed. ^{R15 F4 [CORRECTED]}
- Snapdragon stacks are heavy: Fairphone 6 tree has kernel 6.1.177, 1,478 blob entries, 31 Qualcomm module entries; Pixel needs one adevtool. ^{R15 F4 [CORRECTED]}
- Lock strength depends on OEM firmware: a Qualcomm ABL 'GBL' bug unlocked locked Xiaomi/POCO phones; fixes reached OEMs in March 2026. ^{R15 F4 [CORRECTED]}

NUMBERS

Metric	Value	Basis
Fairphone Gen 6+ US price	\$649 US launch 2026-08-18, Android 16, unlocked; Verizon not guaranteed. R21: Fairphone not sold in India.	reported
Nothing Phone (3) US price	\$799 Snapdragon 8s Gen 4, 12 GB; 5 OS + 7 security years.	reported
Nothing Phone (4a) Pro US price	\$499-599 Snapdragon 7 Gen 4; 3 OS + 6 security years.	reported
Pixel 10a (reference) US price	\$499 Tensor G4, 8 GB; support to Mar 2033.	reported
Fairphone 6 LineageOS tree	kernel 6.1.177; 1,478 blobs; 31 modules Re-measured in real files, 2026-10-02.	sourced
OnePlus 13 LineageOS tree	kernel 6.6.142; 863 blobs; 29 modules Original '~1,000 blobs, ~15 modules' was corrected.	sourced
Android 17 vendor window (FCM levels)	7 to 202604 GrapheneOS mirror, not Google's tree; no boot test.	sourced
2026 smartphone shipments (IDC)	-16.7% Memory crisis; Qualcomm price rise from 2026-09-01; ASP +27.6%.	reported

CANDIDATE DEVICES (US PRICES; INDIA AVAILABILITY UNCHECKED)

Device	SoC / RAM	US price	Relock evidence	Verdict
Fairphone Gen 6+	7s Gen 4 / 12 GB	\$649	Family works (FP3/4/5) by user report; Gen 6+ untested	Candidate; was the launch bet
Fairphone 6	7s Gen 3 / 8 GB	Import; US price unsupported	Same family evidence; LineageOS 23.2 tree	Dev reference; different SoC from Gen 6+
Nothing Phone (3)	8s Gen 4 / 12 GB	\$799	User-reported works	Best high-end Snapdragon candidate
Nothing Phone (4a) Pro	7 Gen 4 / 8-12 GB	\$499-599	(4a) reported; Pro not listed	Price-floor candidate
Motorola Signature 27	8 Elite Extreme Gen 6 / 16 GB	Unannounced	GrapheneOS-confirmed; not shipped	Watch; qualify 2027
Sony Xperia 1 VII	8 Elite / 12 GB	\$1.3-1.5k	1 V/1 VI work; US unlock reportedly blocked	Reference BSP only, not launchable
OnePlus 13/15; Galaxy S26	8 Elite family	\$899-999+	Keys rejected; no OEM unlock	Reject
Pixel 10a (reference)	Tensor G4 / 8 GB	\$499	Vendor-documented; GrapheneOS-proven	v1 device (D17)

US prices from search summaries. Also rejected: ASUS ROG (no new models), Xiaomi (custom keys accepted but not enforced), Honor/Oppo/Vivo (no US certification).

Curated Snapdragon device set and BSP strategy (2/3)

R15superseded

Detail sheet: Snapdragon is a later tier: relock evidence is user reports only

RETAIL-FLASH WITH OEM TERMS VS
MANUFACTURER-BUILT

Factor	Retail-flash + OEM terms	Manufacturer-built
Tamper	Unlock path stays in OEM firmware; yellow boot screen	We own keys; unlock removable; green boot state
Certification	OEM's FCC/PTCRB/carrier approvals stay	Ours: PTCRB \$10k-500k, 12-24 months (vendor figures)
Price	\$499-900 retail	\$50-115 FOB entry silicon (low reliability)
Supply	No MOQ but no allocation; DRAM shortage, OEM exits	MOQ 3-5k+, contract supply
Patches	AOSP side ours; firmware and kernel from OEM monthly	Ours, via ODM contract
Warranty	OEM warranty; software mods may void it (inferred)	Ours

Certification and price rows are US-based; R21 covers India.

SNAPDRAGON VS PIXEL/TENSOR

Area	Snapdragon	Pixel/Tensor
Third-party NPU	LiteRT Qualcomm runtime can ship inside the APK (plausible)	Tensor SDK is beta; GMS-need claim unsupported
On-device LLM	Llama-3.2-3B ~10-28 tok/s on 8 Elite (vendor figure)	Report: not usable; reasoning later found unsupported
Camera and codecs	Qualcomm HW codecs; ISP extras closed	Tensor codec use unproven (R13)
Build effort	~860-1,500 blobs; per-OEM firmware ingest	One adevtool plus Google images
Hardening	TEE keystore; StrongBox/MTE only on rare designs	Titan M2, MTE (not Pixel 11)

Stage 1 AI is cloud; offline LLM is Stage 2.

RISKS

Sev.	Risk	Mitigation
High	OEM firmware redistribution licence unknown; only Sony's Open Devices exists and its licence was never read.	Get written OEM permission first; D15/D16 may shrink exposure because OEM firmware is already on the customer's phone.
High	Relock is user-reported only; an OEM bootloader update can remove it (OnePlus precedent).	Sacrificial-unit relock test per model; hold back each OEM firmware update until it passes.
Medium	Tamper surface: Qualcomm ABL/EDL bugs; lock strength depends on OEM firmware cadence; Fairphone back cover and battery are user-removable.	Test per unit; hold OEM updates until retested; own hardware later could fuse keys (R21).
High	Price and supply: DRAM shortage, Qualcomm price hike (2026-09-01), OnePlus exit, financially fragile small OEMs.	Customer supplies the phone (D15); Pixel stays the v1 device (D17); Snapdragon only as a later tier.
Medium	Android 17 on an older vendor stack risks camera and audio regressions; FP6 and Gen 6+ are separate ports.	Pick one Snapdragon SKU and port it; run the 10-point bring-up checklist before qualifying.

SCOPE

Stage 1: Not required for v1 (D17): Pixel 10a/9a ships first. If a Snapdragon tier starts: pick ONE SKU (FP6 and Gen 6+ are separate ports); take OEM vendor/odm/firmware partitions whole from OEM OTAs, referenced by our vbmeta; system/system_ext/product ours; HSM-held key; factory flow unlock, flash, avb_custom_key, lock, OEM-unlock off. Needs 3 units, a sacrificial relock unit, a 10-point bring-up checklist and written OEM terms.

Stage 2:

- Second Snapdragon SKU (Nothing Phone (3) or (4a) Pro); Signature 27 qualification once it ships.
- On-device speech-to-text, TTS and safety classifier on the NPU, then a 1-3B offline LLM.
- Automated per-OEM firmware ingest with monthly diffs.
- Thundercomm/QCM6490 or 7-series ODM RFQ (R21 now covers own hardware).
- EU variant (Fairphone is native).

OPEN ITEMS

- Written terms, Fairphone first then Nothing, Motorola, Sony: reflash/relock customer phones, OEM firmware as vendor base, early firmware access.
- Sacrificial-unit relock tests for Fairphone Gen 6+ and Nothing Phone (3); custom-key behaviour on newer Qualcomm bootloaders is unknown.
- Does a custom-key-locked unit report our key and locked state on a no-GMS image (Remote Key Provisioning)? Unverified.
- Re-scope the device list for India (D18): list, prices and carriers are US-based; Fairphone is not sold in India (R20 via R21).
- Google's Android 17 tree and Qualcomm/CodeLinaro pages were unreachable; Sony's binary licence is unread.

Curated Snapdragon device set and BSP strategy (3/3)

R15

superseded

Detail sheet: Snapdragon is a later tier: relock evidence is user reports only

DO NOT TAKE AS FACT

Claim	Why
Fairphone Gen 6+ and Nothing Phone (3) 'pass' the relock contract	Refuted as worded: evidence is user reports (avbroot #299, read only via a summarising tool); FP6/6+, (4a) Pro and Xperia 1 VII are not listed.
Samsung removed OEM unlock in One UI 8 in all regions	Press reports from code analysis ('seemingly'); no Samsung statement.
Prices, availability and dates (Gen 6+ \$649 on 2026-08-18; Signature 27 in Nov 2026)	Search summaries only; publisher pages blocked. Prices are US snapshots.
Qualcomm NPU bundling vs Tensor needs Play; Llama-3.2-3B 10-28 tok/s; Whisper latency	Vendor figures or unsupported inference; Hugging Face and Google pages blocked; QAIRT redistribution terms unchecked.
Key Attestation at pairing proves our key and locked state	From memory, unverified; Remote Key Provisioning without GMS is not established.
Light Phone III was built by Foxconn	Unverified; Snapdragon 4 Gen 2 and FCC ID were confirmed, manufacturer was not.
Fairphone 6 \$799-899 US price and T-Mobile/AT&T support	Unsupported: the FP6 is an import; Gen 6+ is Fairphone's first formal US sale.
FCC: software-only change needs no new filing; AT&T IMEI whitelist	Unverified single sources, US-only; counsel must confirm.

Superseded: Core call (Fairphone Gen 6+ as Stage-1 launch device, Pixel 10a as second source) is overturned by D17 (start on Pixel; Snapdragon later); D13 itself is superseded by D17. The price model (\$649 + \$40 + returns = \$899 bundle or ~\$30/mo x 24) is overturned by D15/D16 (we sell no hardware). US carrier gating (T-Mobile/AT&T VoLTE/VoWiFi/E911 MUST, Verizon scope, IMS stack risk, IMEI whitelists) is overturned by D18 (India) and D19 (no cellular calls/SMS). US-only rejections ('no US certification' for Xiaomi/Honor/Oppo/Vivo) rest on A1, superseded by D18. The 25k units/yr ODM gate and PTCRB costs are US items, replaced by R21 India gates (D21).

Minimal Settings app: what stays, goes, is parent-gated (1/3)

R16partly verified

Detail sheet: Child sees about 11 Settings rows; stock AOSP Settings stays hidden as engine room

DECISION

Option D+: new Compose ZuneSettings (about 11 rows) deep-links to stock Wi-Fi, mobile data and Bluetooth screens; stock Settings stays as default-deny engine room. Parent PIN gates risky rows.

Skeptic pass done with corrections, but AOSP primary sources unreachable; checked against GrapheneOS/LineageOS trees; legal claims secondary only.

WHY IT MATTERS

D14 requires standard Wi-Fi and data, the hardest code in Settings. This design reuses it and defaults everything else to off, but leaves known residual gaps.

KEY POINTS

- Stock Settings 17 is huge: 437 manifest entries, 334 exported, 213 ENTRY_FRAGMENTS, 237 Catalyst annotations, 134 config bools (GrapheneOS tree). R16 F1 [CORRECTED]
- Full rewrite rejected: framework and SystemUI hard-reference Settings classes and send 57 distinct Settings.ACTION_* intents. R16 F4 and verification
- Stock Settings is trimmed by a generated sysconfig component-override: about 40 of 437 stay enabled; new upstream ones default to off. R16 F2, section 5
- Two user restrictions are traps: DISALLOW_CONFIG_WIFI and DISALLOW_CONFIG_MOBILE_NETWORKS blank the whole Wi-Fi and mobile pages, contradicting D14. Never set them. R16 F3 [CORRECTED]
- config_show_wifi_settings=false does NOT hide the 'Wi-Fi preferences' row (WEP, Wi-Fi Direct, install certificates); hiding it needs a small Java patch. R16 F2 [CORRECTED]
- Kept Wi-Fi details page still has captive-portal 'Sign in' and venue-website buttons (unguarded ACTION_VIEW); share-QR is ungated without a device lock. R16 F8, Risks [CORRECTED]
- Parent gate: Zune-owned PIN (Keystore HMAC, escalating lockout, portal reset) in Stage 1; Android 17 platform Supervision PIN deferred to Stage 2. R16 sections 1, 5
- In-device factory reset works only if the Device Owner is the sole setter of DISALLOW_FACTORY_RESET; otherwise wipeData throws SecurityException. R16 F9 [CORRECTED]

NUMBERS

Metric	Value	Basis
Stock Settings manifest activity entries	437 (334 exported) XML-parsed recount of GrapheneOS Settings; LineageOS 428	sourced
Stock top-level Settings rows	~23 From top_level_settings.xml	sourced
Child-visible ZuneSettings rows	~11 Design target	estimate
Stock activities kept enabled	~40 of 437 Allowlist design, not yet proven on a device	estimate
Stage 1 effort (Option D+)	8-9 engineer-weeks Breakdown sums to 8.5 weeks; omits per-device QA and accessibility testing	estimate
Rebase cost per upstream drop	2-4 days Likely understated: Catalyst annotations grew 27 to 237 in about 18 months	estimate
New Compose app alternative	4-6 months Rejected option	estimate
Stage 1 Java patch to hide Wi-Fi rows	~30 lines max Skeptic's proposed cap; original ~10 lines for one row; founder decision pending	estimate

SETTINGS APPROACH OPTIONS

Option	Effort	Rebase per drop	Tamper resistance	Verdict
A. Stock Settings, overlays only	Low	Low	Weak (Catalyst/SPA pages, adult homepage)	Part of D+
B. Slim fork, delete screens	3-4 weeks	1-3 weeks	Good	Reject
C. New Compose app	4-6 months	Low, but system-API drift	Best	Reject: worst crash safety
D+. ZuneSettings front, stock engine room	8-9 weeks	2-4 days (likely understated)	Good; strong with Stage 2 gate	Recommend

All effort and rebase figures are unsupported estimates [I].

Minimal Settings app: what stays, goes, is parent-gated (2/3)

R16 partly verified

Detail sheet: Child sees about 11 Settings rows; stock AOSP Settings stays hidden as engine room

WHO CONTROLS WHAT (CONDENSED)

Setting	Class	How enforced
Wi-Fi: join, saved networks, QR, details, hidden network	Child	Stock screens; never DISALLOW_CONFIG_WIFI
Mobile data toggle, usage view, Data Saver, airplane	Child	Stock MobileNetworkActivity, DataUsageSummaryActivity
Data roaming	Child, default off	Parent can lock via DISALLOW_DATA_ROAMING
Data limit, billing cycle, APN, SIM/eSIM management	Parent-gated	Portal via NetworkPolicyManager; carrier config; eSIM via ZuneSetup
Hotspot, VPN, Private DNS, network reset	Hidden / forced	User restrictions; Guard pins strict Private DNS
Developer options, apps, accounts, users, screen lock	Hidden	DISALLOW_* restrictions; component-override
Factory reset	Parent-gated	PIN button in parent area; Device Owner must own restriction
Date, time, time zone	Forced	AUTO_TIME and AUTO_TIME_ZONE=1; portal zone if no NITZ

SKEPTIC PASS: WHAT CHANGED

Original claim	Verdict	Effect
config_show_wifi_settings hides Wi-Fi preferences row	REFUTED	Needs small Java patch; 'zero patches' claim fails
337 exported, ~250 gateway fragments	CORRECTED	334 exported, 213 ENTRY_FRAGMENTS
Platform applies DISALLOW_FACTORY_RESET only with no role holder	CORRECTED	Queries ROLE_SUPERVISION; guarded wipe flow can throw SecurityException
'Help is inert': no URL launches	PARTLY CORRECTED	Captive-portal and carrier-setup URL launches remain in kept screens
SpeechServices licence citation	CORRECTED	Licence is in source repo; model-weight licence unchecked
SystemUI stock tile pool is 36	CORRECTED	29 tiles
Restriction pages blank (was inferred)	CONFIRMED	Behaviour now read in source
sysconfig component-override disables at scan	CONFIRMED	System packages only; Settings can re-enable some at runtime

RISKS

Sev.	Risk	Mitigation
Medium	Nothing checked against Google's android-17.0.0_r1; GrapheneOS and LineageOS trees may differ in Settings and flags.	Re-verify on Google's tag when reachable; week-1 Cuttlefish spike on component-override, restrictions and SystemUI.
Medium	Catalyst screens bypass ENTRY_FRAGMENTS; whether a disabled host activity makes a row a no-op or a crash is unverified.	Mandatory tap-every-row crawl in CI plus a catch-all intent router to prevent ActivityNotFoundException crashes.
Medium	Stage 1 leaves Wi-Fi advanced fields, Wi-Fi preferences row, captive-portal buttons and a Wi-Fi share QR (may expose password) visible.	Allow a Stage-1 Java patch of about 30 lines to hide them; add a gate patch in Stage 2.
Medium	Supervision is new in 17.0 and flag-dependent; time-limit policies flag is only enabled in trunk_staging; platform PIN untested.	Zune-owned PIN in Stage 1; Zune's own product build can set the flag value itself.
Medium	Accessibility: TalkBack and TTS are not in AOSP; GrapheneOS ships prebuilts; TTS model-weight and training-data licence unchecked.	Build TalkBack from google/talkback (Apache-2.0) in Stage 2; check licence before redistributing; decide with counsel.

Minimal Settings app: what stays, goes, is parent-gated (3/3)

R16partly verified

Detail sheet: Child sees about 11 Settings rows; stock AOSP Settings stays hidden as engine room

SCOPE

Stage 1: ZuneSettings (Compose, platform-signed) owns android.settings.SETTINGS: home plus 8 native pages (Sound, Display, Accessibility-lite, Language [English only, D20], Battery, Storage, Emergency, About/Legal) and deep links to stock Wi-Fi, mobile data and Bluetooth. Stock Settings runs default-deny (about 40 of 437 activities enabled via a generated sysconfig component-override); SettingsIntelligence, HTMLViewer and CertInstaller removed. SystemUI trimmed to 6 tiles and a 3-item power menu. ZuneGuardian as Device Owner re-asserts restrictions at every boot. Zune-owned parent PIN gates the Parent area (device reset). Remote settings via signed policy: screen-timeout cap, volume cap, roaming, data limit, location, airplane lock, SOS gesture. Plus a small Java patch to hide the Wi-Fi preferences row (corrected from 'zero patches'). Effort about 8-9 engineer-weeks (estimate).

Stage 2:

- Gate patch (about 150 lines) so a parent-session token unlocks Wi-Fi advanced fields, data limits, APN and eSIM on device
- Platform Supervision PIN spike for a hardware-throttled credential
- Approved-networks-only mode and parent-pushed Wi-Fi credentials from the portal (prototype; needs Device Owner)
- Parent-approved captive-portal sign-in; TalkBack built from google/talkback plus TTS
- Travel time zone, richer emergency card, kid-friendly explainers on Wi-Fi/data pages, more languages (Stage 2 at earliest per D20)

OPEN ITEMS

- Confirm what 'standard' Wi-Fi/data includes; default: stock Wi-Fi, data, roaming, Bluetooth; proxy, VPN, hotspot, APN, eSIM gated or hidden.
- Allow a small Stage-1 Java patch (about 30 lines) to hide Wi-Fi preferences and captive-portal buttons?
- Accept captive-portal Wi-Fi (hotel, school) as unsupported in v1, with parent hotspot as workaround?
- Keep Bluetooth in Stage 1 (headphones) despite arbitrary-device pairing and keyboard surface?
- TalkBack/TTS launch-gating or Stage 2? Parent PIN: Zune-owned (default) vs platform Supervision PIN.

DO NOT TAKE AS FACT

Claim	Why
8-9 engineer-weeks for Stage 1; 2-4 days per upstream rebase	Unsupported estimates; breakdown omits per-device QA and accessibility testing; Catalyst growth suggests rebase cost is understated.
All Settings, SystemUI and supervision behaviour	Checked in GrapheneOS and LineageOS trees, not Google's android-17.0.0_r1 (AOSP hosts blocked).
Supervision flag states in 17.0 (cp2a), including time-limit flag only in trunk_staging	Confirmed in GrapheneOS tree only; Google's release-config map was not read.
Wi-Fi share QR exposes parent's Wi-Fi password	Inferred; the gate skip was read in source but the exposure was not run.
Whether Catalyst XML overlays cannot trim screens; tamper-resistance ratings of options	Inference; consistent with code but not tested.
TTS model-weight and training-data licence	Not checked; this is the real redistribution question.
CVAA (47 CFR Part 14), FCC e-label (47 CFR 2.935), WEA/National Alerts	Secondary sources only (search summaries); US law, written for a US launch that D18 replaced with India.

Superseded: Launch locales en-US and es-US (decision 6, row 21) superseded by D20 (English only). US legal basis (CVAA, FCC e-label, WEA, 911 and 'National Alerts') assumed a US launch: superseded by D18 (India first); Indian equivalents not covered. 'D13 device set' superseded by D17 (Pixel first). 'SMS not readable (D6)' moot under D19 (no SMS).

Selling the Image: Installer, Qualification, Updates, Licensing (1/3)

R17

partly verified

Detail sheet: Every Zune phone relocks with our key; OEM firmware redistribution rights remain unresolved.

DECISION

Web installer (WebUSB, fastboot.js) runs first as Zune's provisioning station; customer self-install later. Relock with our AVB key is mandatory; no OTA with OEM firmware before written OEM terms.

Skeptic pass with corrections; flashing mechanics reproduced from GrapheneOS/LineageOS files, but AOSP and Google licence text unreachable; legal and cost claims secondary or inferred.

WHY IT MATTERS

Relocking resists a child with no computer, but firmware updates must then ship in our own OTA, so OEM redistribution rights become a gating legal dependency.

KEY POINTS

- Relock with Zune's AVB key is mandatory for every 'Zune Ready' device; no unlocked tier; verified by attestation at the station. R17 §1, §4.1
- Custom-key relock is vendor-documented only for Pixel; Fairphone 3/4/5, Nothing, Sony, Motorola g-series are user-reported; Fairphone Gen 6/6+ untested. R17 F3
- Google's Pixel image terms bar modification or redistribution except per an enclosed, unread licence; commercial flashing is unresolved (gate G5). R17 F1, §9 claim 1
- v1 posture: fetch OEM firmware at flash time, flash unmodified, pin digests in Zune's vbmeta; its 'strongest' legal rating is inferred, not evidenced. R17 §3 option b
- Installer follows GrapheneOS's WebUSB fastboot.js (MIT) flow: 21-codename allowlist, unlock, flash, avb_custom_key, lock; Zune adds stricter pre-flight. R17 F5
- Pixel QPR1 firmware (2026-09-15) ships before AOSP source; auto-updated customer phones trip the anti-rollback pre-flight, forcing a monthly vendor backport. R17 §4.1 step 2, §7 risk 7
- Attestation alone is relay-bypassable; claim-gating and per-device seats need station-recorded serial, ID attestation and server-side anomaly checks. R17 F6
- Business model: one-time provisioning fee plus per-child subscription; cloud services are the paywall. Assume the image leaks; protect apps, backend, catalogue. R17 §1, §4.4

NUMBERS

Metric	Value	Basis
GrapheneOS installer device allowlist	21 codenames (incl. tegu = Pixel 9a, stallion = 10a) Read in web-install.js	sourced
LineageOS wiki entries needing OEM firmware first	374 of 743; 188 ship firmware (184 overlap) Recounted at lineage_wiki@902bd57, 2026-09-29	sourced
Android 17 vendor interface levels accepted	5 (FCM 7, 8, 202404, 202504, 202604) Read in GrapheneOS and LineageOS forks, not Google's tree	sourced
v1 station cost per unit	\$25-40 (labour only) R18 puts fully loaded cost at \$85-125	estimate
Self-install support load	20-30% need help; about \$4-6 blended Author's inference	estimate
New-platform qualification	4-6 engineer-weeks plus 3 units (about \$1.2k) Sibling model 1-2 weeks; 2-3 days per monthly release	estimate
US software-only price anchors	Bark \$14/mo or \$99/yr; Pinwheel \$15/mo Search summaries; vendor pages blocked; India anchors in R20	reported
Staged OTA rollout	1/10/50/100% over 7 days, kill switch Design proposal; roll forward, never back	estimate

UNLOCK AND CUSTOM-KEY RELOCK BY OEM (COMMUNITY DATA, 2026-09)

OEM	Unlock	Custom-key relock	Stance
Pixel	OEM toggle, then fastboot flashing unlock; carrier variants locked	Documented	Reference platform; D17 starts here
Fairphone	Per-device website code; two-step unlock	Reported FP3/4/5; Gen 6/6+ unverified	Was commercial candidate; D17 makes it later second source
Nothing (3)/(3a)	Plain fastboot	Reported; Nothing (2a) fails	Stage 2 candidate
Sony Xperia 1 V/VI	Portal plus IMEI code; US/carrier units may never unlock	Reported	Reference BSP only
Motorola	Portal code; g-series	Reported for g-series	Watch
Samsung, OnePlus, Xiaomi, ASUS	Samsung dropped unlock (One UI 8); OnePlus Deep Testing; ASUS tool ended	Newer OnePlus and Xiaomi ignore custom keys	Exclude

User reports on older firmware, not Android 17 tests of candidate phones. Samsung and ASUS are excluded for unlock policy, not failed relock tests.

Selling the Image: Installer, Qualification, Updates, Licensing (2/3)

R17

partly verified

Detail sheet: Every Zune phone relocks with our key; OEM firmware redistribution rights remain unresolved.

WHAT SHIPS IN AN UPDATE PATH

Option	What ships	Verdict
(a) System-only on stock vendor	system, system_ext, product	Dev tool only
(b) OEM firmware fetched at flash time, digests pinned in our vbmeta	Our partitions only	v1 posture; legal strength inferred, relock unproven
(c) OEM blobs inside Zune's OTA (GrapheneOS model)	Everything, one signed OTA	Needs OEM licence before OTA firmware
(d) LineageOS-style tree (Snapdragon)	Blobs plus vendor kernel	Stage 2; about 1 FTE-quarter per device (estimate)

HOW OTHERS DISTRIBUTE AND CHARGE

Project	Model	Charge
GrapheneOS	Pixels only; web and CLI installers; independent resellers	Free; donations
LineageOS	Community maintainers; 293 entries on 23.x; no official Android 17	Free
CalyxOS	Pixel, Fairphone, some Motorola; web installer	Free; membership \$700 year one, then \$10/yr
/e/OS (Murena)	30+ devices DIY; sells Fairphone 6 preinstalled	Free OS; EUR 649 phone; cloud EUR 1.99-24.99/month
iodé	40+ devices; install service only in Toulouse	Free; Premium EUR 3.99-6.99/month
Pinwheel, Troomi, Gabb	Rebadged hardware plus subscription; Pinwheel refuses BYO	Pinwheel \$15/month; Troomi \$24.95-34.95; Gabb from \$29.99 with cellular

No commercial BYO kids' OS vendor found (negative result). Prices are search summaries; CalyxOS confirmed, others not re-checked.

RISKS

Sev.	Risk	Mitigation
High	Google's firmware licence unanswered, Fairphone terms unsigned; R20 treats about 100 devices as the tolerated ceiling.	Re-flash service in v1, then pause; counsel and OEM outreach in parallel (gate G5).
High	Attestation depends on Google's remote key provisioning or a Zune proxy; Google's terms for non-certified OSeS unverified.	Set the RKP host property or run a proxy; confirm Google's terms before launch.
High	Pixel QPR1 firmware ships before AOSP source; auto-updated customer phones fail the anti-rollback pre-flight.	Track Google's latest factory image; budget a vendor-on-r1 backport for every monthly update.
High	AVB key is effectively unrotatable without unlock and wipe; compromise is recall-class.	Offline HSM for the AVB key; separate rotatable OTA signing key.
Medium	Custody of customer phones, support load and bricking liability rest on unvalidated estimates.	Counsel-reviewed custody terms; health-gated A/B OTAs; refund for Zune-caused bricks.

SCOPE

Stage 1: Station-mode web installer for one Pixel platform (9a/10a) with strict pre-flight, relock and attestation check; manual qualification against the Device Support Contract; Zune OTA server (staged rollout, health gating); licence and subscription through the parent portal; written OEM outreach in parallel.

Stage 2:

- Customer self-install web installer plus desktop app (offline bundle, Firefox/Safari users)
- Second platform through the Device Support Contract gates; OEM redistribution deals
- Incremental OTAs, per-build watermarking, published OS-layer patches for trust
- Partner install network, refurbished-device partners, school and IT bulk licences
- EU launch with CRA and digital-content-directive files

OPEN ITEMS

- Written answers from Google on commercial firmware redistribution; Fairphone and Nothing terms unsigned.
- Does the Pixel 10a relock with Zune's key on Android 17? Prototype on a sacrificial unit.
- Portal 'Release device' token is the one sanctioned unlock path; portal account takeover equals device unlock.
- Counsel: CPSIA framing of a returned kids' phone, FCC RF-parameter rule, DMCA exemption expiry (Oct 2027), GPL kernel duty.
- Founder approval of lapse policy (safe mode, security updates 12 more months) and brick policy defaults.

DO NOT TAKE AS FACT

Claim	Why
Option (b) is the strongest legal posture and 'fits personal-device use'	Inferred only; Google's licence text unread and its wording is broader ('used in any way except').
Google factory-image and driver licence wording	Google pages blocked; wording quoted from search summaries.
All Android 17 facts (vendor levels, AVB behaviour, fastboot)	AOSP hosts returned 403; checked only in GrapheneOS and LineageOS forks, not android-17.0.0_r1.
Fairphone Gen 6/6+ relocks with a custom key	Only forum reports; several FP6 threads report unlock problems or soft-bricks.
\$25-40 station cost, \$59 + \$12.99/mo pricing, 20-30% support load, \$300 brick goodwill	Inferred estimates with no source; USD and US-framed, superseded by D18.
Competitor prices (Troomi, Gabb, Bark, Pinwheel, Murena, iodé)	Search summaries; vendor pages blocked; Murena and iodé prices not re-checked.
No commercial BYO kids' OS vendor exists	Negative search result; cannot be proven. Android 17 built-in parental controls announced as free competitor.

Selling the Image: Installer, Qualification, Updates, Licensing (3/3)

R17

partly verified

Detail sheet: Every Zune phone relocks with our key; OEM firmware redistribution rights remain unresolved.

Superseded: US-only launch, US unlocked SKUs and USD pricing (\$59 + \$12.99/mo) (D18); VoLTE, E911, WEA, SMS/MMS qualification and cellular guardian-call lapse rules (D19); Fairphone Gen 6+ as commercial launch device (D17); 100-device beta becomes staged ~200 users (D21), though R20 keeps 100 as the ceiling before Google's written answer.

v1 Flash-and-Deliver Operations ^(1/2)

R18

partly verified

Detail sheet: Flashing takes about 10 minutes; eligibility, phone locks and custody are the real work.

DECISION

Run v1 as an in-person-first scripted service behind a hard eligibility gate, not a mini-factory: 4-phone stations, per-model signing keys, parent-authorised service-unlock, staged cohorts (now ~200 users per D21).

Pixel flash and relock mechanics reproduced from real GrapheneOS/LineageOS files; AOSP unreachable, cost and throughput are unsourced estimates, carrier and legal items secondary.

WHY IT MATTERS

Operations, not engineering, bound the pilot: recruitment, unfixable phones, per-unit cost and phone custody decide whether 200 users are reachable and what each costs.

KEY POINTS

- Unlock and lock each need a physical on-device confirmation and wipe; 'two presses' is Pixel-only (Fairphone adds unlock_critical). Cannot be scripted. R18 F1
- A carrier id on Pixel persist can block bootloader unlock; SIM-unlock differs from OEM unlock, so test the toggle per unit. R18 F2
- Pixel bootloader updates raise an anti-rollback counter; newer-than-bundle phones are blocked pre-unlock. Pixel 10a reportedly excluded from May 2026 increment. R18 F9
- Rotating the AVB key is a recall, not an OTA: each phone needs service-unlock, confirmation and wipe. Keep per-model keys. R18 F7
- With OEM unlocking off, a locked phone cannot be re-flashed over USB; a parent-authorised service-unlock must be built before second batch (G1). R18 F8, §4.5
- Attestation proves lock state and key hash, but needs Google's remote key provisioning, is relayable, and omits the OEM-unlock toggle. R18 §4.2 step 5
- Relocked Snapdragon phones may fall to bootloader exploits (Qualcomm GBL, EDL); keep Pixel the only pilot platform until each model is assessed. R18 F3
- Factory-mode app that creates attested keys and disables OEM unlock is an unmodelled bypass surface unless provably one-way after hand-over. R18 §4.2 Keys

NUMBERS

Metric	Value	Basis
Flash time per phone	5-20 min; about 10 unattended plus 2 presses Search summaries	reported
Hands-on time per phone	50-60 min hands-on; 90-120 min wall Unmeasured; measure in first batch	estimate
Throughput, one technician, 4-up	8-10 phones/day; 15-20 once QA scripted 1,000 devices about 8 weeks with 3 technicians	estimate
Station cell BOM	\$3-5k per 4-phone cell Bare-metal Ubuntu, certified cables, charging rack	estimate
Fully loaded cost per device	~\$122 / ~\$103 / ~\$85 at 10 / 100 / 1,000 USD; India unit cost about INR 1,850 (R20)	estimate
Self-install break-even	2.1-5k devices Corrected from 2.5-5k; \$150-250k build and \$50-70 saving unsourced	estimate
Bare-install price anchors	\$85 mail-in; \$60 remote-guided GrapheneOS installers; price install only, not QA or enrolment	reported
Fixed costs	Counsel \$10-25k; HSM/KMS \$1-5k; insurance \$1-3k/yr Excludes payment fees and cloud cost	estimate

PER-DEVICE COST MODEL (USD)

Line	10 devices	100 devices	1,000 devices
Hands-on minutes	90	55	30
Labour at \$35/h	52	32	18
Shipping, round trip	0	20	36
Support	35	20	12
SIMs, packaging	10	8	6
Failure/RMA reserve	20	18	9
Insurance, overhead	5	5	4
Total per device	~122	~103	~85

Estimates, US-framed. Omits payment fees, cloud cost, rejected applicants, liability payouts (R18 second pass).

RELEASE GATES (COHORT SIZES SUPERSEDED BY D21)

Gate	When	Requirement
G0	Before first external users	Data map, consent method, security programme, retention policy, incident plan
G1	Before second external batch	External portal and pairing review; no-browser and no-YouTube audits; service-unlock built
G2	Before third batch	Processor contracts; breach tabletop exercise
G5 (R17)	Before any OTA carrying OEM firmware	Written OEM firmware redistribution terms

Original 10/25/100 cohorts are replaced by the staged ~200-user phase (R20: 12/42/112/200). Gate content retained.

v1 Flash-and-Deliver Operations (2/2)

R18 partly verified

Detail sheet: Flashing takes about 10 minutes; eligibility, phone locks and custody are the real work.

KEY OPERATING DECISIONS

Decision	Options	R18 pick
Intake	In person; mail-in; new-in-box drop-ship	In person first, drop-ship next, mail-in last
Key custody	Fleet; per-model; per-device	Per-model keys; throwaway key only for staff cohort
Unlock bit after lock	On (recoverable) or off	Off: on lets a child with a PC unlock
OEM firmware	Fetch at flash time, or ship in our OTA	Fetch and pin for v1; OTA needs OEM terms

RISKS

Sev.	Risk	Mitigation
High	Customer phones in unfixable states: carrier id, anti-rollback, FRP, account locks.	Hard eligibility gate; customers self-test with screenshots; new-in-box phones preferred.
High	Service-unlock is unbuilt and untested against DISALLOW_FACTORY_RESET; return-to-stock promise depends on it.	Build before second batch: lift restriction, set toggle, reboot, physical unlock.
High	Attestation QA needs Google remote key provisioning that Zune must host or proxy and must not block.	Ship rkpd, run a reverse proxy, exempt it from the no-YouTube network policy.
High	Google firmware licence and Fairphone terms unanswered; staff-owned phones do not clear it.	Fetch-at-flash-time posture; written OEM answer (gate G5) before wider rollout.
Medium	Custody of customers' phones brings bailment liability, bricking exposure and lithium-battery shipping rules.	Bailee insurance, signed wipe consent, insured drop-off, counsel-reviewed terms; no loaner stock (D15).

SCOPE

Stage 1: Gate-checked intake; 4-phone cell running 'zune-station' (pinned platform-tools, serial-pinned fastboot, signed bundles); factory-mode QA and attestation check; hand-over with parent-portal QR, phone unenrolled with OEM unlock off; OTA from the first external device.

Stage 2:

- Scripted QA, multi-port rack and paid loaners
- Carrier-QR eSIM provisioning
- Remote-guided install, then the customer WebUSB installer (station CLI becomes the engine)
- Reuse for installer: signed manifest, per-model adapter, block-not-warn pre-flight, error-code telemetry
- Self-install only at 250+ devices a month for two months, or over 3 technician FTE, with 98%+ yield

OPEN ITEMS

- Who supplies phones: customer-supplied assumed (D16); company-owned loaners would conflict with D15's no-inventory stance.
- Decide hosting of attestation key provisioning before the first external batch; station QA depends on it.
- Serial-pinned multi-phone wrapper and one-way factory-mode mechanism are unbuilt; hands-on time and yield unmeasured.
- Return-to-stock promise needs service-unlock, which is scheduled after the first batch; resolve the timing conflict.
- Counsel before first external users: agreement, insurance, Google licence, carrier terms, lithium shipping; add Indian bailment terms (R19).

DO NOT TAKE AS FACT

Claim	Why
\$85-125 per-device cost, 50-60 min hands-on, throughput and yield	Estimates with no source; omit payment fees, cloud cost, rejected applicants, liability payouts.
Self-install break-even at 2.1-5k devices	Both inputs (\$150-250k build, \$50-70 saved per device) are unsourced.
Custom-key relock works on Pixel, Fairphone, Motorola, Nothing, Sony	User reports on older firmware (avbroot issue 299), not Android 17 tests; only Pixel is vendor-documented.
Verizon, AT&T, T-Mobile unlock policies	Search summaries only; carrier pages blocked; US-specific and moot for India.
Warranty effects (Google, Fairphone, Motorola, Nothing) and Magnuson-Moss	Search summaries; 'Pixel unlock is fine' is vendor policy, not law.
Any check against android-17.0.0_r1	AOSP hosts returned 403; forks of GrapheneOS and LineageOS were used instead.
Parallel fastboot hangs; lock each call	Rests on a 2013 bug report; unverified on current phones.
Attested verified-boot key equals sha256(avb_pkmd.bin)	Inferred; not computed because no public GrapheneOS avb_pkmd.bin exists.

Superseded: US-only framing, USD costs and 10/25/100 cohorts (D18, D21: staged ~200); US carrier and SIM matrix, VoLTE and inbound-SMS QA, 'parents see SMS' onboarding (D19); COPPA and CPSIA clauses (DPDP, R19); Fairphone-first ordering (D17: Pixel first).

India: Law and Regulation ^(1/3)

R19 partly verified

Detail sheet: India's child-data law has no clear exemption for parent-visible monitoring, Zune's core feature.

DECISION

Run an invite-only, parent-contracted pilot through an Indian entity with India-region hosting, Rule 10 parent verification at the flash visit, purpose-limited monitoring and 112 plus SOS, after Indian counsel sign-off.

Skeptic pass re-read DPDP Act, Rules, Telecom Act, POCSO judgment via GitHub mirrors; other Indian sources blocked (secondary); some items unverified.

WHY IT MATTERS

Rs 200 crore is the maximum penalty for breaching DPDP s.9 (children), and parent message review is the feature at legal risk. Indian counsel's opinion gates the pilot.

KEY POINTS

- DPDP s.9(3) bans tracking or behavioural monitoring of children; the Fourth Schedule has no parental-control or messaging exemption. [R19 §2.1](#)
- Defensible reading: the parent is part of the Data Principal (s.2(j)(i)) and consents, so Zune is a conduit. Unsettled; counsel gate G1. [R19 §1](#), [§10 Q1](#)
- Child means under 18. Rule 10 verifies parents via held or provided details or a DigiLocker-type token; cards are not named. [R19 §2.1](#)
- Rule 12 never lifts s.9(2), which bars processing likely to harm a child's well-being; it bites on the AI assistant and messenger. [R19 §2.1](#)
- Dates: s.9 and Rule 10 bind 13 May 2027; MeitY floated 13 Nov 2026 (ungazetted). Behave as if in force now. [R19 §2.1](#)
- Reported, unverified: Centre will bar under-18 social-media accounts via IT Rules. Assume Zune's messenger is a social media intermediary; stay closed-contact. [R19 §2.4](#)
- POCSO ss.19-21: intermediaries lose s.79 safe harbour without reporting to Indian police; the person in charge risks up to one year. [R19 §2.4](#)
- Replacing the OS drops the vendor's triple-press SOS (2016 panic-button rules, unread); Zune rebuilds it and keeps a 112-only path (A9). [R19 §2.2](#)

NUMBERS

Metric	Value	Basis
Maximum penalty, DPDP s.9	Rs 200 crore Security failures Rs 250 crore; breach-notification failures Rs 200 crore	sourced
Child threshold	Under 18 Act s.2(f); age bands do not reduce duties	sourced
s.9 and Rule 10 effective	13 May 2027 One source says 14 May; 16 Dec 2025 corrigendum unread	sourced
Breach reporting	Within 72 hours (detailed report to Board) Prompt first notice first (Rule 7); notice to each affected principal	sourced
Mandatory retention	At least 1 year Rule 8(3): personal data, traffic data, logs; scope ambiguous	sourced
CERT-In incident duty	6-hour reporting; 180-day logs in India Direction of 28 Apr 2022	reported
IT Rules 2026 takedown	3 hours (was 36); 2 hours for intimate-image/CSAM In force 20 Feb 2026; 2-hour figure partly unchecked	reported
RBI e-mandate	No re-authentication up to Rs 15,000; 24-hour pre-debit notice From 21 Apr 2026; notice duty is the issuer's	reported

RISK-RANKED OBLIGATIONS

Obligation	Applies	Severity	Design implication
DPDP s.9(3)	May 2027 (maybe Nov 2026)	Critical, Rs 200 crore	Strict-reading design plus counsel opinion
Rule 10 verification	Same	Critical	DigiLocker token or staff-checked ID; consent ledger
Under-18 account rule (proposed)	Unknown	High	Closed-contact positioning; MeitY outreach
POCSO ss.19-21; 3 h/2 h takedown	Now	High	Runbook, evidence vault, 24x7 on-call
Security, breach, CERT-In	Now / May 2027	High, Rs 250 crore	Encryption, logs, breach runbooks
Telecom Act ss.3, 42; TIUE; interception	Now, unclear	High	Counsel view; verified-parent traceability
2026 DoT authorisation and user-ID rules	In force Jun-Aug 2026	Unknown until read	Counsel reads before G1; may reach messenger, calls, phone custody
Panic button, 112, IMEI	Now	Medium	SOS, 112 path, IMEI untouched

R19 matrix has 12 rows; consumer law, RBI/GST/DLT and BIS/WPC (own hardware, D21) omitted here.

India: Law and Regulation (2/3)

R19 partly verified

Detail sheet: India's child-data law has no clear exemption for parent-visible monitoring, Zune's core feature.

US/CELLULAR PLAN VERSUS INDIA (D18-D21)

Item	US/cellular plan	India
Child privacy	COPPA: under 13; about \$53k per violation	DPDP s.9: under 18; up to Rs 200 crore
Parent consent method	Card, ID or KBA	ID or DigiLocker token; card does not count
Handset law	FCC, CPSIA	BIS, WPC, TEC only for own hardware (D21)
OS age signal	California AB 1043	None; IT Rules age proposal instead
CSAM reporting	NCMEC; 18 USC 2258A	POCSO: Indian police or cyber-crime portal
Parent reads messages	State wiretap and stored-communications law	Telecom Act s.42, IT Act s.69, DPDP s.9(3)
Cellular legal tree (R12)	Carrier of record, CPNI, CALEA, STIR/SHAKEN, SMS vault	Gone under D19
Emergency	911, E911, WEA	112 path and triple-press SOS

MINIMUM GATES BEFORE THE FIRST ~200-USER PHASE

Gate	Requirement
G1	Written Indian counsel opinion: s.9(3), messenger class, parent access versus interception
G2	Indian entity, India hosting, published grievance and DPDP contacts
G3	Rule 10 consent, withdrawal and erasure flows live
G4	Parent-only contract, notices, custody terms, warranty disclosure, insurance
G5	Security baseline, CERT-In and breach runbooks, 180-day logs
G6	POCSO and takedown on-call; tabletop drilled
G7	Vendor contracts, India-region inference, Anthropic written confirmation for child use
G8-G10	112 field test and SOS shipped; payments and DLT ready or free pilot; invite-only list, kill switch

RISKS

Sev.	Risk	Mitigation
High	s.9(3) has no guidance or Board ruling; parent message review may count as prohibited monitoring.	Purpose-limited monitoring; written counsel opinion; seek s.9(4), s.9(5) or s.17(5) exemption dialogue.
High	Reported under-18 social-media rule and Karnataka under-16 plan could capture a 1:1 child messenger.	Closed-contact positioning: no discovery, handles, feeds or groups; MeitY outreach after counsel.
High	Number-free child messenger is policy-adverse given DoT SIM-binding direction (Nov 2025).	Invite-only, parent-linked identity; each child traces to a verified adult.
High	2026 DoT authorisation and radio-equipment-possession rules unread; may reach voice/video service or phone custody.	Counsel reads them before gate G1 (1-2 weeks).
High	Whether 112 works on an AOSP-built phone with data-only or no SIM is untested.	Field-test 112 on Jio, Airtel, Vi, BSNL; tell parents emergency calling is not guaranteed.

SCOPE

Stage 1: Invite-only ~200-family pilot: Indian private limited; AWS Mumbai (Hyderabad DR); Indian NTP and 180-day logs; one verified adult per family (DigiLocker token or staff-checked ID) plus mobile OTP; consent ledger and both-family consent per contact pair; parent-directed, purpose-bound monitoring; 12-month retention vault; no analytics SDKs; 112-only path and triple-press SOS; POCSO and CERT-In runbooks.

Stage 2:

- Significant Data Fiduciary readiness (India DPO, DPIA, audit)
- Automated DigiLocker and consent-manager option; CSAM hash-matching
- Parent-key end-to-end encryption only if counsel clears IT Act s.69
- MeitY and Data Protection Board exemption dialogue; on-shore models (Sarvam or self-hosted)
- Indic notices; BIS, WPC, MTCTE, CEIR, ITSAR for own hardware (R21)

OPEN ITEMS

- Counsel must read MeitY's 16 Dec 2025 DPDP corrigendum; every Rule wording cited may change.
- Tiered monitoring (full content under 13, excerpts plus metadata at 13+) deviates from 'parent-visible' D7/D11; founder decides (A8 open).
- Hindi parent notices: ss.5(3)/6(3) give principals English or Eighth Schedule choice; needs founder exception to D20.
- Free invite-only first phase until payments, GST and DLT are ready; conflicts with R20's priced funnel.
- Is re-flashing a certified phone 'manufacturing' or a 'product service' (BIS, panic-button rules, ITSAR)?

DO NOT TAKE AS FACT

Claim	Why
Centre told Supreme Court on 28 Sep 2026 it will bar under-18 accounts (tiers 8-12, 12-16, 16-18)	Unverified: sources blocked, event four days old; treat as a reported development.
DoT SIM-binding extended to 31 Dec 2026; July 2026 username notices to WhatsApp, Telegram, Signal	Unverified; sources blocked. Only the Nov 2025 direction is corroborated.
Telecom licensing is not the main risk; OTT likely not authorisable	Inferred from a Sept 2024 TRAI item; the 2026 authorisation rules were never read.
Panic Button and GPS Rules 2016 (three-press panic call)	Name, scope and dates from memory; the PIB link was never opened.
Anthropic in-country inference via Bedrock Mumbai/Hyderabad	One unread trade article; Anthropic's own post does not say it.
Data Protection Board chair appointed June 2026	REFUTED: two sources say no chair or members appointed as of 1 Aug 2026.
No rule requires cellular voice on an India-sold handset; payment-aggregator and GST thresholds	Negative search or memory only; counsel to confirm.

India: Law and Regulation (3/3)

R19

partly verified

Detail sheet: India's child-data law has no clear exemption for parent-visible monitoring, Zune's core feature.

Superseded: Tiered-visibility default (excerpts plus metadata at 13+) conflicts with 'parent-visible' in D7/D11 and is not adopted until the founder decides; Hindi parent notices need an explicit exception to D20. Otherwise written against D18-D21.

India: Market, Connectivity, Language and Pricing (1/2)

R20

partly verified

Detail sheet: 200 Indian families are reachable; the INR 43-48k Pixel is the binding constraint.

DECISION

Default: launch in Bengaluru, in person, four phases to ~200 families; Wi-Fi first, optional nano-SIM; telephony deny-all except 112; English UI; Pixel 9a/10a only; pricing unvalidated.

Skeptic pass corrected many items; market and legal facts mostly search summaries, code checks use GrapheneOS/LineageOS forks not Google's tree, and unit economics are unsourced.

WHY IT MATTERS

The funnel is the real risk: families must own or buy a Pixel at INR 43-48k, and unit economics at INR 399/month are thin once cloud cost is counted.

KEY POINTS

- Pixel 10a: INR 49,999 launch, 47,999 now. The 39,999 sale price needs a bank offer plus exchange; budget INR 43-48k per unit. R20 §2.1
- Keep stock Telecom, IMS and CarrierSettings (Jio is VoLTE-only; 112 needs IMS) but deny everything except 112. R20 §1.3, §4.2
- Platform emergency numbers come from the modem, so '112 always works' is unproven; enforce 112-only in Zune's screen and field-test per carrier. R20 §2.2
- Child's SIM is an ordinary KYC'd SIM in the parent's name; moved to another phone, it escapes all OS controls. R20 §4.2
- Language (D20): English UI; assistant understands Hinglish but replies in simple English; a 300-prompt Indian-English/Hinglish safety eval gates release. R20 §4.4
- Cell-broadcast alerts: keep CellBroadcastReceiver and Indic fonts; overlay link_method=none and enable_text_copy=false so alert URLs cannot become links (D1/D2). R20 §1.4, §2.2
- Content: NCERT-aligned tags, never NCERT text; StoryWeaver (CC BY 4.0), ISRO, PhET, NASA; StoryWeaver bulk ingestion and ISRO rewrites need written permission. R20 §2.4
- Weather: Open-Meteo (USD 29/month) plus NDMA SACHET CAP feed and CPCB AQI; IMD API needs an agreement and a fixed-IP server proxy. R20 §2.5

NUMBERS

Metric	Value	Basis
Pixel 10a price in India	INR 49,999 launch; INR 47,999 now Flipkart Big Billion Days 39,999 needs bank offer plus exchange; about 43,999 without trade-in	reported
Planning cost per phone	INR 43-48k Refurbished Pixel 9a about INR 29.8-37.2k; 8a about 25.9k (Cashify)	estimate
Proposed price (unvalidated)	INR 2,499 activation + INR 399/month or 3,999/year GST-inclusive; USD/INR about 95.97 (RBI reference 2026-09-28)	estimate
In-person unit cost	about INR 1,850 Courier adds INR 800-3,000 round trip; omits payment fees and cloud cost	estimate
Cloud cost per child	INR 180-200/month INR 199 founding price is below this	estimate
Monthly net at 200 families, full price	about INR 26-30k After GST, cloud cost and about 2% gateway fees; original INR 68k was GST-only	estimate
Waitlist funnel	60,000 reach, 1,200 waitlist, ~300 eligible, ~200 paid Unsourced assumptions; test with a waitlist page	estimate
Open-Meteo Standard plan	USD 29/month, 1M calls Commercial licence; free tier is non-commercial	reported

STAGED PLAN TO ~200 FAMILIES

Stage	Families (cumulative)	How	Gate to next
C0	12 (12)	Staff and friends, own Pixels, in person, free	Carrier matrix passes; no open child-safety P1; counsel signs terms
C1	30 (42)	Bengaluru school-parent community; founding price	95%+ first-pass flash; 85%+ daily use; NPS 40+
C2	70 (112)	Bengaluru plus courier to Hyderabad, Pune	OTA 98% in 72 h; written Google firmware answer; 90%+ payments
C3	90 (200)	Three cities	75%+ eight-week retention; activation fee covers cost

Arithmetic sums to 202 (make C3 88). R19 wins on legal posture: run C0-C1 invite-only and free until counsel gates clear.

GATE G0 CARRIER MATRIX

Carrier	What is known	Gate G0 test
Jio	VoLTE-only, no 2G/3G fallback; custom-ROM data plausible (old posts)	Data attach, 5G, SIM and SIM-less 112, cell broadcast
Airtel	5G SA rolling out in phases; VoNR in limited areas	Same, plus correct fallback to 112 when on SA
Vi	NSA only	Same matrix
BSNL	4G about 96k sites on air	Same matrix; ask about data-card SIMs

'India has no consumer data-only SIM' is unverified; ask Airtel and BSNL first.

India: Market, Connectivity, Language and Pricing (2/2)

R20partly verified

Detail sheet: 200 Indian families are reachable; the INR 43-48k Pixel is the binding constraint.

CONTENT SOURCES AND LICENCE VERDICTS

Source	Verdict
StoryWeaver	CC BY 4.0; get Pratham Books' written yes before bulk bundling
ISRO	Free if accurate and acknowledged; rewrites not clearly covered
NCERT textbooks	Commercial use needs written permission; do not bundle
NDLI	Non-commercial only; exclude
DIKSHA	Platform MIT; content mixed CC licences; check per item
PhET, NASA, Wikipedia	Clean sources; Khan and TED are non-commercial (R06)
Public-domain epics	Verify each edition, e.g. Griffith's Ramayana

RISKS

Sev.	Risk	Mitigation
High	Funnel: INR 35-48k phone for a child; Pixel is small in India (about 2% of premium, weak source).	Waitlist page to test demand; refurbished after eligibility test; Pixel 8a only if price-sensitive.
High	Carrier behaviour on a custom OS (Jio IMS, SIM-less 112, Vi, BSNL) is untested.	Gate G0 carrier matrix before any family; promise no SIM-less 112 until tested.
High	Google firmware redistribution unanswered; blocks beyond about 100 devices.	Written Google answer gates phase C2; Indian counsel and Google outreach now.
High	DPDP parental visibility versus 'behavioural monitoring'; messenger may count as social media; Karnataka under-16 plan.	Run C0-C1 invite-only and free until counsel clears R19 gates G1-G10.
Medium	INR 199 founding price is below per-child cloud cost; courier price loses money past about INR 1,540 round trip.	Rework founding price; test pricing on the waitlist before charging; measure real cloud cost.

SCOPE

Stage 1: Four staged phases from Bengaluru. Pre-flash checks: GST invoice, IMEI via Sanchar Saathi KYM (SMS 14422), reject grey and US-SKU units, OEM-unlock and bootloader reads, battery health, nano-SIM tray. 112-only Emergency screen with read-aloud helpline cards (child cannot dial them: 'tell a grown-up or press SOS'). Carrier matrix G0; English UI; SACHET plus Open-Meteo weather.

Stage 2:

- Hindi and regional languages: Indic IME and fonts, ASR/TTS (AI4Bharat), Devanagari and Hinglish evals, translated content
- eSIM LPA and UPI AutoPay
- IMD data agreement; NCERT permission
- Nothing and Motorola India devices after relock tests; own hardware (R21)
- School mode, exam-focus schedule, sibling profiles, parent-opted location (counsel: DPDP bars tracking children)

OPEN ITEMS

- Confirm first city: Bengaluru default meets Karnataka's under-16 social-media announcement (6 Mar 2026); counsel to weigh.
- Test 112 with and without SIM on Jio, Airtel, Vi, BSNL; verify whether any consumer data-only SIM exists.
- Founder to confirm age range (A8); R20 proposes NEP stages, ages 6-14.
- Reserve of 4-6 company-owned replacement phones is a narrow D15 exception; founder approval needed.
- Reconcile pricing with R19: free invite-only first phase until payments, GST and DLT are ready.

DO NOT TAKE AS FACT

Claim	Why
India has no consumer data-only SIM	Unverified: none found, none ruled out; conflicts with A10's data-only SIM assumption.
AOSP cell-broadcast extreme/severe alerts are always on for MCC 404/405	Cited GrapheneOS commit disables always_on parsing; unproven for upstream AOSP.
AOSP always treats 112 as an emergency number, with or without SIM	True only in the legacy fallback; on Pixel the modem and ECC database supply the list.
Jio is VoLTE-only; data works on custom ROMs	Old forum posts and a help FAQ; untested for Zune's build.
Unit cost INR 1,850; 60,000-to-200 funnel; INR 2,499 + 399/month pricing	Unsourced assumptions; omit payment fees, cloud cost, rejected applicants, liability payouts.
Hinglish moderation classifier AUC about 0.75	Unverified; the paper covers toxicity, not child safety.
Nothing builds the (3a) and (3) in Chennai; OEM unlock available on Indian Pixel SKUs	Both unverified; test every unit for the unlock toggle.

Superseded: Nothing overturned by a D-number (written against D17-D21). Its public priced funnel and INR 199 founding price yield to R19's free invite-only first phase until counsel gates clear; Hindi and regional languages stay Stage 2 (D20).

Own-hardware roadmap for India ^(1/3)

R21 unverified

Detail sheet: Own hardware is gated, not scheduled; first device is a rebadged ODM phone

DECISION

Recommended: run the ~200-user Pixel phase first (D21); send paper RFQs now, commit money only at gates H1-H4. First own device: rebadged ODM 4G phone, voice off except 112.

No skeptic pass; evidence is mostly WebSearch summaries because fetches were blocked. Economics and timelines are the author's inference.

WHY IT MATTERS

Own hardware could widen the market below the INR 40k+ Pixel price and remove every unlock path, but adds patching, memory-cost and key-custody risk.

KEY POINTS

- Case is price and tamper-proofing, not margin: Pixel 10a is INR 47,999; an own device at INR 15-18k widens the funnel. ^{R21 s1}
- First device: rebadged ODM-platform 4G LTE phone on Snapdragon 4 Gen 2 or 6-series; precedents HMD Fusion X1 and Light Phone III. ^{R21 s1, 2.1}
- Patching is the central risk: entry SoCs lack Google long-support, so we own kernel, firmware and HAL patching; promise 3 years. ^{R21 s1, 2.1}
- Memory shock: mobile DRAM roughly doubled in Q2 2026 and is 43-45% of a low-end BoM; Nothing cancelled CMF Phone 3 Pro. ^{R21 2.5}
- No Indian incentive fits a startup: PLI ended 31 Mar 2026; MPMS needs INR 1,000 crore turnover (Indian brands), 10,000 crore otherwise. ^{R21 2.4}
- Own hardware can fuse our root key, disable debug and drop the unlock command; reject Unisoc (BootROM bug CVE-2022-38694). ^{R21 2.2}
- Compliance: roughly 3-5 months in parallel with DVT (estimate): BIS, WPC ETA, IMEI/TAC, SAR, EPR. Settle the product name first. ^{R21 2.3}
- Wi-Fi-only is the real regulatory simplification; a data-only cellular phone is not, because the modem, IMEI and TAC stay. ^{R21 s3}

NUMBERS

Metric	Value	Basis
Pixel 10a India price	INR 47,999 (~39,999 on sale) From report 20; USD 1 about INR 96.	reported
Assumed landed cost per own phone	USD 115 (range 95-150) Author assumption.	estimate
Assumed MRP / hardware contribution	INR 16,999 / ~INR 2,070 (USD 21) After 6% warranty reserve and 3% payments/logistics; GST creditable.	estimate
Net subscription per month	INR 140 (INR 399 gross) After cloud ~INR 190 and payment fee 8; 2.5 paying years = USD 44.	estimate
Contribution per unit	~USD 65 Hardware USD 21 plus subscription USD 44.	estimate
Breakeven volume	~8.5k (rebadge) / ~39k (ODM design) Vendor ODM guide claims ~22k units; low reliability.	estimate
Mobile DRAM shock	~2x in Q2 2026; sub-INR 10k phones +32% LPDDR4X 4 GB +75%; budget shipments -65% in H1 2026.	reported
BIS CRS (IS 13252)	INR 53,000 fee + lab INR 10-80k 30-45 working days; valid 2 years; licence in factory's name.	reported

UNIT ECONOMICS BY LIFETIME VOLUME (USD, ALL ESTIMATES)

Lifetime units	Path	Fixed cost	Team (3 yrs)	Peak working capital	Contribution	Net
1,000	(a) pilot lot	0.25M	0.30M	0.12M	0.07M	-0.49
10,000	(a) lots of 2-3k	0.25M	0.30M	0.3-1.2M	0.65M	About +0.1M (break) ~-8.5M
100,000	(b) ODM design	1.5M	1.2M	3-5M	7M	+4.3M (break) ~-39k

All inferred by the author. Supplier terms about 30% deposit, 70% before shipment (memory, unchecked): cash leaves ~100 days before first sale. Pixel path for comparison: ~INR 2,000 contribution in year one, ~1,700 a year after, no capex.

Own-hardware roadmap for India (2/3)

R21 unverified

Detail sheet: Own hardware is gated, not scheduled; first device is a rebadged ODM phone

OPTIONS FOR THE FIRST OWN DEVICE

Option	Updates and patches	Tamper and keys	Cost and time	Verdict
(a) Rebadged ODM platform (Dixtel, Optimus, Tinno-class)	A14/A15 vendor under our A17 system; 3-year patch promise	ODM fuses our key hash	About USD 0.25M fixed; 12-16 months RFQ to pilot	First device
(b) ODM-designed kids device (SD 7-series, QCM6490, Dimensity)	4-year BSP typical, negotiable	Full control; PTT/SOS keys, sealed SIM	About USD 1.5M fixed; 24-36 months	Only at 10k+/yr visibility
(c) Wi-Fi (+LTE) device on MediaTek Genio 510/700	4-year BSP; 10-year chip lifecycle	Avoid Unisoc; verify MediaTek BROM	Lower BoM; no IMEI/TAC/cellular approvals; conflicts with D14	Stage-2 home SKU
(d) OEM partnership (Fairphone-style, Nothing, Lava, HMD)	OEM cadence	OEM rarely cedes keys (inferred)	Fairphone not in India; HMD uses own software	Fallback

Costs and times are author estimates; report 02 says 12-24 months to pilot.

HARDWARE GATES FROM THE 200-FAMILY PIXEL PHASE

Gate	When	Pass criteria
H1: start RFQ negotiation	End of C1 (~M5)	D30 retention 85%+; first-pass flash 95%+
H1 (cont.)	(same)	Price top lost-family reason for 40%+ of drop-offs; 3,000 waitlist sign-ups
H2: commit EVT funds (USD 0.5M)	End of C2 (~M9)	OTA success 98% in 72 h; zero unrecovered bricks
H2 (cont.)	(same)	Median patch lag under 30 days for 3 straight months
H2 (cont.)	(same)	Zero open no-browser/no-YouTube bypass; counsel gates closed; SHC signed
H3: first purchase order	After DVT (~M15)	BIS, WPC, TAC and EPR done; memory allocation secured
H3 (cont.)	(same)	Independent tamper test passed on 300-500 pilot units
H4: scale to option (b)	1,000 units in field	W8 retention 75%+; returns under 6%; support under INR 150/device/month

Also H2: key ceremony rehearsed; under 2 support contacts per family per week. H4: patch SLA met; 10k units/yr visible. C1/C2 are cohorts of the 200-family phase (report 20). Thresholds are proposals awaiting founder approval.

RISKS

Sev.	Risk	Mitigation
High	BSP lag and patch ownership: entry SoCs have no Google long-support; 3-year promise needs unverified ODM SLAs.	Secure Hardware Contract: patch SLA (critical fix to OTA within 30 days), 4-year BSP, source escrow.
High	Memory cost and allocation: first PO may price 25%+ above today's quotes or go unfilled; small brands get no priority.	Gate H3 requires memory allocation secured before first PO; Pixel path stays the base.
High	Key custody: a fused key cannot be rotated; loss or leak is a fleet recall.	HSM with two-person custody before any ODM contact; ODM gets only the public hash.
Medium	Certification delays and brand-specific filings; BIS licence sits in the factory's name.	Settle product name by M6; hire BIS and EPR consultants; run certification parallel to DVT.
Medium	ODM trust and warranty: backdoor precedent (Adups 2016, from memory), battery recalls, no Indian service network.	Third-party audit of vendor blobs; no ODM pre-installs; 6% warranty reserve; build Indian service network.

Own-hardware roadmap for India (3/3)

R21

unverified

Detail sheet: Own hardware is gated, not scheduled; first device is a rebadged ODM phone

SCOPE

Stage 1: Not v1 (D21). Now, with no hardware spend: device-agnostic vendor/zune layer; HSM-held hardware root key with two-person custody before any ODM contact; OTA that updates modem, bootloader and vendor partitions; Secure Hardware Contract as the RFQ spec; paper RFQs (Dixtel, Optimus, a Tier-2 house, a Shenzhen design house); Indian Pvt Ltd, IEC code, trademark; optional cheap-tier retail phone test (~USD 30-60k, estimate).

Stage 2:

- Timeline (estimate, months from Oct 2026): EVT M10-12, DVT and certification M12-15, pilot lot M15-16, 1,000-unit limited sale M16-20.
- M20-30 decide option (b) custom design; first custom units about M30-36.
- Wi-Fi-only 'home' SKU on MediaTek Genio; custom device with hardware PTT/SOS keys, charge-only USB, camera and mic switches.
- Sealed SIM or eSIM with our LPA; own RKP attestation backend; kernel LTS bump at year 3; second ODM; repair network.
- EU/US variants (CPSIA, RED, CRA).

OPEN ITEMS

- Founder has not approved gates H1-H4, paper RFQs, silicon choice or the 3-year support promise; report defaults only.
- Counsel/BIS consultant must confirm: cellular ETA route, whether a voice-less device is a 'handset' under panic-button Rules, ITSAR, BCD rate.
- Jio/Airtel/Vi data-only attach is untested for our build (Jio is VoLTE-only; data without VoLTE reportedly works).
- Google Remote Key Provisioning for a non-GMS OS is unverified; own hardware needs its own attestation backend or a Google agreement.
- Product name must be settled by M6; BIS, TAC, WPC and EPR filings are brand-specific.

DO NOT TAKE AS FACT

Claim	Why
All market, regulatory and vendor facts (BIS, WPC, TEC, MPMS, ODM names, pricing)	WebSearch summaries only; WebFetch blocked for BIS, TEC, MediaTek and Indian press; a BIS consultant and counsel must confirm.
Unit economics: USD 115 landed, INR 16,999 MRP, USD 65 contribution, breakevens 8.5k and 39k	Author inference from stated assumptions; not sourced; sensitive to memory prices and subscription uptake.
Timelines: 12-16 months to pilot lot; first custom units M30-36	Author inference; optimistic against report 02's 12-24 months.
Memory shock figures (DRAM ~2x, 43-45% of BoM, sub-INR 10k +32%, shipments -65%)	Secondary press summaries; small brands' allocation priority is inferred.
BCD on finished phones 15% vs 20%; BIS fee INR 53,000	Conflicting secondary sources; the report itself says verify.
Supplier terms 30% deposit; Adups 2016 backdoor; Genio has no integrated modem	From memory, unchecked.
Smartphones absent from MTCTE phases I-VI; 83 ITSAR requirements only proposed	Per phase lists and a secondary source; confirm with TEC.
Android 17 accepts old vendors (FCM 7 to 202604)	Read in a GrapheneOS mirror, not Google's tree (403); no boot test.

Glossary

Terms an Android and infrastructure reader will want pinned down

Term	Meaning
AOSP	Android Open Source Project: Google's open base of Android
AVB	Android Verified Boot; a custom key lets the bootloader trust only our images
Re-lock	Locking the bootloader again after flashing, so nothing else can be installed
A/B + OTA	Dual-slot system with over-the-air updates and rollback
adevtool	GrapheneOS tool that builds Pixel vendor modules from Google's stock images
Cuttlefish	Google's virtual Android device, used for hardware-free testing
SPL	Security patch level, e.g. 2026-06-05
GKI	Generic Kernel Image: Google's common Android kernel
Mainline / APEX	Updatable system modules; with no Google Play they update only through our OTA
WebView	System web renderer that apps embed; a browsing risk we must contain

Term	Meaning
Supervision framework	Android 17 SupervisorManager / ROLE_SUPERVISION for parental controls
FGS	Foreground service: Android's rule for background audio, camera and mic
LiveKit / SFU	Open-source real-time media server for voice, video and push-to-talk
DSC	Device Support Contract: our checklist a phone must pass to be qualified
DPDP Act	India's Digital Personal Data Protection Act 2023 (child = under 18)
COPPA	US children's privacy law (under 13)
BIS / WPC	Indian product-registration and radio-approval regimes
IMD	India Meteorological Department
WEA	US Wireless Emergency Alerts; India uses cell-broadcast alerts
GMS	Google Mobile Services (Play, FCM...): not used in Zune